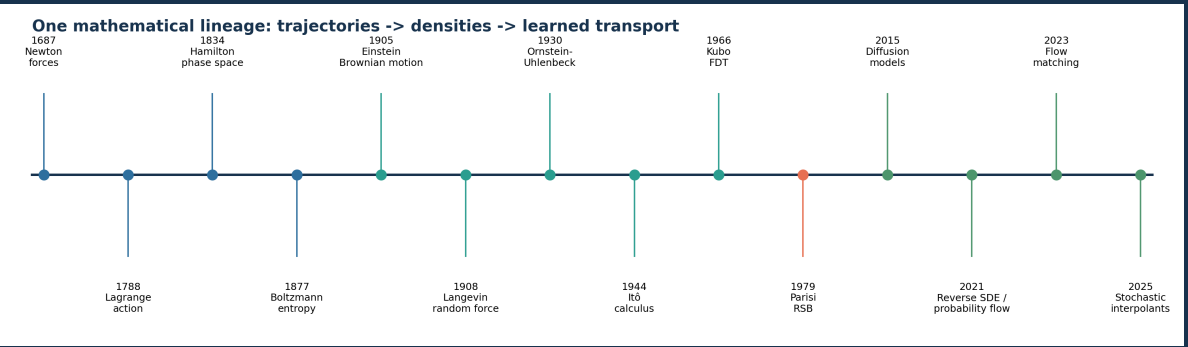
From Newton to Diffusion Models

Mechanics, statistical physics, stochastic dynamics, and generative flows from first principles

Derivations, intuition, numerical experiments, tables, and an executable notebook



Reproducible research package, September 2026

Prepared for Giovanni Mantovani

Preface

This book is designed to answer a single long question: how do we travel, without conceptual gaps, from Newton's equation for one trajectory to a neural model that transports noise into data? The route is not a sequence of unrelated formalisms. Each new theory appears because the previous description becomes inefficient, incomplete, or unstable at a new scale.

Newtonian mechanics predicts trajectories. The stationary-action principle reorganizes the same dynamics around variations of whole paths. Lagrangian mechanics makes constraints and symmetries transparent; Hamiltonian mechanics turns evolution into geometry in phase space. Statistical mechanics replaces an impossible microscopic state by a probability distribution over states. The continuity and Liouville equations describe how such distributions move. Stochastic mechanics adds unresolved degrees of freedom as noise, producing Itô calculus and the Fokker–Planck equation. Non-equilibrium statistical mechanics explains response, dissipation, and fluctuation. Spin glasses show what happens when a probability landscape becomes frustrated and decomposes into many states. Quantum mechanics retains Hamiltonian structure while replacing classical probabilities by amplitudes.

Brownian motion then supplies the historical and mathematical bridge to modern diffusion models. Langevin dynamics samples a Boltzmann energy landscape. Score matching learns its gradient without knowing the partition function. A diffusion model learns one such score at every noise level, then integrates a reverse stochastic differential equation or an equivalent deterministic probability-flow ordinary differential equation. Flow matching begins from the continuity equation and learns the velocity of a chosen probability path directly.

The treatment is formal enough for doctoral study but intentionally reconstructs every central equation from elementary objects. Boxes labelled "Intuition" explain what the symbols are doing; "Derivation" boxes isolate computations; "Common pitfall" boxes distinguish statements that are often conflated. Every major transition is paired with an experiment in the companion code.

What is in the ZIP package?

The PDF is the mathematical spine. The directory `src/physicsdiff/` contains reusable implementations; `experiments/` produces all figures and metrics; `notebooks/from_newton_to_diffusion_walkthrough.executed.ipynb` is the executable tour; and `interactive/index.html` opens the browser-based Plotly experiments. Static versions of all visuals appear in the PDF because ordinary PDF readers do not reliably execute interactive JavaScript.

Conventions

We set Boltzmann's constant $k_B = 1$ in most derivations and restore it when its physical meaning matters. Vectors are bold; a microstate is usually z; a probability density is ρ or p; the score of a density is $s_t(x) = \nabla_x \log p_t(x)$. Repeated indices are summed only when explicitly stated. Stochastic differentials are Itô unless marked otherwise.

A reader's map

Goal	Read first	Then compute
Classical mechanics from scratch	Chapters 1–4	Experiments 01–05
Equilibrium statistical mechanics	Chapters 5–9	Experiments 06–10
Stochastic and non-equilibrium dynamics	Chapters 10–12	Experiments 10–12, 16–19
Spin glasses and replicas	Chapter 13	Experiment 13
Quantum bridge	Chapter 14	Experiment 14
Diffusion and flow models	Chapters 15–23	Experiments 20–27

Intuition

The unifying object is not "energy" alone. It is the evolution of a distribution under a vector field and, when unresolved degrees of freedom are present, under diffusion. Hamiltonian mechanics, Liouville's theorem, Langevin sampling, score-based diffusion, continuous normalizing flows, and flow matching are different answers to the same structural question: what equation transports mass in state space?

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Part I

From trajectories to ensembles

1

Newtonian mechanics from scratch

1.1 What a mechanical theory must predict

A classical mechanical state is the information required to predict future motion. For a point particle in three-dimensional Euclidean space this state is its position $q(t) \in \mathbb{R}^3$ and velocity $\dot{q}(t)$. Newtonian mechanics postulates a second-order differential equation

$$m\ddot{q}(t) = F(q(t), \dot{q}(t), t), \tag{1.1}$$

where $m > 0$ is inertial mass and F is the total force. Given $q(t_0)$ and $\dot{q}(t_0)$, a sufficiently regular force field determines a local trajectory. This is the first important structural fact: a second-order equation needs two initial conditions.

Equation (1.1) is often presented as a formula to memorize. It is better understood as three claims. First, motion is deterministic at the level of the model. Second, interactions add as vectors: $F = \sum_a F_a$. Third, acceleration, not velocity, is the immediate response to force. The latter separates Newtonian dynamics from overdamped stochastic dynamics, where friction removes momentum so rapidly that force controls velocity instead.

Intuition

Mass is resistance to changing velocity. Two bodies subject to the same force acquire accelerations in inverse proportion to their masses. A force does not "cause motion"; a force changes momentum. Uniform motion requires no net force.

1.2 Momentum, impulse, and conservation

Define momentum $p = m\dot{q}$. Then Newton's equation is

$$\dot{p} = F. \tag{1.2}$$

Integrating over a time interval gives the impulse–momentum relation

$$p(t_1) - p(t_0) = \int_{t_0}^{t_1} F(q(t), \dot{q}(t), t)\, \mathrm{d}t. \tag{1.3}$$

If $F = 0$, momentum is conserved. For N interacting particles, suppose internal forces obey Newton's third-law form $F_{ij} = -F_{ji}$. Summing $m_i\ddot{q}_i = F_i^{\text{ext}} + \sum_{j\neq i} F_{ij}$ cancels all internal pairs and gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{N} p_i = \sum_{i=1}^{N} F_i^{\text{ext}}. \tag{1.4}$$

Thus total momentum is conserved when the external force vanishes. The deeper Lagrangian explanation will be translation symmetry; Newtonian cancellation is its concrete particle-level manifestation.

Angular momentum about the origin is $L = q \times p$. Differentiating,

$$\dot{L} = \dot{q} \times p + q \times \dot{p} = q \times F = \tau, \tag{1.5}$$

because $\dot{q} \times m\dot{q} = 0$. Hence zero torque implies conservation of angular momentum. A central force $F(q) = f(\|q\|)q$ has zero torque, so the trajectory remains in the plane perpendicular to the constant vector L.

1.3 Work and energy

The kinetic energy of one particle is

$$T = \frac{1}{2} m \|\dot{q}\|^2. \tag{1.6}$$

Taking a derivative and using Newton's equation,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = m\dot{q} \cdot \ddot{q} = F \cdot \dot{q}. \tag{1.7}$$

Therefore the change in kinetic energy is the work performed along the path:

$$T(t_1) - T(t_0) = \int_{t_0}^{t_1} F \cdot \dot{q}\, \mathrm{d}t = \int_{q(t_0)}^{q(t_1)} F \cdot \mathrm{d}q. \tag{1.8}$$

A force is conservative on a simply connected domain if $F = -\nabla V$ for a scalar potential V. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}(T + V) = F \cdot \dot{q} + \nabla V \cdot \dot{q} = 0, \tag{1.9}$$

so the mechanical energy $E = T + V$ is conserved. Conservative forces make the work path-independent; non-conservative forces such as viscous drag $F = -\gamma \dot{q}$ dissipate energy:

$$\dot{E} = -\gamma \|\dot{q}\|^2 \le 0. \tag{1.10}$$

Derivation: projectile motion

For a projectile close to Earth's surface, $F = (0, -mg)$. With initial state $q(0) = (0, h)$ and $\dot{q}(0) = (v_0 \cos\theta, v_0 \sin\theta)$,

$$x(t) = v_0 \cos\theta\, t, \tag{1.11}$$

$$y(t) = h + v_0 \sin\theta\, t - \frac{1}{2} g t^2. \tag{1.12}$$

Eliminating $t = x/(v_0 \cos\theta)$ produces a parabola. The horizontal momentum is conserved because gravity is vertical; the total energy $\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + mgy$ is constant until impact.

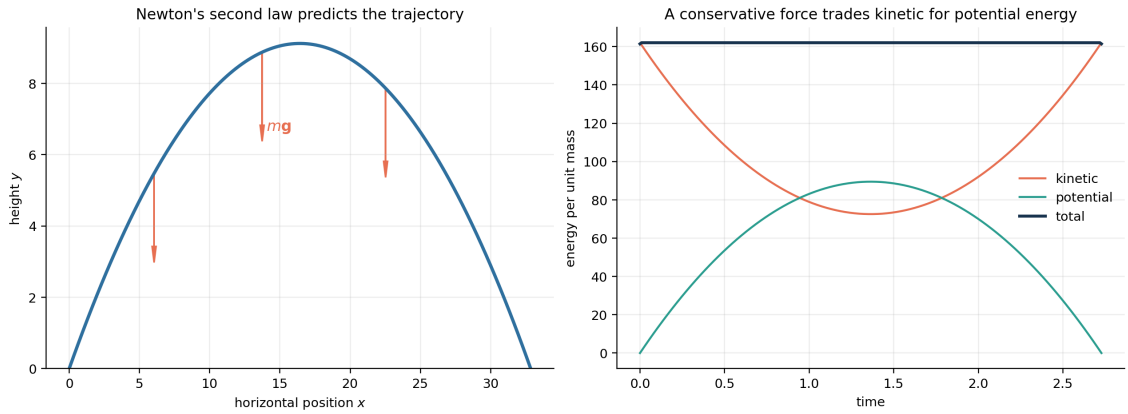


Figure 1.1: Newton's second law integrated for a projectile. The velocity changes by a constant vertical acceleration while the horizontal component remains fixed.

1.4 From one particle to configuration space

For N particles in three dimensions, collect all coordinates into

$$q = (q_1, \dots, q_N) \in \mathbb{R}^{3N}. \tag{1.13}$$

This $3N$-dimensional space is configuration space. A single point q describes an entire arrangement of particles. Newton's equations become

$$M\ddot{q} = F(q, \dot{q}, t), \tag{1.14}$$

with a diagonal mass matrix M. Constraints reduce the number of independent coordinates. A particle constrained to a circle in the plane appears to require (x, y), but the condition $x^2 + y^2 = R^2$ leaves only one degree of freedom, naturally represented by an angle θ.

This motivates generalized coordinates $q^1, \dots, q^n$ adapted to the geometry. Newtonian vector balance can handle constraints by adding unknown reaction forces, but this becomes cumbersome: the constraint forces must be solved even when they do no virtual work. Lagrangian mechanics will eliminate many of them automatically.

1.5 Initial-value problems and stability

Write the second-order equation as a first-order system for $z = (q, v)$:

$$\dot{z} = \begin{pmatrix} v \\ M^{-1}F(q, v, t) \end{pmatrix} = f(z, t). \tag{1.15}$$

If f is locally Lipschitz in z, the Picard–Lindelöf theorem gives local existence and uniqueness. Deterministic does not mean numerically predictable forever. Nearby initial conditions can separate exponentially in chaotic systems, while discretization error accumulates under numerical integration. A theory of dynamics therefore includes both its differential equation and an understanding of which numerical method respects its structure.

1.6 Why naive numerical integration can mislead

Consider the harmonic oscillator

$$\ddot{q} + \omega^2 q = 0, \qquad E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 q^2. \tag{1.16}$$

The exact orbit is an ellipse in (q, v) and has constant energy. Explicit Euler updates

$$q_{n+1} = q_n + hv_n, \qquad v_{n+1} = v_n - h\omega^2 q_n. \tag{1.17}$$

The update matrix has determinant $1 + h^2\omega^2 > 1$, so it expands phase-space area and injects energy. Symplectic Euler instead uses the new velocity in the position update,

$$v_{n+1} = v_n - h\omega^2 q_n, \qquad q_{n+1} = q_n + hv_{n+1}, \tag{1.18}$$

and its determinant is one. Velocity Verlet is second-order and time-reversible. Its energy is not exactly constant at each step, but its error remains bounded over long times because it follows the exact dynamics of a nearby Hamiltonian.

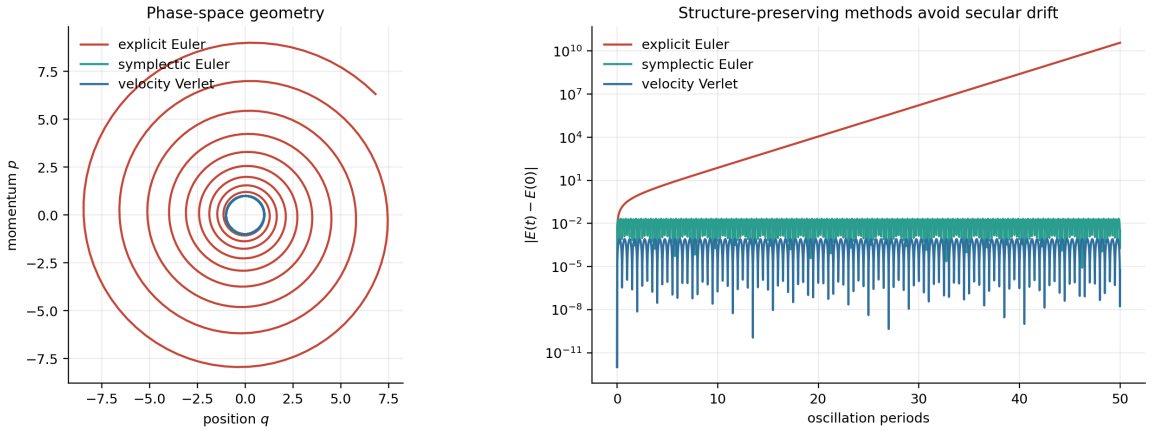


Figure 1.2: The same oscillator integrated by different schemes. Explicit Euler creates spurious energy, whereas symplectic schemes preserve the qualitative geometry and bounded energy error.

Computational experiment

Run `experiments/generate_all.py` or the corresponding notebook section. At the chosen step size, the measured maximum energy errors are approximately 3.80×10^{10} for explicit Euler, 2.08×10^{-2} for symplectic Euler, and 8.00×10^{-4} for velocity Verlet. The point is not merely that one method is more accurate: the successful methods encode Hamiltonian geometry.

1.7 Where Newton's formulation becomes awkward

Newton's laws are complete for classical point particles, but they hide several structures:

1. constraints and curved coordinates require explicit reaction forces;
2. symmetries are not systematically connected to conserved quantities;
3. fields and relativistic theories are naturally expressed as variational principles;
4. the equation focuses on local acceleration rather than the geometry of whole paths.

The stationary-action principle addresses all four. It does not replace Newton's law with mysticism: for ordinary conservative particles it derives the same equation from a scalar functional of the entire trajectory [20, 49].

Bridge to the next theory

Newton asks: given the force at this instant, what is the acceleration? Variational mechanics asks: among nearby admissible paths with fixed endpoints, which path makes the action stationary? The two questions produce the same motion when the action is chosen correctly, but the second reveals constraints, symmetry, and geometry.

2

Stationary action: local dynamics from global variations

2.1 A functional is a function of a path

An ordinary function maps a number to a number. A functional maps an entire function to a number. Given a path $q : [t_0, t_1] \to \mathbb{R}^n$, define the action

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t)\, \mathrm{d}t, \tag{2.1}$$

where L is the Lagrangian. Hamilton's principle states that the physical path makes S stationary under infinitesimal variations that leave the endpoints fixed.

Stationary does not always mean globally minimal. It means that the first-order change vanishes. The action may be a minimum for sufficiently short intervals, a maximum, or a saddle. The correct name is therefore the principle of stationary action, not universally "least action."

2.2 The first variation

Perturb the path by

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t), \tag{2.2}$$

where $\eta(t_0) = \eta(t_1) = 0$. The first variation is

$$\delta S[q;\eta] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0}. \tag{2.3}$$

Using the chain rule,

$$\delta S = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q}\cdot\eta + \frac{\partial L}{\partial \dot{q}}\cdot\dot{\eta}\right]\mathrm{d}t \tag{2.4}$$

$$= \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right]\cdot\eta\,\mathrm{d}t + \left[\frac{\partial L}{\partial \dot{q}}\cdot\eta\right]_{t_0}^{t_1}. \tag{2.5}$$

The boundary term vanishes because the endpoints are fixed. If $\delta S = 0$ for every smooth test variation η, the fundamental lemma of the calculus of variations implies

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0, \qquad i = 1, \dots, n.} \tag{2.6}$$

These are the Euler–Lagrange equations.

Recovering Newton's law

For a particle in a potential, choose

$$L(q, \dot{q}) = T - V = \frac{1}{2}m\|\dot{q}\|^2 - V(q). \tag{2.7}$$

Then $\partial L/\partial \dot{q} = m\dot{q}$ and $\partial L/\partial q = -\nabla V$. Equation (2.6) gives

$$m\ddot{q} = -\nabla V = F. \tag{2.8}$$

Thus the familiar difference $T - V$ is not arbitrary: its velocity derivative creates momentum, and its position derivative creates conservative force.

2.3 Landau's route to the free-particle Lagrangian

Landau and Lifshitz begin from symmetry rather than guessing $T - V$ [36]. For a free particle, homogeneity of space and time imply that the Lagrangian cannot depend explicitly on q or t; isotropy implies dependence on velocity only through v^2. Hence $L = L(v^2)$.

Now change to an inertial frame moving at infinitesimal constant velocity ε. Galilean relativity sends $v \mapsto v + \varepsilon$. To first order,

$$L((v+\varepsilon)^2) - L(v^2) = 2L'(v^2)v \cdot \varepsilon + O(\varepsilon^2). \tag{2.9}$$

Two Lagrangians that differ by a total derivative $\mathrm{d}f(q,t)/\mathrm{d}t$ generate the same Euler–Lagrange equations. Galilean invariance therefore permits the change above only if its velocity dependence is linear and can be written as $\mathrm{d}(a \cdot q)/\mathrm{d}t = a \cdot v$. This requires $2L'(v^2)$ to be a constant, conventionally m. Integrating,

$$L_{\text{free}} = \frac{1}{2}mv^2 + C. \tag{2.10}$$

The additive constant does not affect the equations. For noninteracting particles, additivity gives $L = \sum_i \frac{1}{2}m_i v_i^2$. Interactions that depend only on configuration enter as $-V(q)$.

Intuition

The kinetic term is fixed by the symmetries of nonrelativistic spacetime. It is not chosen because "energy should be in the action." The coefficient called mass appears as the response required by Galilean transformations.

2.4 Why total derivatives do not matter

Let

$$L'(q,\dot{q},t) = L(q,\dot{q},t) + \frac{\mathrm{d}}{\mathrm{d}t}F(q,t). \tag{2.11}$$

Then

$$S'[q] = S[q] + F(q(t_1),t_1) - F(q(t_0),t_0). \tag{2.12}$$

For fixed endpoints the added term is constant under variations, so $\delta S' = \delta S$. This "gauge freedom" in the Lagrangian is essential. It explains why a symmetry need only preserve L up to a total derivative and anticipates gauge potentials in electromagnetism and phase transformations in quantum mechanics.

2.5 Discrete action and a computable variational principle

Let $t_k = t_0 + kh$, $k = 0, \ldots, N$, and approximate a one-dimensional free-particle action by

$$S_h(q_0, \ldots, q_N) = \sum_{k=0}^{N-1} \frac{m}{2h}(q_{k+1} - q_k)^2. \tag{2.13}$$

The endpoints $q_0 = a$ and $q_N = b$ are fixed; the interior points are variables. Differentiating with respect to q_k gives

$$\frac{\partial S_h}{\partial q_k} = \frac{m}{h}(2q_k - q_{k-1} - q_{k+1}). \tag{2.14}$$

Stationarity implies $q_{k+1} - q_k = q_k - q_{k-1}$, so the increments are constant and the path is a straight line. The discrete variational equation is the finite-difference form of $\ddot{q} = 0$.

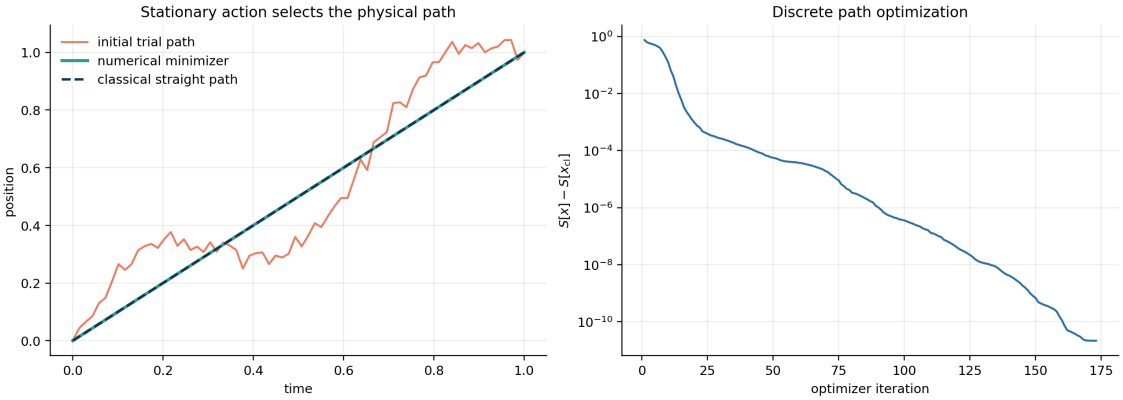


Figure 2.1: A random path with fixed endpoints is optimized by reducing its discrete action. The solution approaches the constant-velocity classical path.

Computational experiment

The supplied optimizer starts from a rough path whose discrete action is about 3.3293. It converges to 0.50000000002, matching the analytic classical action $m(b-a)^2/[2(t_1-t_0)] = 0.5$. This is a direct numerical demonstration of a functional becoming stationary.

2.6 The second variation and stability

Stationarity determines candidate paths. Stability is encoded by the second variation. For

$$L = \frac{1}{2}m\dot{q}^2 - V(q), \tag{2.15}$$

the second-order change around a classical path is

$$\delta^2 S[\eta] = \int_{t_0}^{t_1} \left(m\dot{\eta}^2 - \eta^\top \nabla^2 V(q_{\mathrm{cl}}(t))\eta \right) \mathrm{d}t. \tag{2.16}$$

Positive definiteness means a local minimum. A zero mode often signals a continuous symmetry; a negative mode signals a saddle direction. In semiclassical quantum mechanics, fluctuations of this quadratic form control the prefactor around a classical path.

2.7 Maupertuis and fixed-energy paths

There are several "action principles." Hamilton's principle fixes the initial and final times. Maupertuis' principle fixes the energy and makes stationary the abbreviated action

$$W = \int p \cdot \mathrm{d}q. \tag{2.17}$$

For a natural system $H = p^2/(2m) + V(q) = E$, one can eliminate the speed and obtain a geometric path principle. The trajectory is a geodesic of the Jacobi metric

$$\mathrm{d}s_J^2 = 2m(E - V(q))\,\mathrm{d}q^2. \tag{2.18}$$

This already hints that dynamics can be geometry. Hamiltonian mechanics will make that statement precise in phase space [5, 46].

2.8 Noether's theorem in its simplest form

Suppose an infinitesimal transformation $q \mapsto q + \varepsilon X(q,t)$ changes the Lagrangian by a total derivative $\varepsilon \mathrm{d}B/\mathrm{d}t$. Along a solution,

$$J = \frac{\partial L}{\partial \dot{q}} \cdot X - B \tag{2.19}$$

is conserved. For translations, X is constant and J is momentum. For rotations, $X = \omega \times q$ and J is angular momentum projected along ω. For time translation, the conserved quantity is energy. Noether's theorem turns conservation laws from isolated tricks into consequences of symmetry [51].

Common pitfall

The variational principle does not imply that a particle examines every possible path and chooses one. It is a mathematical characterization of the trajectory, not a microscopic decision process. Likewise, the action has units of energy times time; it is not itself energy.

Bridge to the next theory

The Euler–Lagrange equation is coordinate-covariant: we can use angles, lengths, or any smooth generalized coordinates. This is the practical power of Lagrangian mechanics and the natural next step.

3

Lagrangian mechanics: coordinates, constraints, and symmetry

3.1 Generalized coordinates

A mechanical configuration need not be represented by Cartesian particle positions. If the admissible configurations form an n-dimensional manifold Q, choose local coordinates $q = (q^1, \dots, q^n)$. A velocity is a tangent vector $\dot{q} \in T_q Q$, and the Lagrangian is a scalar function on the tangent bundle,

$$L : TQ \times \mathbb{R} \to \mathbb{R}, \qquad (q, \dot{q}, t) \mapsto L(q, \dot{q}, t). \tag{3.1}$$

This geometric language simply records that position and velocity transform differently under coordinate changes.

For particles with Cartesian positions $r_a = r_a(q, t)$, the velocities are

$$\dot{r}_a = \sum_i \frac{\partial r_a}{\partial q^i} \dot{q}^i + \frac{\partial r_a}{\partial t}. \tag{3.2}$$

Substituting into the kinetic energy automatically produces the coordinate-dependent mass matrix. For time-independent constraints,

$$T = \frac{1}{2} \sum_{i,j} M_{ij}(q) \dot{q}^i \dot{q}^j, \tag{3.3}$$

where $M(q)$ is positive definite. The Euler–Lagrange equations include the inertial terms generated by curved coordinates.

3.2 Example: the simple pendulum

A pendulum bob of mass m is constrained by $x^2 + y^2 = \ell^2$. Set

$$x = \ell \sin\theta, \qquad y = -\ell \cos\theta. \tag{3.4}$$

Then $T = \tfrac{1}{2} m \ell^2 \dot{\theta}^2$ and $V = mg\ell(1 - \cos\theta)$, so

$$L = \frac{1}{2} m \ell^2 \dot{\theta}^2 - mg\ell(1 - \cos\theta). \tag{3.5}$$

The Euler–Lagrange equation is

$$\ddot{\theta} + \frac{g}{\ell} \sin\theta = 0. \tag{3.6}$$

The unknown tension never appears because it is normal to the allowed virtual displacement and performs no virtual work.

For $|\theta| \ll 1$, $\sin\theta \approx \theta$, giving a harmonic oscillator with frequency $\omega_0 = \sqrt{g/\ell}$. At finite amplitude the frequency depends on energy and the separatrix distinguishes oscillations from rotations.

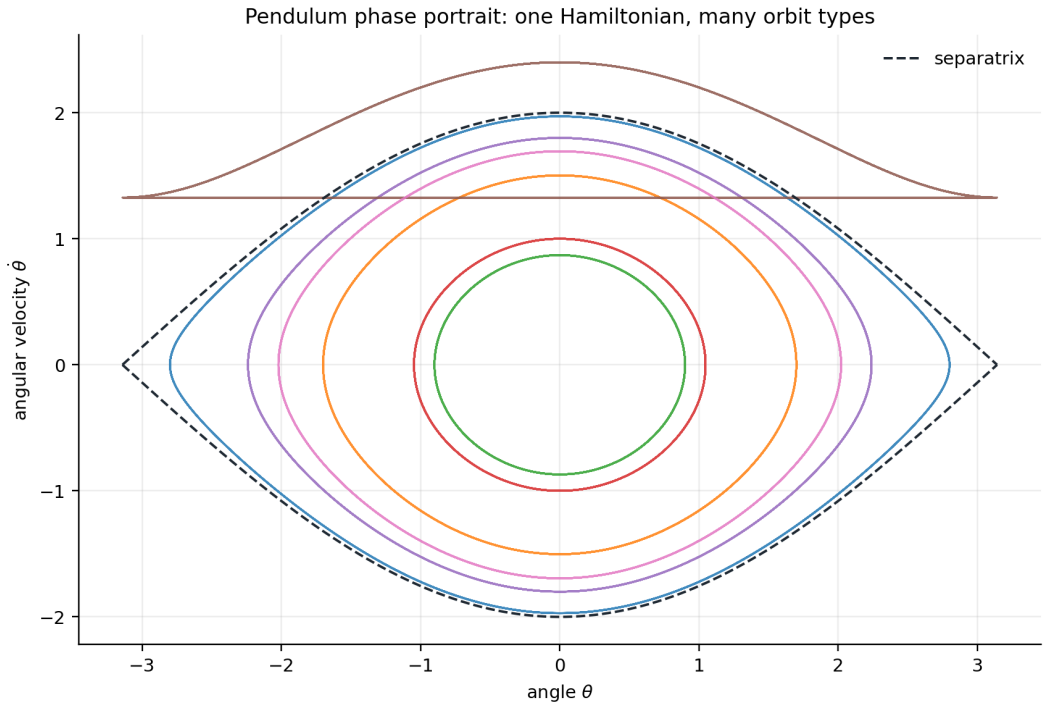


Figure 3.1: Pendulum trajectories in $(\theta, \dot\theta)$ space. Closed curves are librations, open curves are rotations, and the separatrix passes through the unstable upright equilibrium.

3.3 Holonomic constraints and Lagrange multipliers

A holonomic constraint has the form

$$f_a(q,t) = 0, \qquad a = 1, \dots, k. \tag{3.7}$$

When eliminating it by coordinates is inconvenient, extremize

$$\widetilde{S}[q,\lambda] = \int \left[L(q,\dot q,t) + \sum_a \lambda_a(t) f_a(q,t) \right] \mathrm{d}t. \tag{3.8}$$

Variation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot q^i} - \frac{\partial L}{\partial q^i} = \sum_a \lambda_a \frac{\partial f_a}{\partial q^i}, \tag{3.9}$$

plus the constraints. The right-hand side is the generalized reaction force. Multipliers therefore have physical content: they are whatever forces are required to remain on the constraint surface.

3.4 Cyclic coordinates and conserved momenta

The generalized momentum conjugate to q^i is

$$p_i = \frac{\partial L}{\partial \dot q^i}. \tag{3.10}$$

If L does not depend on q^i, then

$$\dot p_i = \frac{\partial L}{\partial q^i} = 0. \tag{3.11}$$

Such a coordinate is cyclic. For a central potential in plane polar coordinates,

$$L = \frac{1}{2} m(\dot r^2 + r^2 \dot\phi^2) - V(r), \tag{3.12}$$

and ϕ is cyclic, giving $p_\phi = mr^2\dot{\phi}$, the angular momentum. The radial dynamics becomes

$$\frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r) = E, \qquad V_{\text{eff}}(r) = V(r) + \frac{p_\phi^2}{2mr^2}. \tag{3.13}$$

The centrifugal barrier is not a new force inserted by hand; it arises after reducing a conserved coordinate.

3.5 Energy as the Noether charge of time translation

Define

$$E_L = \sum_i \dot{q}^i \frac{\partial L}{\partial \dot{q}^i} - L. \tag{3.14}$$

Along an Euler–Lagrange trajectory,

$$\frac{\mathrm{d}E_L}{\mathrm{d}t} = \sum_i \ddot{q}^i \frac{\partial L}{\partial \dot{q}^i} + \sum_i \dot{q}^i \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial t} - \sum_i \frac{\partial L}{\partial q^i}\dot{q}^i - \sum_i \frac{\partial L}{\partial \dot{q}^i}\ddot{q}^i \tag{3.15}$$

$$= -\frac{\partial L}{\partial t}. \tag{3.16}$$

Thus E_L is conserved when L has no explicit time dependence. For $L = T - V$ with T quadratic in velocities, Euler's homogeneous-function theorem gives $\sum_i \dot{q}^i \partial T/\partial \dot{q}^i = 2T$, hence $E_L = T + V$.

Common pitfall

Generalized momentum need not equal mass times a coordinate velocity. In polar coordinates, $p_\phi = mr^2\dot{\phi}$. In an electromagnetic field, canonical momentum is $p = m\dot{q} + eA(q,t)$ and differs from kinetic momentum $m\dot{q}$.

3.6 Electromagnetism and velocity-dependent potentials

For charge e in scalar and vector potentials (φ, A),

$$L(q,\dot{q},t) = \frac{1}{2}m\dot{q}^2 + eA(q,t)\cdot\dot{q} - e\varphi(q,t). \tag{3.17}$$

The Euler–Lagrange equation yields the Lorentz force

$$m\ddot{q} = e\left(E + \dot{q}\times B\right), \tag{3.18}$$

where $E = -\nabla\varphi - \partial_t A$ and $B = \nabla\times A$. Under the gauge transformation $A \mapsto A + \nabla\chi$, $\varphi \mapsto \varphi - \partial_t\chi$, the Lagrangian changes by $e\,\mathrm{d}\chi/\mathrm{d}t$, so the motion is unchanged.

3.7 Small oscillations and normal modes

Near a stable equilibrium q_0, write $q = q_0 + \xi$. To quadratic order,

$$L \approx \frac{1}{2}\dot{\xi}^{\top}M\dot{\xi} - \frac{1}{2}\xi^{\top}K\xi, \tag{3.19}$$

where $K = \nabla^2 V(q_0)$. Looking for $\xi(t) = a\mathrm{e}^{\mathrm{i}\omega t}$ gives the generalized eigenvalue problem

$$Ka = \omega^2 Ma. \tag{3.20}$$

The eigenvectors are normal modes and the eigenvalues are squared frequencies. A coupled many-body system becomes a set of independent harmonic oscillators at quadratic order. This is the basis of phonons, molecular vibrations, Gaussian field theories, and the harmonic thermal bath used later.

3.8 Dissipation and the limits of a standard Lagrangian

Linear viscous forces $Q_i^{\text{diss}} = -\sum_j C_{ij}\dot{q}^j$ can be included through Rayleigh's dissipation function

$$\mathcal{R}(\dot{q}) = \frac{1}{2}\dot{q}^\top C\dot{q}, \tag{3.21}$$

leading to

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} + \frac{\partial \mathcal{R}}{\partial \dot{q}^i} = 0. \tag{3.22}$$

This is useful but not a closed microscopic explanation: friction usually represents eliminated degrees of freedom. Langevin dynamics will add the corresponding fluctuating force, with fluctuation and dissipation tied by temperature.

Bridge to the next theory

The generalized momenta $p_i = \partial L/\partial \dot{q}^i$ suggest replacing velocities by momenta as independent variables. The Legendre transform performs that change and converts second-order motion on configuration space into first-order flow on phase space.

4

Hamiltonian mechanics: geometry in phase space

4.1 The Legendre transform

Suppose the velocity Hessian

$$W_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \tag{4.1}$$

is nonsingular. Then $p_i = \partial L/\partial \dot{q}^i$ can be inverted to give $\dot{q} = \dot{q}(q, p, t)$. Define the Hamiltonian by the Legendre transform

$$H(q, p, t) = p \cdot \dot{q} - L(q, \dot{q}, t). \tag{4.2}$$

The transform replaces slope variables by their conjugates. For a convex one-dimensional function $f(v)$,

$$f^*(p) = \sup_v \{pv - f(v)\}, \qquad p = f'(v). \tag{4.3}$$

The tangent line of slope p touches f at the maximizing v. For $f(v) = mv^2/2$, $p = mv$ and $f^*(p) = p^2/(2m)$.

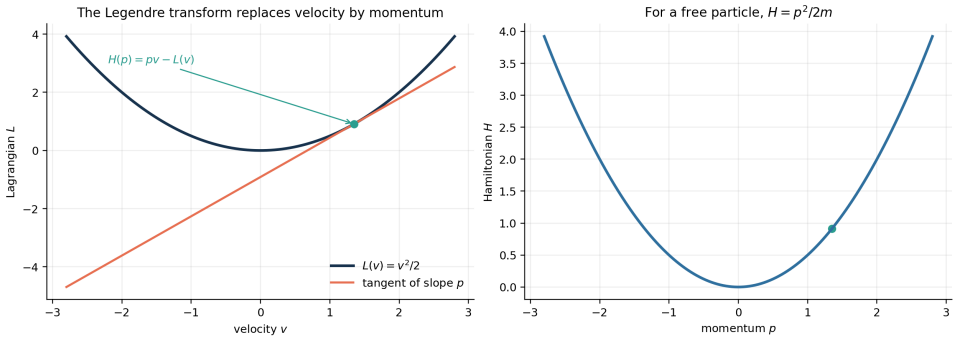


Figure 4.1: Geometric construction of a Legendre transform. The conjugate variable is a slope; the transformed value is the intercept encoded by the supporting tangent.

Taking a differential of Equation (4.2),

$$\mathrm{d}H = \dot{q}^i \mathrm{d}p_i + p_i \mathrm{d}\dot{q}^i - \frac{\partial L}{\partial q^i}\mathrm{d}q^i - \frac{\partial L}{\partial \dot{q}^i}\mathrm{d}\dot{q}^i - \frac{\partial L}{\partial t}\mathrm{d}t \tag{4.4}$$

$$= \dot{q}^i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q^i - \frac{\partial L}{\partial t}\mathrm{d}t, \tag{4.5}$$

where $p_i = \partial L/\partial \dot{q}^i$ and the Euler–Lagrange equation were used. Comparing coefficients gives Hamilton's equations

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.} \tag{4.6}$$

Also $\partial_t H = -\partial_t L$ after the change of variables.

4.2 Phase space and the symplectic matrix

The pair $z = (q, p) \in \mathbb{R}^{2n}$ is a point in phase space. Define

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}. \tag{4.7}$$

Then Hamilton's equations are

$$\dot{z} = J\nabla H(z). \tag{4.8}$$

The matrix J rotates a gradient into a vector tangent to energy level sets because

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \nabla H^\top J \nabla H = 0 \tag{4.9}$$

for time-independent H, using $J^\top = -J$. Hamiltonian motion is therefore not gradient descent. Gradient descent follows $-\nabla H$ and decreases energy; Hamiltonian flow follows the skew-gradient $J\nabla H$ and conserves energy.

Intuition

A potential gradient points uphill in configuration space. Multiplication by the symplectic matrix couples positions and momenta and turns that gradient sideways in phase space. The result circulates along constant-energy surfaces instead of descending them.

4.3 Poisson brackets

For observables $A(q, p, t)$ and $B(q, p, t)$ define

$$\{A, B\} = \sum_i \left(\frac{\partial A}{\partial q^i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q^i} \right) = \nabla A^\top J \nabla B. \tag{4.10}$$

Along a trajectory,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{4.11}$$

Thus A is conserved when it has no explicit time dependence and $\{A, H\} = 0$. Canonical coordinates satisfy

$$\{q^i, q^j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q^i, p_j\} = \delta^i_j. \tag{4.12}$$

The bracket is antisymmetric and obeys the Jacobi identity. In quantum mechanics, commutators will play the analogous algebraic role.

4.4 Canonical transformations

A coordinate change $(q, p) \mapsto (Q, P)$ is canonical if it preserves the symplectic form

$$\omega = \sum_i \mathrm{d}q^i \wedge \mathrm{d}p_i. \tag{4.13}$$

Equivalently, its Jacobian M satisfies $M^\top J M = J$. Canonical transformations preserve Poisson brackets and Hamilton's form of the equations. They are generated by functions whose derivatives relate old and new variables. The Hamilton–Jacobi equation seeks a transformation that makes the new Hamiltonian especially simple:

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0. \tag{4.14}$$

Here S is Hamilton's principal function. In semiclassical quantum mechanics, the phase of a wavefunction approximately satisfies this same equation.

4.5 Phase-volume preservation

The divergence of a Hamiltonian vector field is

$$\nabla_z \cdot (J\nabla_z H) = \sum_i \left(\frac{\partial^2 H}{\partial q^i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q^i} \right) = 0. \tag{4.15}$$

Therefore an infinitesimal phase-space volume is neither compressed nor expanded. This is the geometric core of Liouville's theorem, which we will rederive for probability densities in chapter 9. It also explains why a deterministic Hamiltonian flow cannot transform a simple density into a target density by changing local volume alone: it can stretch and fold, but its Jacobian determinant remains one.

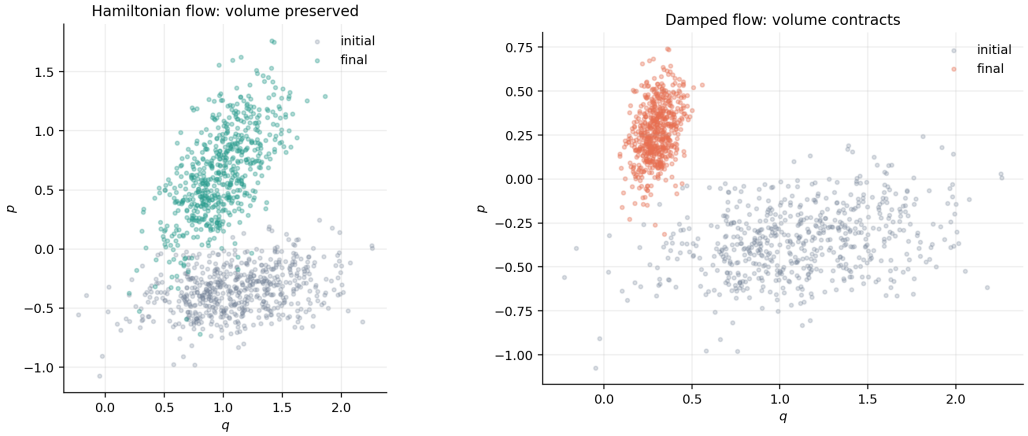


Figure 4.2: A cloud transported by conservative harmonic flow preserves area, whereas a damped flow contracts it. The measured conservative area ratio is one to numerical precision; damping reduces it to about 0.151.

4.6 Symplectic integration

A numerical map Φ_h is symplectic if its Jacobian satisfies

$$D\Phi_h(z)^\top J D\Phi_h(z) = J. \tag{4.16}$$

Symplectic Euler, Verlet, and leapfrog satisfy this property. They preserve phase volume exactly and approximately conserve a nearby modified Hamiltonian over long intervals. This is why their energy error oscillates rather than drifting systematically. Molecular dynamics, Hamiltonian Monte Carlo, and long-time celestial mechanics all rely on this structure.

4.7 The harmonic oscillator as a complete example

For

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2, \tag{4.17}$$

Hamilton's equations give $\dot{q} = p/m$ and $\dot{p} = -m\omega^2 q$. The exact flow is

$$\begin{pmatrix} q(t) \\ p(t) \end{pmatrix} = \begin{pmatrix} \cos \omega t & \frac{1}{m\omega} \sin \omega t \\ -m\omega \sin \omega t & \cos \omega t \end{pmatrix} \begin{pmatrix} q(0) \\ p(0) \end{pmatrix}. \tag{4.18}$$

The matrix determinant is one and the energy ellipses are invariant. In action–angle coordinates,

$$I = \frac{E}{\omega}, \qquad \theta(t) = \theta(0) + \omega t, \tag{4.19}$$

so the dynamics is uniform rotation. Integrable systems admit such coordinates locally; chaotic systems generally do not.

4.8 From a trajectory to a distribution

Hamiltonian mechanics still assumes exact knowledge of one microstate $z(t)$. For $N \sim 10^{23}$ particles this is neither measurable nor useful. Instead, assign a density $\rho(z,t)$ over phase space. Every point follows Hamilton's equations, and the density is transported by that flow. Because the flow is incompressible, density is constant along trajectories. This is the entry point to statistical mechanics.

Bridge to the next theory

Hamiltonian mechanics describes a $6N$-dimensional deterministic flow. Statistical mechanics asks which macroscopic predictions remain possible when the precise initial point is unknown and only coarse constraints such as energy, particle number, or temperature are available.

5

Why statistical mechanics exists

5.1 The failure is practical, not mechanical

A macroscopic sample contains on the order of Avogadro's number of microscopic constituents. If N classical particles move in three dimensions, an exact state requires $6N$ coordinates and momenta. Hamilton's equations are formally valid, but specifying, integrating, and interpreting approximately 10^{24} numbers is impossible. More importantly, macroscopic questions do not ask for them. Pressure, temperature, magnetization, and heat capacity are reproducible despite microscopic uncertainty.

Statistical mechanics changes the object of prediction. Instead of one microstate z, it uses a probability measure $\rho(z)$ constrained by the macroscopic information available. A macroscopic observable $A(z)$ is predicted through

$$\langle A \rangle_\rho = \int A(z)\rho(z)\,\mathrm{d}z. \tag{5.1}$$

The theory succeeds when fluctuations around this average are small on the macroscopic scale, or when their distribution can itself be calculated.

Intuition

Statistics is not inserted because Newton's equations become random. It is inserted because almost all microscopic detail is unobserved and irrelevant to the quantities we can control. The remarkable fact is that large systems often become more predictable macroscopically as the number of microscopic variables grows.

5.2 Microstates, macrostates, and multiplicity

Consider N independent two-state variables $s_i \in \{-1, +1\}$. A microstate is the full vector $s = (s_1, \ldots, s_N)$, of which there are 2^N. A macrostate may retain only the magnetization

$$M = \sum_{i=1}^{N} s_i. \tag{5.2}$$

If $n_+ = (N + M)/2$ spins are positive, the multiplicity is

$$\Omega(M) = \binom{N}{n_+}. \tag{5.3}$$

Many microstates correspond to the same macrostate. For large N, the multiplicity is overwhelmingly concentrated near $M = 0$. Using Stirling's approximation,

$$\log N! = N \log N - N + O(\log N), \tag{5.4}$$

we obtain, for magnetization density $m = M/N$,

$$\frac{1}{N} \log \Omega(Nm) \approx -\frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2}. \tag{5.5}$$

The logarithm converts an exponentially large multiplicity into an extensive quantity. This is the combinatorial origin of entropy.

5.3 Boltzmann and Gibbs entropy

For a macrostate with multiplicity Ω, Boltzmann's entropy is

$$S_B = k_B \log \Omega. \tag{5.6}$$

For a discrete probability distribution p_i, Gibbs' entropy is

$$S_G[p] = -k_B \sum_i p_i \log p_i. \tag{5.7}$$

If $p_i = 1/\Omega$ on the accessible microstates, the two coincide. For a continuous density, $-\int \rho \log \rho$ depends on the reference measure and coordinates, so physically meaningful statements involve phase-space cells, relative entropy, or entropy differences.

The additivity of entropy follows from multiplicativity of independent state counts. If systems A and B are independent, $\Omega_{AB} = \Omega_A \Omega_B$, hence

$$S_{AB} = k_B \log(\Omega_A \Omega_B) = S_A + S_B. \tag{5.8}$$

This explains a recurrent structural principle: independent probabilities multiply, while the corresponding energies or negative log-probabilities add.

5.4 The microcanonical ensemble

An isolated system has fixed energy E, particle number N, and volume V. The microcanonical postulate assigns equal weight to accessible states in a thin energy shell:

$$\rho_{\mathrm{mc}}(z) = \frac{\delta_\Delta(H(z) - E)}{\Omega(E, \Delta)}, \tag{5.9}$$

where δ_Δ denotes a shell of width Δ and

$$\Omega(E, \Delta) = \int \delta_\Delta(H(z) - E)\, \mathrm{d}z. \tag{5.10}$$

The thermodynamic entropy is $S(E, V, N) = k_B \log \Omega(E, V, N)$ up to conventions that become equivalent in the thermodynamic limit for ordinary short-range systems.

Equal a priori probability is not derived from mechanics alone. It is a modelling postulate justified by symmetry, typicality, coarse graining, and empirical success. Ergodicity can connect long-time averages to ensemble averages, but strict ergodicity is neither universally true nor always required.

5.5 The canonical ensemble as constrained inference

A system in thermal contact with a much larger environment can exchange energy. The canonical distribution is

$$p_i = \frac{\mathrm{e}^{-\beta E_i}}{Z(\beta)}, \qquad Z(\beta) = \sum_i \mathrm{e}^{-\beta E_i}, \tag{5.11}$$

where $\beta = (k_B T)^{-1}$. One derivation counts bath states and is developed in chapter 6. A complementary derivation maximizes Gibbs entropy subject to normalization and a fixed mean energy.

Introduce multipliers α and β and extremize

$$\mathcal{J}[p] = -\sum_i p_i \log p_i - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - U \right). \tag{5.12}$$

The stationarity condition is

$$-\log p_i - 1 - \alpha - \beta E_i = 0, \tag{5.13}$$

so $p_i \propto \mathrm{e}^{-\beta E_i}$. The multiplier β is identified thermodynamically as $1/(k_B T)$. Jaynes' maximum-entropy view makes explicit that the canonical law is the least committed distribution compatible with the specified expectation [28].

5.6 Concentration and self-averaging

Suppose $A_N = N^{-1} \sum_{i=1}^{N} X_i$ with independent variables of finite variance. Then

$$\mathrm{Var}(A_N) = \frac{\mathrm{Var}(X_1)}{N}. \tag{5.14}$$

The standard deviation shrinks as $N^{-1/2}$. Extensive quantities fluctuate as $\sqrt{N}$, but relative fluctuations shrink as $N^{-1/2}$. This is why a kilogram of matter has a sharply defined energy density even though it continuously exchanges microscopic energy with its surroundings.

Interactions complicate independence, especially near critical points where correlation lengths grow. Yet for short-range systems away from phase transitions, additivity and concentration remain powerful. Large-deviation theory refines the statement:

$$\mathbb{P}(A_N \approx a) \asymp \mathrm{e}^{-N I(a)}, \tag{5.15}$$

where $I(a) \geq 0$ is a rate function. Entropy and free energy are closely related to such exponential rates.

5.7 Ensembles as different experimental controls

Table 5.1: The main equilibrium ensembles.

Ensemble	Fixed controls	Fluctuating quantity	Statistical weight
Microcanonical	E, V, N	none of these	uniform on energy shell
Canonical	T, V, N	energy	$\mathrm{e}^{-\beta H}/Z$
Grand canonical	T, V, μ	energy and particle number	$\mathrm{e}^{-\beta(H-\mu N)}/\Xi$
Isothermal–isobaric	T, P, N	energy and volume	$\mathrm{e}^{-\beta(H+PV)}/\Delta$

Different ensembles are Legendre-related descriptions adapted to different constraints. For additive short-range systems in the thermodynamic limit, they often predict the same local observables. They can be inequivalent for long-range interactions, nonconcave entropy, finite systems, or metastable regimes.

Bridge to the next theory

The canonical exponential is not magic. It follows by expanding the logarithm of the number of bath states when a small subsystem borrows energy. That calculation also identifies temperature as the bath's logarithmic sensitivity to energy.

6

A thermal bath and the Boltzmann distribution

6.1 System plus bath is isolated

Let a small system S interact weakly with a much larger bath B. The composite system is isolated with fixed total energy

$$E_{\text{tot}} = E_s + E_b. \tag{6.1}$$

Assume the interaction energy is negligible for counting states. If the subsystem is in microstate i with energy E_i, the bath must have energy $E_{\text{tot}} - E_i$. Under the microcanonical postulate, the probability of i is proportional to the number of compatible bath microstates:

$$p_i \propto \Omega_B(E_{\text{tot}} - E_i). \tag{6.2}$$

Take logarithms and expand the bath entropy $S_B(E) = k_B \log \Omega_B(E)$ around E_{tot}:

$$\frac{1}{k_B} S_B(E_{\text{tot}} - E_i) = \frac{1}{k_B} S_B(E_{\text{tot}}) - \frac{E_i}{k_B} \left.\frac{\partial S_B}{\partial E}\right|_{E_{\text{tot}}} + O\left(\frac{E_i^2}{C_B T^2}\right). \tag{6.3}$$

Thermodynamics defines

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}, \tag{6.4}$$

so

$$\Omega_B(E_{\text{tot}} - E_i) \approx \Omega_B(E_{\text{tot}}) \mathrm{e}^{-E_i/(k_B T)}. \tag{6.5}$$

After normalization,

$$\boxed{p_i = \frac{\mathrm{e}^{-\beta E_i}}{Z}, \qquad \beta = \frac{1}{k_B T}, \qquad Z = \sum_i \mathrm{e}^{-\beta E_i}.} \tag{6.6}$$

Intuition

The subsystem loses probability at high energy because taking energy from the bath sharply reduces the bath's multiplicity. Temperature measures how rapidly the bath's log multiplicity changes with energy. A hot bath has small β, so it penalizes subsystem energy weakly.

6.2 Why the exponential is structurally inevitable

For two weakly coupled subsystems, additive energy requires a factorized weight:

$$w(E_A + E_B) = w(E_A) w(E_B). \tag{6.7}$$

Under mild regularity, the positive solutions are exponentials $w(E) = \mathrm{e}^{-\beta E}$. Thus additive conserved quantities naturally generate exponential-family distributions. This is the same log-linear structure used by energy-based machine learning models.

6.3 The partition function is a generating object

The canonical partition function

$$Z(\beta) = \sum_i \mathrm{e}^{-\beta E_i} \tag{6.8}$$

normalizes the distribution, but it does much more. Differentiating,

$$U = \langle E \rangle = -\frac{\partial}{\partial \beta} \log Z, \tag{6.9}$$

and

$$\mathrm{Var}(E) = \frac{\partial^2}{\partial \beta^2} \log Z = -\frac{\partial U}{\partial \beta}. \tag{6.10}$$

Because $\partial\beta/\partial T = -1/(k_B T^2)$,

$$C_V = \frac{\partial U}{\partial T} = \frac{\mathrm{Var}(E)}{k_B T^2} \geq 0. \tag{6.11}$$

Heat capacity is therefore an equilibrium fluctuation. This is an elementary example of fluctuation–response relations.

The Helmholtz free energy is

$$F(T, V, N) = -k_B T \log Z. \tag{6.12}$$

Using $p_i = \mathrm{e}^{-\beta E_i}/Z$ in the Gibbs entropy gives $S = (U - F)/T$, hence $F = U - TS$.

6.4 A finite Einstein-solid bath

An Einstein solid distributes q indistinguishable energy quanta among N distinguishable oscillators. Its multiplicity is

$$\Omega(N, q) = \binom{q + N - 1}{q}. \tag{6.13}$$

Take a small subsystem with discrete energy $n = 0, 1, \ldots$ quanta and a bath with N_B oscillators and total q_{tot} quanta. The exact subsystem probability is

$$p_{\mathrm{exact}}(n) = \frac{\Omega_S(N_S, n)\Omega_B(N_B, q_{\mathrm{tot}} - n)}{\sum_m \Omega_S(N_S, m)\Omega_B(N_B, q_{\mathrm{tot}} - m)}. \tag{6.14}$$

For a one-state subsystem at each n, the ratio of neighboring bath multiplicities is

$$\frac{\Omega_B(N_B, q-1)}{\Omega_B(N_B, q)} = \frac{q}{q + N_B - 1} \approx \mathrm{e}^{-\beta\epsilon}, \tag{6.15}$$

which becomes nearly constant when the bath is large compared with exchanged energy.

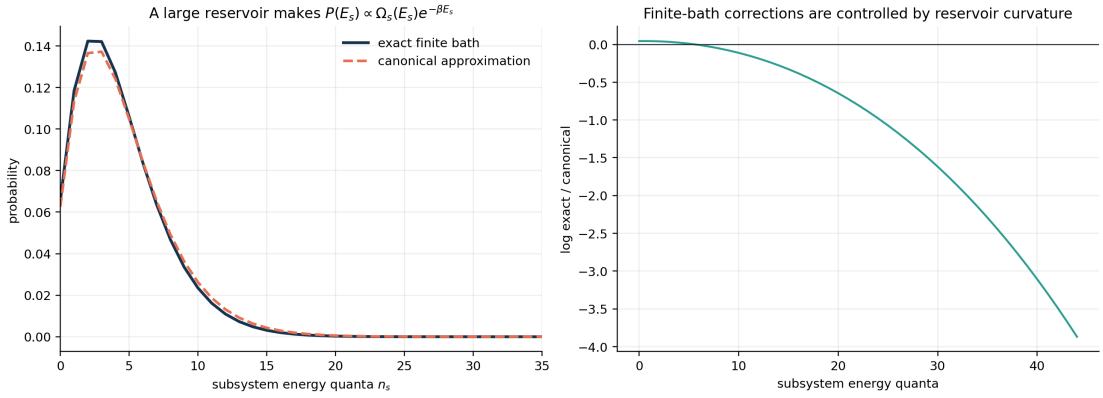


Figure 6.1: Exact energy distribution induced by a finite combinatorial bath compared with its canonical approximation. The bath's log multiplicity is nearly linear over the subsystem energy range.

Computational experiment

The finite-bath experiment estimates $\beta \approx 0.5058$ from the local entropy slope. The Kullback–Leibler divergence between the exact finite-bath law and its canonical approximation is about 2.61×10^{-3}. Increasing bath size while holding its energy density fixed reduces finite-size curvature and improves the exponential approximation.

6.5 Classical phase-space form

For a classical Hamiltonian $H(q, p)$, the canonical density is

$$\rho_\beta(q,p) = \frac{1}{Z} e^{-\beta H(q,p)}, \qquad Z = \frac{1}{h^{3N} N!} \int e^{-\beta H(q,p)} \, dq \, dp. \tag{6.16}$$

The factors h^{3N} and $N!$ make the classical state count dimensionless and correct indistinguishability; quantum mechanics ultimately supplies their natural interpretation.

For $H = \sum_i p_i^2/(2m_i) + V(q)$, momentum and position factorize. Each quadratic momentum component is Gaussian,

$$p(p_j) \propto \exp\left(-\beta \frac{p_j^2}{2m_j}\right), \tag{6.17}$$

and equipartition gives

$$\left\langle \frac{p_j^2}{2m_j} \right\rangle = \frac{1}{2} k_B T. \tag{6.18}$$

Every independent quadratic term contributes $k_B T/2$ to the mean energy.

6.6 Canonical typicality and equivalence

The bath derivation assumes weak coupling and a smooth, extensive bath entropy. If the bath has negative heat capacity, long-range interactions, or comparable size to the system, the expansion may fail. At phase coexistence, finite-size corrections and metastability can be important. The canonical ensemble is therefore a controlled approximation under identifiable assumptions, not an unconditional law.

For ordinary matter, the relative energy fluctuation follows from Equation (6.11):

$$\frac{\sqrt{\mathrm{Var}(E)}}{U} \sim N^{-1/2}. \tag{6.19}$$

Thus the canonical energy is sharply concentrated around the microcanonical one in the thermodynamic limit, explaining ensemble equivalence for many observables [19, 30, 55].

Bridge to the next theory

The partition function converts microscopic energies into thermodynamic potentials. We now derive the laws of thermodynamics as statements about entropy maximization, Legendre transforms, and equilibrium response.

7

Thermodynamics and equilibrium statistical mechanics

7.1 Thermodynamics as a theory of state functions

Thermodynamics describes equilibrium macrostates without tracking microscopic trajectories. For a simple compressible system, the internal energy is a function $U(S, V, N)$. Its differential is the fundamental relation

$$\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V + \mu\,\mathrm{d}N. \tag{7.1}$$

This defines the intensive conjugates

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N}, \quad P = -\left(\frac{\partial U}{\partial V}\right)_{S,N}, \quad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}. \tag{7.2}$$

Equivalently, in the entropy representation $S(U, V, N)$,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{U,N}, \quad -\frac{\mu}{T} = \left(\frac{\partial S}{\partial N}\right)_{U,V}. \tag{7.3}$$

7.2 Zeroth and first laws

The zeroth law states that thermal equilibrium is transitive, allowing temperature to be defined. The first law is energy conservation with heat and work distinguished by transfer mechanism:

$$\mathrm{d}U = \delta Q + \delta W. \tag{7.4}$$

For quasistatic pressure–volume work done on the system, $\delta W = -P\mathrm{d}V$, and for a reversible process $\delta Q_{\mathrm{rev}} = T\mathrm{d}S$, recovering Equation (7.1) at fixed N. The symbol δ emphasizes that heat and work are path-dependent transfers, not state functions. Their sum changes the state function U.

7.3 The second law from multiplicity

Put two isolated subsystems in weak thermal contact with fixed total energy. The total multiplicity is

$$\Omega_{\mathrm{tot}}(E_A) = \Omega_A(E_A)\Omega_B(E_{\mathrm{tot}} - E_A), \tag{7.5}$$

so total entropy is additive. Equilibrium maximizes

$$S_{\mathrm{tot}}(E_A) = S_A(E_A) + S_B(E_{\mathrm{tot}} - E_A). \tag{7.6}$$

The stationarity condition is

$$\frac{\partial S_A}{\partial E_A} = \frac{\partial S_B}{\partial E_B}, \tag{7.7}$$

which gives $T_A = T_B$. Concavity ensures stability. A spontaneous macroscopic energy transfer proceeds toward overwhelmingly larger multiplicity; entropy-decreasing fluctuations are possible for finite systems but exponentially unlikely in system size.

The Clausius inequality is

$$\Delta S \geq \int \frac{\delta Q}{T_{\text{boundary}}}, \tag{7.8}$$

with equality for a reversible process. For an isolated system, $\Delta S \geq 0$. Microscopic Hamiltonian dynamics remains reversible; macroscopic irreversibility enters through coarse graining, special low-entropy initial conditions, and the practical loss of microscopic correlations.

Common pitfall

The second law does not say microscopic entropy must increase under exact Hamiltonian flow. Fine-grained Gibbs entropy is constant by Liouville's theorem. Thermodynamic entropy increases after coarse graining or when reduced variables discard correlations with unresolved degrees of freedom.

7.4 Legendre transforms and thermodynamic potentials

Different experimental controls require different potentials. Starting from $U(S, V, N)$:

$$F(T, V, N) = U - TS, \qquad \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N, \tag{7.9}$$

$$H(S, P, N) = U + PV, \qquad \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N, \tag{7.10}$$

$$G(T, P, N) = U - TS + PV, \qquad \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N, \tag{7.11}$$

$$\Omega(T, V, \mu) = U - TS - \mu N, \qquad \mathrm{d}\Omega = -S\mathrm{d}T - P\mathrm{d}V - N\mathrm{d}\mu. \tag{7.12}$$

At fixed T, V, N, equilibrium minimizes F; at fixed T, P, N, it minimizes G. Statistical mechanics identifies

$$F = -k_B T \log Z, \qquad \Omega = -k_B T \log \Xi. \tag{7.13}$$

These are log normalizers, exactly as cumulant-generating functions are in probability theory.

7.5 Maxwell relations

Equality of mixed derivatives produces measurable identities. From $\mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V$ at fixed N,

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \qquad P = -\left(\frac{\partial F}{\partial V}\right)_T, \tag{7.14}$$

therefore

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V. \tag{7.15}$$

Thermodynamic potentials organize response coefficients through geometry of derivatives.

7.6 Free energy as energy–entropy competition

For any trial distribution q_i over microstates with energies E_i, define the nonequilibrium free-energy functional

$$\mathcal{F}[q] = \sum_i q_i E_i + k_B T \sum_i q_i \log q_i. \tag{7.16}$$

Let $p_i = \mathrm{e}^{-\beta E_i}/Z$. A direct rearrangement gives

$$\mathcal{F}[q] = F + k_B T D_{\mathrm{KL}}(q \| p) \geq F. \tag{7.17}$$

Thus the Boltzmann distribution uniquely minimizes free energy, and the excess free energy is a relative entropy. This identity connects equilibrium thermodynamics, variational inference, and energy-based learning.

Derivation: Equation (7.17)

Since $\log p_i = -\beta E_i - \log Z$,

$$k_B T D_{\mathrm{KL}}(q\|p) = k_B T \sum_i q_i \log q_i + \sum_i q_i E_i + k_B T \log Z \tag{7.18}$$

$$= \mathcal{F}[q] - F. \tag{7.19}$$

Nonnegativity of KL divergence proves the variational principle.

7.7 Phase transitions and nonanalyticity

For finite N, Z is usually analytic in temperature. A sharp phase transition emerges in the thermodynamic limit when zeros or competing exponential contributions make the free-energy density nonanalytic. An order parameter such as magnetization distinguishes phases. Near a continuous transition, correlation length and fluctuations grow; naive independence and mean-field approximations fail.

For an Ising model,

$$H(s) = -\sum_{\langle i,j\rangle} J_{ij} s_i s_j - h \sum_i s_i, \tag{7.20}$$

magnetization is $m = N^{-1} \sum_i \langle s_i \rangle$. Ferromagnetic couplings favor alignment. Random signs will instead create frustration and spin-glass behavior in chapter 13.

7.8 Equilibrium as detailed balance

A Markov process with transition rates $W_{i\to j}$ has stationary distribution π if

$$\sum_j \left(\pi_j W_{j\to i} - \pi_i W_{i\to j}\right) = 0. \tag{7.21}$$

Detailed balance is the stronger pairwise condition

$$\pi_i W_{i\to j} = \pi_j W_{j\to i}. \tag{7.22}$$

It implies zero probability current between every pair and time-reversal symmetry at stationarity. Equilibrium Langevin dynamics satisfies detailed balance under appropriate conditions. Non-equilibrium steady states can have a stationary density while sustaining nonzero circulating currents.

Bridge to the next theory

Equilibrium averages can be calculated from an ensemble. To replace ensemble averages by measurements along one long trajectory, we need to understand ergodicity, recurrence, mixing, and their limitations.

8

Ergodicity, mixing, and typicality

8.1 Time averages versus ensemble averages

An experiment often observes one system over time, while statistical mechanics computes an average over many possible microstates. For an observable A and flow Φ^t, the time average is

$$\overline{A}(z_0) = \lim_{T\to\infty} \frac{1}{T} \int_0^T A(\Phi^t z_0)\, \mathrm{d}t. \tag{8.1}$$

The ensemble average under invariant measure μ is

$$\langle A \rangle_\mu = \int A(z)\, \mu(\mathrm{d}z). \tag{8.2}$$

A measure-preserving system is ergodic if every invariant measurable set has measure zero or one. Birkhoff's ergodic theorem states that the time average exists almost everywhere and equals the conditional expectation onto invariant sets. If the system is ergodic,

$$\overline{A}(z_0) = \langle A \rangle_\mu \tag{8.3}$$

for almost every initial state [6, 32].

8.2 What ergodicity does and does not mean

Ergodicity does not mean that a trajectory visits every point. In a continuous phase space, individual points have measure zero. It means that the trajectory samples measurable regions in proportion to their invariant measure in the infinite-time limit. It also says nothing about the time required. A formally ergodic system can mix so slowly that experiments remain trapped in metastable regions.

Mixing is stronger. A measure-preserving flow is mixing if

$$\mu(\Phi^{-t} A \cap B) \to \mu(A)\mu(B) \tag{8.4}$$

as $t \to \infty$. Equivalently, correlations decay:

$$\langle f(\Phi^t z) g(z) \rangle - \langle f \rangle \langle g \rangle \to 0. \tag{8.5}$$

Mixing implies ergodicity, but not conversely. A rigid irrational rotation on a circle is ergodic but not mixing.

8.3 Integrability versus chaos

An integrable Hamiltonian system with n independent conserved quantities in involution can be represented in action–angle variables:

$$\dot{I} = 0, \qquad \dot{\theta} = \omega(I). \tag{8.6}$$

Trajectories remain on invariant n-tori in a $2n$-dimensional phase space and cannot explore the entire energy surface. Integrable systems are therefore not generally ergodic on that surface.

Nonlinear perturbations can destroy some tori and produce chaotic regions with sensitive dependence on initial conditions. Yet chaos is not identical to ergodicity: phase space may contain a mixture of regular islands and chaotic seas. The Kolmogorov–Arnold–Moser picture explains why many invariant tori survive sufficiently small perturbations.

8.4 The standard map experiment

The area-preserving standard map is

$$p_{n+1} = p_n + K \sin q_n \pmod{2\pi}, \tag{8.7}$$

$$q_{n+1} = q_n + p_{n+1} \pmod{2\pi}. \tag{8.8}$$

For small K, invariant curves dominate; for larger K, broad chaotic regions emerge while regular islands can remain.

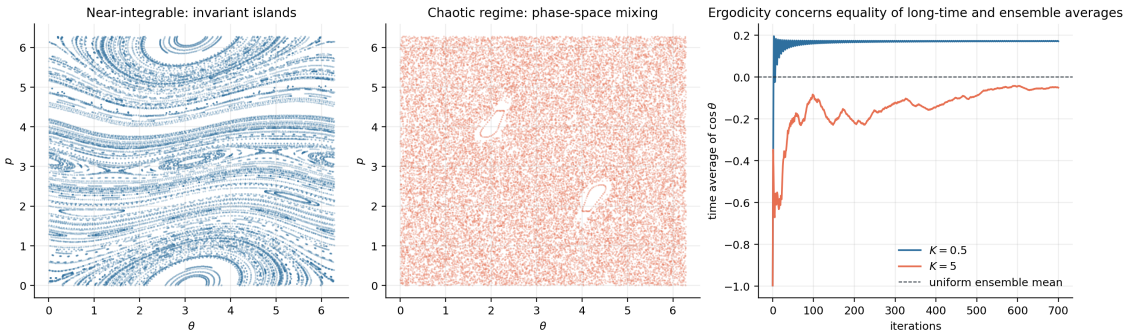


Figure 8.1: Standard-map phase portraits at different nonlinearities. Ordered invariant curves give way to chaotic transport, but surviving islands show why "chaotic" need not mean globally ergodic.

Computational experiment

Open `interactive/pendulum_phase_portrait.html` and the notebook's ergodicity section. Varying initial conditions shows that finite-time averages can depend strongly on which invariant component is sampled. This is the computational reason Monte Carlo diagnostics must assess mixing rather than assume it.

8.5 Poincaré recurrence and irreversibility

A measure-preserving system with finite accessible volume returns arbitrarily close to its initial condition for almost every initial state. This Poincaré recurrence theorem seems to conflict with irreversible relaxation. There is no contradiction: recurrence times for macroscopic systems are typically astronomical, and macroscopic relaxation concerns coarse observables that ignore precise correlations. A gas can return microscopically, but the recurrence probability on laboratory timescales is negligible.

Loschmidt's reversibility objection asks why entropy rises if velocities can be reversed. The answer is not that reversal is forbidden. It is that low-entropy macrostates occupy an extraordinarily small fraction of phase space, and exact velocity reversal requires exponentially precise control of correlations. The arrow of time also relies on a low-entropy boundary condition in the past.

8.6 Typicality without full ergodicity

Many equilibrium predictions require less than global ergodicity. If almost every microstate in an energy shell gives nearly the same value of a macroscopic observable, then one typical microstate suffices. Concentration of measure can make thermodynamic observables stable even when dynamics explores only a subset of the shell. Weak coupling between many degrees of freedom and locality can support equilibration of small subsystems without complete exploration of the global state space.

This distinction matters in glasses. A system may rapidly equilibrate within one metastable basin but take exponentially long to move between basins. The physically observed state is then a restricted or history-dependent ensemble. Replica symmetry breaking formalizes a decomposition into many pure states.

8.7 Ergodicity in stochastic processes

Noise can restore exploration. An overdamped Langevin process

$$\mathrm{d}X_t = -\nabla U(X_t)\mathrm{d}t + \sqrt{2T}\mathrm{d}W_t \tag{8.9}$$

may be ergodic under confining and regularity conditions, with invariant density proportional to $\mathrm{e}^{-U/T}$. Irreducibility lets noise reach different regions; a Lyapunov drift condition prevents escape to infinity. Yet high barriers produce metastable mixing times scaling roughly as $\exp(\Delta U/T)$. The invariant distribution can be correct while finite simulations remain biased.

Common pitfall

Do not use "ergodic" as a synonym for "random-looking." It is a precise property relative to an invariant measure. Also distinguish stationarity, which says a distribution does not change with time, from ergodicity, which says time averages identify ensemble averages.

Bridge to the next theory

To state invariance and transport precisely, we need the continuity equation for a density. Applying it to Hamiltonian phase flow produces the Liouville equation and the preservation of fine-grained entropy.

9

Continuity equations and Liouville's theorem

9.1 Conservation of probability in state space

Let $\rho(x,t)$ be a density transported by a deterministic velocity field $v(x,t)$ in $\mathbb{R}^d$. For any fixed region Ω, conservation says that mass changes only through the boundary:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho \,\mathrm{d}x = -\int_{\partial\Omega} \rho v \cdot n \,\mathrm{d}S. \tag{9.1}$$

Using the divergence theorem,

$$\int_\Omega \left[\partial_t \rho + \nabla\cdot(\rho v)\right]\mathrm{d}x = 0. \tag{9.2}$$

Because this holds for every region,

$$\boxed{\partial_t \rho + \nabla\cdot(\rho v) = 0.} \tag{9.3}$$

Expanding the divergence gives the material-derivative form

$$\frac{D\rho}{Dt} := \partial_t\rho + v\cdot\nabla\rho = -\rho\nabla\cdot v. \tag{9.4}$$

Density increases along trajectories where the flow compresses and decreases where it expands.

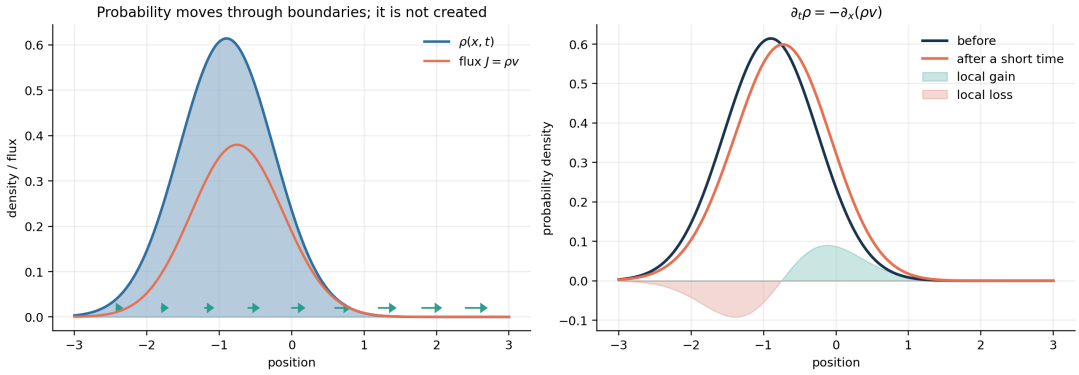


Figure 9.1: Mass conservation under an advecting vector field. Local density changes are determined by flux divergence, not by particles being created or destroyed.

9.2 Characteristics and Jacobians

Let $X_t(x_0)$ solve

$$\frac{\mathrm{d}}{\mathrm{d}t}X_t(x_0) = v(X_t(x_0),t), \qquad X_0(x_0) = x_0. \tag{9.5}$$

The flow map pushes the initial density forward. Conservation of mass gives

$$\rho_t(X_t(x_0))\det(DX_t(x_0)) = \rho_0(x_0). \tag{9.6}$$

Jacobi's formula yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\det(DX_t) = \nabla\cdot v(X_t,t). \tag{9.7}$$

Combining this with Equation (9.6) recovers Equation (9.4). This identity is the foundation of continuous normalizing flows: log density changes by minus the integrated divergence.

9.3 Liouville equation for Hamiltonian ensembles

For phase-space coordinates $z = (q,p)$ and Hamiltonian velocity

$$v_H(z) = \begin{pmatrix} \nabla_p H \\ -\nabla_q H \end{pmatrix} = J\nabla H, \tag{9.8}$$

Equation (4.15) gives $\nabla_z \cdot v_H = 0$. The continuity equation becomes

$$\partial_t \rho + v_H \cdot \nabla_z \rho = 0. \tag{9.9}$$

Using the Poisson bracket,

$$\boxed{\partial_t \rho = \{H,\rho\} = -\{\rho,H\}.} \tag{9.10}$$

Along a Hamiltonian trajectory,

$$\frac{D\rho}{Dt} = 0. \tag{9.11}$$

The density is stirred but not locally compressed. Equivalently, the flow Jacobian has determinant one [38, 42].

9.4 Stationary distributions

Any density that depends only on conserved quantities is stationary. If $\rho(z) = f(H(z))$, then

$$\{\rho,H\} = f'(H)\{H,H\} = 0. \tag{9.12}$$

Both the microcanonical shell and the canonical density $\rho \propto \mathrm{e}^{-\beta H}$ are invariant under isolated Hamiltonian dynamics. Hamiltonian dynamics alone does not select a temperature; temperature enters through preparation or coupling to a bath.

9.5 Fine-grained entropy conservation

Define the Gibbs entropy

$$S_G[\rho] = -k_B \int \rho \log \rho \,\mathrm{d}z. \tag{9.13}$$

For a continuity equation with suitable boundary conditions,

$$\frac{\mathrm{d}S_G}{\mathrm{d}t} = -k_B \int (1+\log\rho)\partial_t\rho \,\mathrm{d}z \tag{9.14}$$

$$= k_B \int (1+\log\rho)\nabla\cdot(\rho v)\,\mathrm{d}z \tag{9.15}$$

$$= k_B \int \rho \nabla\cdot v \,\mathrm{d}z. \tag{9.16}$$

For Hamiltonian flow, $\nabla\cdot v_H = 0$, so $\mathrm{d}S_G/\mathrm{d}t = 0$. A filamented density can look increasingly uniform at coarse resolution while retaining the same fine-grained entropy in ever finer structures.

9.6 Liouville operator and Koopman viewpoints

Define the Liouville operator acting on densities by

$$\mathcal{L}^*\rho = -\nabla \cdot (\rho v), \tag{9.17}$$

and the Koopman generator acting on observables by

$$\mathcal{L}A = v \cdot \nabla A. \tag{9.18}$$

They are adjoint:

$$\int A(\mathcal{L}^*\rho)\,\mathrm{d}x = \int (\mathcal{L}A)\rho\,\mathrm{d}x \tag{9.19}$$

up to boundary terms. One can evolve states forward, densities forward, or observables backward. This duality reappears for Markov semigroups, Fokker–Planck equations, and backward Kolmogorov equations.

9.7 Reduced descriptions and the BBGKY idea

The full N-particle density obeys Liouville's equation in $6N$ dimensions. Define a one-particle marginal by integrating out the others. Its evolution depends on the two-particle marginal; the two-particle equation depends on the three-particle marginal, and so on. This is the BBGKY hierarchy. Kinetic theories such as the Boltzmann equation close the hierarchy by assumptions about weak correlations. Irreversible equations arise after such reduction and approximation, even though the full microscopic equation is reversible.

9.8 Numerical volume experiment

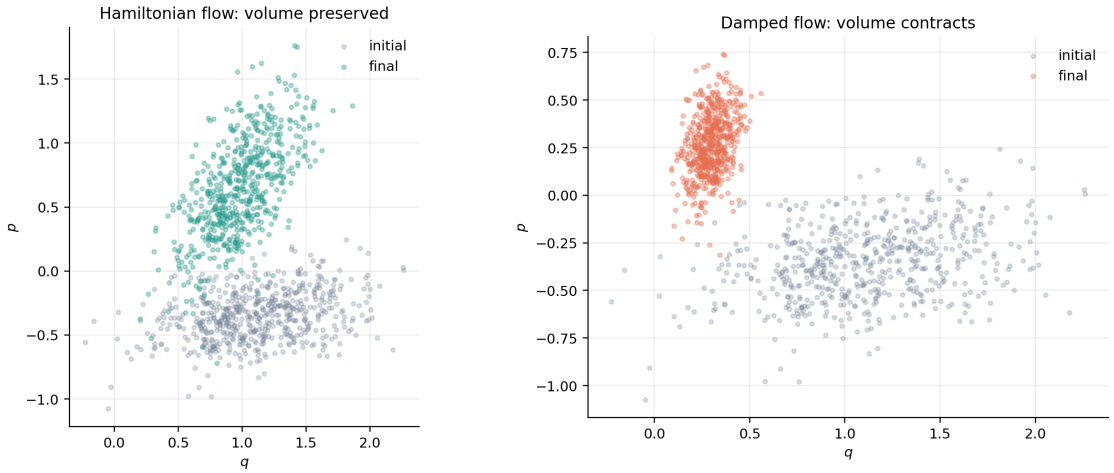


Figure 9.2: Evolution of the same initial cloud under Hamiltonian and damped oscillator flows. Hamiltonian transport rotates and shears without changing area; friction makes the divergence negative and contracts volume.

For the damped oscillator $\dot q = p$, $\dot p = -q - \gamma p$, the divergence is $-\gamma$. Hence

$$\det(DX_t) = \mathrm{e}^{-\gamma t}. \tag{9.20}$$

The experiment's convex-hull area ratio agrees with this contraction, while the Hamiltonian ratio is one to floating-point precision.

Bridge to the next theory

A deterministic continuity equation only advects density. Unresolved collisions add random increments, whose cumulative second-order effect creates a diffusion term. The resulting transport law is the Fokker–Planck equation.

10

From random increments to the Fokker–Planck equation

10.1 Why add stochasticity?

A small set of resolved variables interacts with many unresolved degrees of freedom. Their combined effect can often be decomposed into systematic drift and rapidly fluctuating forcing. The effective state $X_t \in \mathbb{R}^d$ is then modeled by an Itô stochastic differential equation (SDE)

$$\mathrm{d}X_t = a(X_t, t)\,\mathrm{d}t + B(X_t, t)\,\mathrm{d}W_t, \tag{10.1}$$

where $W_t \in \mathbb{R}^m$ is Brownian motion. Over a short interval Δt,

$$\Delta X \approx a\,\Delta t + B\sqrt{\Delta t}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_m). \tag{10.2}$$

The random increment is of order $\sqrt{\Delta t}$, larger than the deterministic increment of order Δt at very short times. Its mean is zero but its square is order Δt, so second derivatives survive in the density evolution.

10.2 The infinitesimal generator

For a smooth test function φ, Itô's lemma gives

$$\mathrm{d}\varphi(X_t, t) = \left(\partial_t \varphi + a_i \partial_i \varphi + \frac{1}{2} D_{ij} \partial_i \partial_j \varphi\right) \mathrm{d}t + \partial_i \varphi B_{ik} \mathrm{d}W_t^k, \tag{10.3}$$

where $D = BB^{\top}$. Taking conditional expectations identifies the backward generator

$$\mathcal{L}\varphi = a_i \partial_i \varphi + \frac{1}{2} D_{ij} \partial_i \partial_j \varphi. \tag{10.4}$$

The expectation $u(x, t) = \mathbb{E}[\varphi(X_T) \mid X_t = x]$ satisfies the backward Kolmogorov equation

$$\partial_t u + \mathcal{L} u = 0, \qquad u(x, T) = \varphi(x). \tag{10.5}$$

10.3 Adjoint derivation of Fokker–Planck

Let $p(x, t)$ be the density of X_t. Since

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[\varphi(X_t)] = \int (\mathcal{L}\varphi)(x)\, p(x, t)\,\mathrm{d}x, \tag{10.6}$$

and also

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[\varphi(X_t)] = \int \varphi(x)\, \partial_t p(x, t)\,\mathrm{d}x, \tag{10.7}$$

integration by parts moves derivatives from φ to p and gives

$$\boxed{\partial_t p = -\partial_i (a_i p) + \frac{1}{2} \partial_i \partial_j (D_{ij} p).} \tag{10.8}$$

This is the Fokker–Planck, or forward Kolmogorov, equation. It is a continuity equation

$$\partial_t p = -\nabla \cdot J, \tag{10.9}$$

with probability current, for constant diffusion matrix D,

$$J = ap - \frac{1}{2} D \nabla p. \tag{10.10}$$

Drift advects probability; diffusion moves it down density gradients.

Intuition

The deterministic continuity equation transports a sharp point into a sharp point. Diffusion turns one initial point into a cloud. The Fokker–Planck equation describes the cloud exactly at the distribution level, even though individual sample paths are nowhere differentiable.

10.4 Kramers–Moyal expansion

For a Markov process, define conditional jump moments

$$M^{(n)}(x) = \lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \mathbb{E}[(X_{t+\Delta t} - X_t)^{\otimes n} \mid X_t = x]. \tag{10.11}$$

A formal expansion of the master equation yields

$$\partial_t p = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \partial_{i_1} \cdots \partial_{i_n} \left(M^{(n)}_{i_1 \cdots i_n} p \right). \tag{10.12}$$

For a diffusion, only the first two infinitesimal moments survive: $M^{(1)} = a$ and $M^{(2)} = D$, recovering Equation (10.8). Pawula's theorem constrains consistent finite truncations: beyond the first term, a positive transition density generally requires truncation after the second or retaining the full series.

10.5 The Ornstein–Uhlenbeck density

Consider

$$\mathrm{d}X_t = -\theta X_t \mathrm{d}t + \sigma \mathrm{d}W_t, \qquad \theta > 0. \tag{10.13}$$

Its Fokker–Planck equation is

$$\partial_t p = \theta \partial_x (xp) + \frac{\sigma^2}{2} \partial_x^2 p. \tag{10.14}$$

The exact solution remains Gaussian. Starting from deterministic $X_0 = x_0$,

$$m_t = x_0 \mathrm{e}^{-\theta t}, \qquad v_t = \frac{\sigma^2}{2\theta} (1 - \mathrm{e}^{-2\theta t}). \tag{10.15}$$

The stationary law is $\mathcal{N}(0, \sigma^2/(2\theta))$.

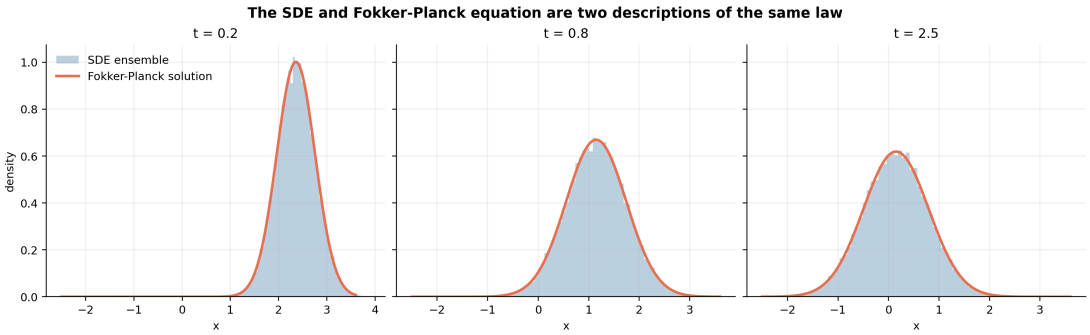


Figure 10.1: Evolution of an Ornstein–Uhlenbeck density. Drift pulls mass toward the origin while diffusion broadens it; their balance produces a stationary Gaussian.

10.6 Stationarity, detailed balance, and currents

A stationary density solves $\mathcal{L}^* p_* = 0$, equivalently $\nabla \cdot J_* = 0$. Equilibrium detailed balance usually requires the stronger condition $J_* = 0$. For constant isotropic diffusion $D = 2TI$ and gradient drift $a = -\nabla U$,

$$J_* = -p_* \nabla U - T \nabla p_*. \tag{10.16}$$

Setting $J_* = 0$ gives

$$\nabla \log p_* = -\frac{1}{T} \nabla U, \tag{10.17}$$

so

$$p_*(x) = \frac{1}{Z} \mathrm{e}^{-U(x)/T}. \tag{10.18}$$

The Boltzmann law emerges dynamically from drift–diffusion balance.

A non-equilibrium steady state may satisfy $\nabla \cdot J_* = 0$ with $J_* \neq 0$. Probability then circulates persistently. Stationarity alone is not equilibrium.

10.7 Underdamped Kramers equation

For position and momentum,

$$\mathrm{d}Q_t = \frac{P_t}{m} \mathrm{d}t, \tag{10.19}$$

$$\mathrm{d}P_t = -\nabla V(Q_t) \mathrm{d}t - \gamma P_t \mathrm{d}t + \sqrt{2\gamma m k_B T} \mathrm{d}W_t, \tag{10.20}$$

the phase-space density satisfies Kramers' equation

$$\partial_t f = -\frac{p}{m} \cdot \nabla_q f + \nabla_p \cdot [(\nabla V + \gamma p) f] + \gamma m k_B T \, \Delta_p f. \tag{10.21}$$

Its stationary density is

$$f_*(q, p) \propto \exp\left[-\beta \left(\frac{p^2}{2m} + V(q)\right)\right]. \tag{10.22}$$

Friction alone would collapse momenta; noise alone would heat indefinitely. The Einstein relation fixes their amplitudes so that the desired temperature is stationary [18, 33, 57].

Bridge to the next theory

The second derivative in Fokker–Planck comes from the non-vanishing quadratic variation of Brownian paths. Ordinary calculus misses this term. Itô's lemma is the corrected chain rule.

11

Brownian motion and Itô's lemma

11.1 Brownian motion as a mathematical object

A standard Brownian motion $(W_t)_{t\geq 0}$ satisfies:

1. $W_0 = 0$ almost surely;
2. increments over disjoint intervals are independent;
3. $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $t > s$;
4. paths are continuous almost surely.

It is self-similar: $(W_{ct})_{t\geq 0}$ has the same law as $(\sqrt{c}W_t)_{t\geq 0}$. Typical increments therefore scale as $\sqrt{\Delta t}$, making a derivative scale as $1/\sqrt{\Delta t}$ and diverge. Brownian paths are almost surely nowhere differentiable.

11.2 Quadratic variation

For a partition $0 = t_0 < \cdots < t_n = t$, define

$$[W]_t^{(n)} = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2. \tag{11.1}$$

As the mesh tends to zero,

$$[W]_t^{(n)} \to t \tag{11.2}$$

in probability, and almost surely along suitable refining partitions. By contrast, a differentiable path has zero quadratic variation. Symbolically,

$$(\mathrm{d}W_t)^2 = \mathrm{d}t, \qquad \mathrm{d}W_t\,\mathrm{d}t = 0, \qquad (\mathrm{d}t)^2 = 0. \tag{11.3}$$

These are bookkeeping rules for orders in mean square, not ordinary algebraic identities.

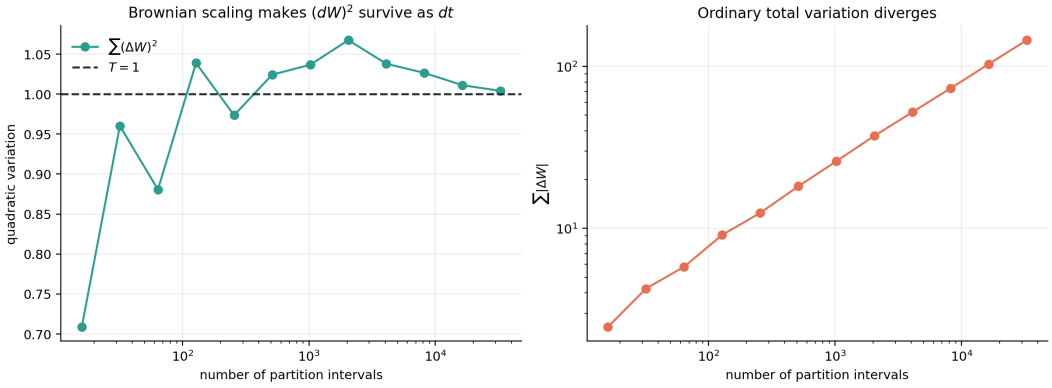


Figure 11.1: Quadratic variation along increasingly fine Brownian partitions approaches elapsed time, while the corresponding quantity for a smooth path tends to zero.

11.3 Derivation of the one-dimensional Itô formula

Let

$$\mathrm{d}X_t = a(X_t, t)\mathrm{d}t + b(X_t, t)\mathrm{d}W_t \tag{11.4}$$

and let $f(x,t)$ be twice differentiable in x and once in t. Expand to second order:

$$\mathrm{d}f = f_t \mathrm{d}t + f_x \mathrm{d}X + \frac{1}{2} f_{xx} (\mathrm{d}X)^2 + \text{higher orders}. \tag{11.5}$$

Using Equation (11.3),

$$(\mathrm{d}X)^2 = (a\mathrm{d}t + b\mathrm{d}W)^2 = b^2 \mathrm{d}t. \tag{11.6}$$

Therefore

$$\boxed{\mathrm{d}f(X_t, t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) \mathrm{d}t + b f_x \mathrm{d}W_t.} \tag{11.7}$$

The extra Hessian term is the entire difference from the ordinary chain rule [26].

Example: geometric Brownian motion

Let $\mathrm{d}X_t = \mu X_t \mathrm{d}t + \sigma X_t \mathrm{d}W_t$ and choose $f(x) = \log x$. Since $f' = 1/x$ and $f'' = -1/x^2$,

$$\mathrm{d} \log X_t = \left(\mu - \frac{1}{2} \sigma^2 \right) \mathrm{d}t + \sigma \mathrm{d}W_t. \tag{11.8}$$

Integrating,

$$X_t = X_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right]. \tag{11.9}$$

The $-\sigma^2/2$ correction would be missed by ordinary calculus.

11.4 Multidimensional Itô formula

For Equation (10.1), $f : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}$ satisfies

$$\mathrm{d}f = \left(\partial_t f + a \cdot \nabla f + \frac{1}{2} \operatorname{tr}(B B^\top \nabla^2 f) \right) \mathrm{d}t + \nabla f^\top B \mathrm{d}W_t. \tag{11.10}$$

The drift term is $(\partial_t + \mathcal{L}) f$. For a vector-valued function, apply the formula componentwise. This is the local route to the generator used in chapter 10.

11.5 Itô integrals and martingales

The stochastic integral

$$\int_0^t H_s \, \mathrm{d}W_s \tag{11.11}$$

is defined as a mean-square limit of sums using left endpoints. For adapted square-integrable H_s, it has mean zero and satisfies Itô's isometry

$$\mathbb{E} \left[\left(\int_0^t H_s \mathrm{d}W_s \right)^2 \right] = \mathbb{E} \left[\int_0^t H_s^2 \mathrm{d}s \right]. \tag{11.12}$$

It is a martingale: its conditional future increment has mean zero. This permits expectation equations to be obtained by dropping the stochastic integral when integrability conditions hold.

11.6 Itô versus Stratonovich

The Stratonovich integral uses midpoint-like sums and is denoted $\circ \mathrm{d}W_t$. For

$$\mathrm{d}X_t = a_S(X_t) \mathrm{d}t + b(X_t) \circ \mathrm{d}W_t, \tag{11.13}$$

the equivalent Itô equation is

$$\mathrm{d}X_t = \left[a_S(X_t) + \frac{1}{2}b(X_t)b'(X_t)\right]\mathrm{d}t + b(X_t)\mathrm{d}W_t. \tag{11.14}$$

In multiple dimensions the correction is $\frac{1}{2}\sum_{j,k} B_{jk}\partial_j B_{ik}$. Stratonovich calculus obeys the ordinary chain rule and often arises as the limit of smooth colored noise with vanishing correlation time. Itô calculus makes adaptedness and martingale structure explicit. Neither convention is intrinsically more physical; the correct drift depends on how the white-noise limit is obtained [29, 69].

11.7 Euler–Maruyama

For step h, the basic numerical scheme is

$$X_{n+1} = X_n + a(X_n, t_n)h + B(X_n, t_n)\sqrt{h}\,\xi_n, \qquad \xi_n \sim \mathcal{N}(0, I). \tag{11.15}$$

It has strong order $1/2$ and weak order 1 under standard conditions. Strong accuracy concerns sample paths; weak accuracy concerns expectations of test functions. In diffusion-based generative modeling, distributional or weak accuracy can matter more than matching a particular noise realization.

Common pitfall

Writing $\dot{W}_t$ is useful physics shorthand for white noise but not an ordinary function. White noise is a generalized random distribution. The mathematically defined object is the integrated SDE involving $\mathrm{d}W_t$.

Bridge to the next theory

Itô calculus tells us how fluctuations propagate. Non-equilibrium statistical mechanics asks how the same microscopic fluctuations are tied to macroscopic dissipation and linear response.

12

Non-equilibrium statistical mechanics and fluctuation–dissipation

12.1 Beyond a stationary Boltzmann law

Equilibrium statistical mechanics predicts stationary distributions and static fluctuations. Non-equilibrium statistical mechanics studies currents, relaxation, driven systems, entropy production, and response to perturbations. A central idea is that spontaneous equilibrium fluctuations encode how the system responds to a weak external force.

Consider an equilibrium Hamiltonian perturbed by a small field $h(t)$ coupled to observable B:

$$H_h = H_0 - h(t)B. \tag{12.1}$$

For another observable A, linear response has the form

$$\delta\langle A(t)\rangle = \int_{-\infty}^{t} R_{AB}(t-s)h(s)\,\mathrm{d}s + O(h^2), \tag{12.2}$$

where causality requires $R_{AB}(\tau) = 0$ for $\tau < 0$.

12.2 A simple Langevin derivation

Take the overdamped harmonic process

$$\gamma\dot{x} = -kx + h(t) + \sqrt{2\gamma k_B T}\,\eta(t). \tag{12.3}$$

Averaging removes the zero-mean noise:

$$\gamma\frac{\mathrm{d}}{\mathrm{d}t}\langle x\rangle = -k\langle x\rangle + h(t). \tag{12.4}$$

The causal response function is

$$R_{xx}(t) = \frac{1}{\gamma}\mathrm{e}^{-kt/\gamma}\Theta(t). \tag{12.5}$$

At equilibrium, the unperturbed correlation is

$$C_{xx}(t) = \langle x(t)x(0)\rangle = \frac{k_B T}{k}\mathrm{e}^{-k|t|/\gamma}. \tag{12.6}$$

For $t > 0$,

$$\boxed{R_{xx}(t) = -\frac{1}{k_B T}\frac{\mathrm{d}}{\mathrm{d}t}C_{xx}(t).} \tag{12.7}$$

The rate at which equilibrium memory decays determines susceptibility to an applied field.

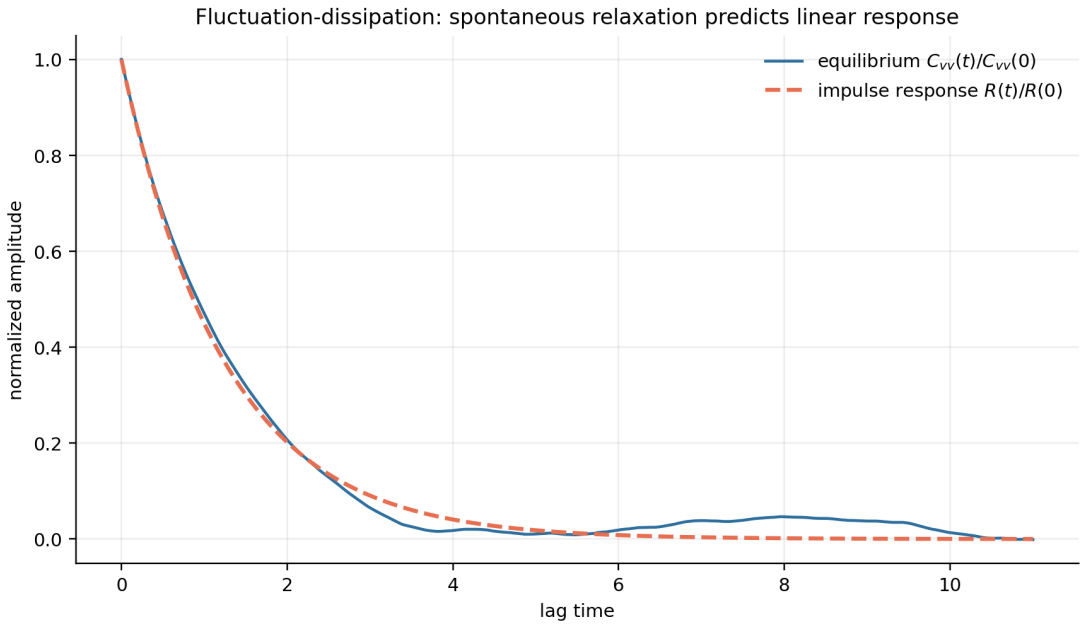


Figure 12.1: Numerical equilibrium correlation and independently measured impulse response for a harmonic Langevin system. The fluctuation–dissipation prediction aligns the two curves.

Computational experiment

The simulated response curve and the derivative of the unperturbed correlation have a root-mean-square discrepancy of about 2.18×10^{-2} at finite sample size and time step. Increasing trajectory count reduces the statistical error.

12.3 Kubo formula

Under equilibrium and time-reversal assumptions, the classical Kubo relation can be written, for suitable conjugate variables, as

$$R_{AB}(t) = \beta \Theta(t) \langle \dot{B}(0) A(t) \rangle_{\mathrm{eq}}. \tag{12.8}$$

Equivalent forms differ by integration by parts and parity under time reversal. In frequency space, the dissipative part of a susceptibility is related to the fluctuation spectrum. The quantum version introduces $\hbar\omega$ factors and becomes the classical formula at low frequency or high temperature [9, 34, 35].

12.4 Green–Kubo transport coefficients

Macroscopic transport coefficients are integrals of equilibrium current correlations. For example, a diffusion coefficient can be written as

$$D = \frac{1}{d} \int_0^\infty \langle v(t) \cdot v(0) \rangle \, \mathrm{d}t. \tag{12.9}$$

Viscosity is related to stress autocorrelation and conductivity to electrical-current autocorrelation. These formulas are striking: transport under weak forcing can be measured from an unforced equilibrium simulation [21].

For a free underdamped Langevin particle,

$$m\dot{v} = -\gamma v + \sqrt{2\gamma k_B T}\,\eta, \tag{12.10}$$

we have $\langle v(t)v(0) \rangle = (k_B T/m)\mathrm{e}^{-\gamma t/m}$. Equation (12.9) gives

$$D = \frac{k_B T}{\gamma}, \tag{12.11}$$

Einstein's relation between diffusion and mobility.

12.5 Onsager reciprocity and entropy production

Near equilibrium, thermodynamic fluxes J_i respond linearly to forces X_j:

$$J_i = \sum_j L_{ij} X_j. \tag{12.12}$$

Microscopic reversibility implies Onsager reciprocity $L_{ij} = L_{ji}$, with sign modifications for variables odd under time reversal or in magnetic fields [52]. The entropy production rate is

$$\dot{S}_{\mathrm{prod}} = \sum_i J_i X_i \geq 0, \tag{12.13}$$

so the symmetric part of L must be positive semidefinite.

For a diffusion with current J and diffusion matrix D, a common steady-state entropy-production form is

$$\dot{S}_{\mathrm{prod}} = \int \frac{J(x)^{\top} D^{-1} J(x)}{p(x)}\, \mathrm{d}x \geq 0, \tag{12.14}$$

up to convention-dependent constants. Detailed balance makes $J = 0$ and eliminates steady entropy production.

12.6 Fluctuation theorems far from equilibrium

Linear response applies near equilibrium. Exact fluctuation relations reach farther. If a system starts in canonical equilibrium and is driven by a protocol, Jarzynski's equality states

$$\left\langle \mathrm{e}^{-\beta W} \right\rangle = \mathrm{e}^{-\beta \Delta F}. \tag{12.15}$$

Jensen's inequality gives $\langle W \rangle \geq \Delta F$. Crooks' relation compares forward and time-reversed work distributions:

$$\frac{P_F(W)}{P_R(-W)} = \mathrm{e}^{\beta (W - \Delta F)}. \tag{12.16}$$

These identities quantify the probability of apparent second-law violations in small fluctuating systems [12, 27, 62].

12.7 Non-equilibrium free-energy decay

For overdamped Langevin dynamics targeting $\pi \propto \mathrm{e}^{-U/T}$, the density obeys

$$\partial_t p = \nabla \cdot (p \nabla U + T \nabla p) = T \nabla \cdot \left(p \nabla \log \frac{p}{\pi} \right). \tag{12.17}$$

Define

$$\mathcal{F}[p] - F = T D_{\mathrm{KL}}(p \| \pi). \tag{12.18}$$

Then, under suitable boundary conditions,

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(p_t \| \pi) = -T \int p_t \left\| \nabla \log \frac{p_t}{\pi} \right\|^2 \mathrm{d}x \leq 0. \tag{12.19}$$

The right-hand side is relative Fisher information. Langevin relaxation is a gradient flow of free energy in probability space. Modern score models learn the gradients appearing in this dissipation law.

Bridge to the next theory

Near-equilibrium theory assumes relaxation around one thermodynamic state. Spin glasses violate that simple picture: frustration creates many competing valleys, long memory, broken ergodicity, and a nontrivial distribution of overlaps between states.

13

Spin glasses, the replica method, and broken symmetry

13.1 Disorder plus frustration

An Ising spin glass has variables $s_i \in \{-1, +1\}$ and random couplings

$$H_J(s) = -\sum_{(i,j)} J_{ij} s_i s_j - h \sum_i s_i. \tag{13.1}$$

Positive J_{ij} favors alignment; negative J_{ij} favors anti-alignment. On a loop whose coupling signs multiply to -1, not all bonds can be satisfied simultaneously. This is frustration. The couplings are quenched: they are drawn once and then remain fixed while spins equilibrate.

Quenched disorder differs from annealed disorder. The physical free energy is

$$F = -\frac{1}{\beta} \mathbb{E}_J[\log Z_J], \qquad Z_J = \sum_s \mathrm{e}^{-\beta H_J(s)}. \tag{13.2}$$

The easier quantity $-\beta^{-1} \log \mathbb{E}_J[Z_J]$ is annealed and generally differs because $\mathbb{E} \log Z \neq \log \mathbb{E} Z$.

13.2 Edwards–Anderson and Sherrington–Kirkpatrick models

The Edwards–Anderson model places random short-range couplings on a lattice [15]. The mean-field Sherrington–Kirkpatrick (SK) model couples every pair:

$$H_J(s) = -\frac{1}{\sqrt{N}} \sum_{i<j} J_{ij} s_i s_j - h \sum_i s_i, \qquad J_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1). \tag{13.3}$$

The $1/\sqrt{N}$ scaling makes the local random field order one and the total energy extensive after optimization [63].

Ordinary magnetization can vanish in a spin-glass phase. The Edwards–Anderson order parameter detects frozen local moments:

$$q_{\mathrm{EA}} = \frac{1}{N} \sum_i \langle s_i \rangle^2. \tag{13.4}$$

A more informative object is the overlap between two independent equilibrium replicas a, b with the same disorder:

$$q_{ab} = \frac{1}{N} \sum_{i=1}^{N} s_i^{(a)} s_i^{(b)}. \tag{13.5}$$

Its distribution $P_J(q)$ reveals whether Gibbs measure is concentrated in one pair of symmetry-related pure states or decomposes into many incongruent states.

13.3 The replica trick

The logarithm in Equation (13.2) can be represented formally as

$$\mathbb{E}_J[\log Z_J] = \lim_{n \to 0} \frac{\mathbb{E}_J[Z_J^n] - 1}{n}. \tag{13.6}$$

For positive integer n, Z_J^n is the partition function of n copies sharing the same disorder:

$$Z_J^n = \sum_{\{s^a\}} \exp\left[\frac{\beta}{\sqrt{N}} \sum_{a=1}^{n} \sum_{i<j} J_{ij} s_i^a s_j^a\right]. \tag{13.7}$$

Averaging Gaussian J_{ij} produces interactions between replicas. Since $\mathbb{E}e^{tJ} = e^{t^2/2}$,

$$\mathbb{E}_J[Z_J^n] = \sum_{\{s^a\}} \exp\left[\frac{\beta^2}{2N} \sum_{i<j} \left(\sum_a s_i^a s_j^a\right)^2\right] \tag{13.8}$$

$$\approx \sum_{\{s^a\}} \exp\left[\frac{\beta^2 N}{4} \sum_{a,b} q_{ab}^2 + O(N^0)\right]. \tag{13.9}$$

Thus the overlap matrix $Q = (q_{ab})$ becomes the order parameter. Delta constraints enforcing its definition lead, at large N, to a saddle-point integral over Q.

Common pitfall

The replica trick is not justified merely by writing $n \to 0$. One computes at integer n, finds a formula, analytically continues it, and takes a non-integer limit. The continuation may be nonunique. In several mean-field models the final Parisi formula has rigorous support, but the heuristic steps should remain explicit.

13.4 Replica symmetry

The simplest ansatz assumes all off-diagonal overlaps are equal:

$$q_{aa} = 1, \qquad q_{ab} = q \quad (a \neq b). \tag{13.10}$$

At zero external field, the SK replica-symmetric self-consistency equation is

$$q = \int \mathcal{D}z \, \tanh^2(\beta\sqrt{q}\, z), \qquad \mathcal{D}z = \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz. \tag{13.11}$$

At high temperature the solution $q = 0$ is stable. Below the critical temperature, a nonzero solution appears, but the replica-symmetric saddle becomes unstable in a region identified by the de Almeida–Thouless condition. It predicts pathologies such as negative entropy at very low temperature.

13.5 Replica symmetry breaking

Parisi's solution replaces one off-diagonal value by a hierarchy. In one-step replica symmetry breaking, replicas form blocks: overlap is q_1 within a block and q_0 across blocks. Full replica symmetry breaking iterates this structure continuously and is encoded by a nondecreasing function $q(x)$, $x \in [0, 1]$ [48, 54, 70].

The physical interpretation is that equilibrium measure decomposes into many pure states α with weights w_α. Two replicas first choose states and then configurations within them. Their overlap distribution is

$$P(q) = \sum_{\alpha,\beta} w_\alpha w_\beta \delta(q - q_{\alpha\beta}). \tag{13.12}$$

In the full mean-field solution, state organization is ultrametric: for three states, the two smallest pairwise overlaps are equal. The energy landscape metaphor becomes a precise probabilistic hierarchy.

13.6 TAP and cavity viewpoints

The Thouless–Anderson–Palmer equations describe local magnetizations m_i:

$$m_i = \tanh\left[\beta\left(\frac{1}{\sqrt{N}}\sum_{j\neq i} J_{ij}m_j - \beta(1-q)m_i + h\right)\right], \qquad q = \frac{1}{N}\sum_i m_i^2. \tag{13.13}$$

The term $-\beta(1-q)m_i$ is the Onsager reaction correction: a spin influences its neighbors and must subtract the feedback returning through them [71].

The cavity method removes one variable, solves the remaining system, and computes the effective random field when the variable is reinserted. On sparse locally tree-like graphs, this becomes belief propagation. In dense weakly coupled models, the cavity expansion leads to TAP equations and approximate message passing. Mézard and Montanari develop this unifying perspective across inference, coding, random constraint problems, and spin glasses [47].

13.7 Exact finite-size experiment

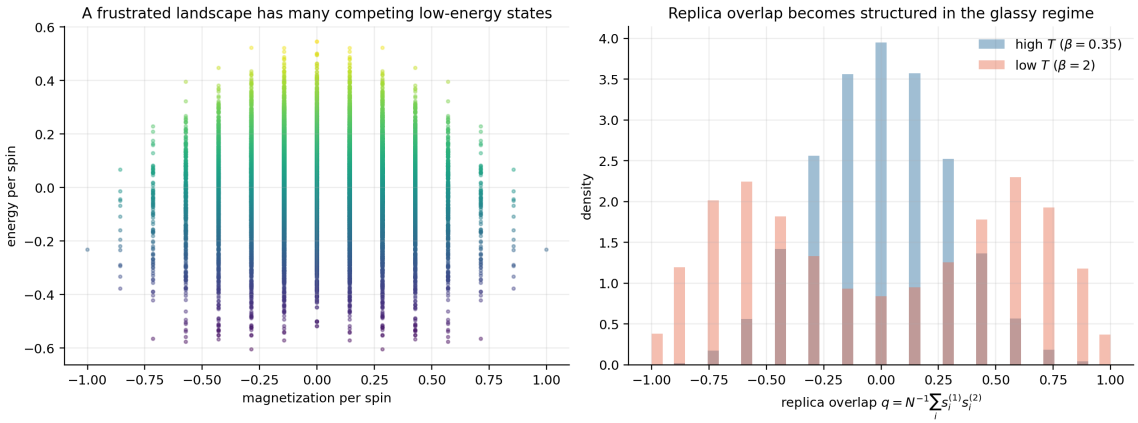


Figure 13.1: Exact enumeration of a small SK instance: rugged energies and overlap statistics change sharply between high and low temperature. Finite N shows the mechanism without pretending to reproduce the thermodynamic-limit Parisi solution.

Computational experiment

For the supplied $N = 16$ SK instance, all 2^{16} states are enumerated. The ground-state energy per spin is approximately -0.6046. The Boltzmann effective number of states $\exp(S)$ falls from about 1.11×10^4 at high temperature to about 1.10×10^2 at low temperature. Run the notebook to resample overlaps and inspect sample-to-sample disorder variability.

13.8 Why spin glasses matter beyond magnets

Random optimization, neural associative memory, compressed sensing, community detection, random satisfiability, and high-dimensional Bayesian inference can all generate glassy landscapes. Replica and cavity calculations predict thresholds, free energies, metastable complexity, and algorithmic barriers. But equilibrium landscape complexity and optimization difficulty are not identical: an algorithm may exploit structure invisible to a generic thermal dynamics, or fail before an information-theoretic threshold.

Bridge to the next theory

Classical statistical mechanics uses probabilities over phase-space points. Quantum mechanics changes the state object to a complex amplitude, but Hamiltonian generators, conservation laws, thermal operators, and even imaginary-time diffusion preserve much of the same mathematical architecture.

14

Quantum mechanics and the Schrödinger equation from first principles

14.1 Why classical trajectories are insufficient

Interference and diffraction show that microscopic systems cannot be represented by a classical probability distribution over definite trajectories alone. A quantum pure state is a normalized vector $|\psi\rangle$ in a complex Hilbert space. In the position representation it is a wavefunction

$$\psi(x,t) = \langle x|\psi(t)\rangle, \qquad \int |\psi(x,t)|^2 \mathrm{d}x = 1. \tag{14.1}$$

Born's rule interprets $|\psi(x,t)|^2$ as position probability density [7]. The complex phase is essential because amplitudes add before probabilities are squared, producing interference.

14.2 From classical energy to a wave equation

For a plane wave

$$\psi(x,t) = A\mathrm{e}^{\mathrm{i}(k\cdot x-\omega t)}, \tag{14.2}$$

quantum relations associate momentum and energy with wave number and frequency:

$$p = \hbar k, \qquad E = \hbar\omega. \tag{14.3}$$

The differential operators

$$\widehat{p} = -\mathrm{i}\hbar\nabla, \qquad \widehat{E} = \mathrm{i}\hbar\partial_t \tag{14.4}$$

return the corresponding eigenvalues on a plane wave. Promoting the classical Hamiltonian $H = p^2/(2m) + V(x,t)$ to an operator gives

$$\boxed{\mathrm{i}\hbar\partial_t\psi(x,t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x,t)\right]\psi(x,t).} \tag{14.5}$$

This is a motivated construction, not a derivation from classical mechanics: the Schrödinger equation is a quantum dynamical postulate whose form is constrained by linearity, locality, spatial symmetries, and the classical dispersion relation [14, 60].

14.3 Unitary time evolution

For a time-independent self-adjoint Hamiltonian $\widehat{H}$,

$$|\psi(t)\rangle = \mathrm{e}^{-\mathrm{i}\widehat{H}t/\hbar}|\psi(0)\rangle. \tag{14.6}$$

The propagator is unitary, so inner products and total probability are conserved. Expanding in energy eigenstates $\widehat{H}|n\rangle = E_n|n\rangle$,

$$|\psi(t)\rangle = \sum_n c_n \mathrm{e}^{-\mathrm{i}E_n t/\hbar}|n\rangle. \tag{14.7}$$

A stationary state changes only by a global phase and has time-independent measurement probabilities.

14.4 Probability continuity equation

Let $\rho = |\psi|^2$. Multiply Equation (14.5) by ψ^* and its conjugate by ψ, then subtract. For real V,

$$\partial_t \rho + \nabla \cdot j = 0, \tag{14.8}$$

where

$$j = \frac{\hbar}{m} \operatorname{Im}(\psi^* \nabla \psi). \tag{14.9}$$

Thus quantum probability obeys a continuity equation, but its current depends on phase as well as density. Two wavefunctions with the same $|\psi|^2$ can have different currents.

Writing $\psi = \sqrt{\rho} \mathrm{e}^{\mathrm{i}S/\hbar}$ gives

$$\partial_t \rho + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) = 0, \tag{14.10}$$

$$\partial_t S + \frac{|\nabla S|^2}{2m} + V - \frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} = 0. \tag{14.11}$$

The second equation is the Hamilton–Jacobi equation plus a quantum potential. In the formal $\hbar \to 0$ limit, the classical equation is recovered where the approximation remains regular.

14.5 Canonical commutation and Poisson brackets

Position and momentum satisfy

$$[\widehat{x}_i, \widehat{p}_j] = \mathrm{i}\hbar \delta_{ij}. \tag{14.12}$$

Heisenberg evolution is

$$\frac{\mathrm{d}\widehat{A}}{\mathrm{d}t} = \frac{\mathrm{i}}{\hbar} [\widehat{H}, \widehat{A}] + \frac{\partial \widehat{A}}{\partial t}. \tag{14.13}$$

Compare this with the classical equation $\dot{A} = \{A, H\} + \partial_t A$. The correspondence $[A, B]/(\mathrm{i}\hbar) \leftrightarrow \{A, B\}$ is structural, though operator ordering makes quantization nontrivial.

The commutator implies uncertainty:

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}. \tag{14.14}$$

This is not measurement noise added to a hidden exact phase point; it is a constraint on variances in any quantum state.

14.6 Free Gaussian wavepacket

For $V = 0$, Fourier components evolve as

$$\widetilde{\psi}(k, t) = \widetilde{\psi}(k, 0) \mathrm{e}^{-\mathrm{i}\hbar k^2 t/(2m)}. \tag{14.15}$$

A Gaussian packet remains Gaussian but spreads because different wave numbers accumulate different phases. For initial width σ_0,

$$\sigma_t^2 = \sigma_0^2 + \frac{\hbar^2 t^2}{4m^2 \sigma_0^2}. \tag{14.16}$$

Unlike ordinary diffusion, this variance grows quadratically at late times and the evolution remains reversible.

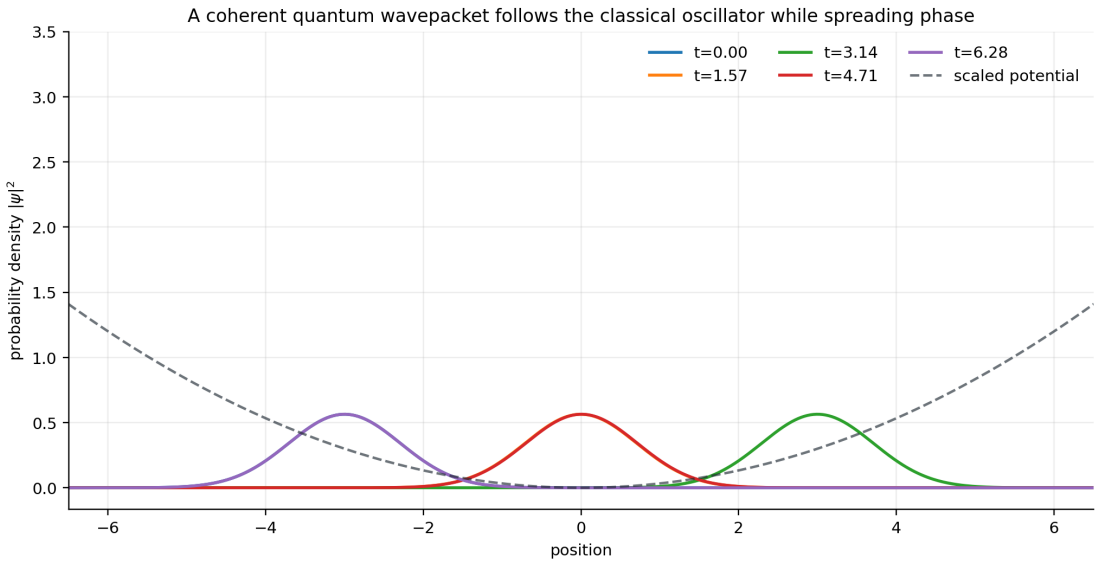


Figure 14.1: Split-step Fourier evolution of a quantum wavepacket. Interference and dispersion arise from phase evolution; the numerical norm is conserved to about 4×10^{-13}.

14.7 Split-step Fourier method

For $\widehat{H} = \widehat{T} + \widehat{V}$, a second-order Strang splitting is

$$e^{-ih(\widehat{T}+\widehat{V})/\hbar} = e^{-ih\widehat{V}/(2\hbar)} e^{-ih\widehat{T}/\hbar} e^{-ih\widehat{V}/(2\hbar)} + O(h^3). \tag{14.17}$$

Potential evolution is diagonal in position space; kinetic evolution is diagonal in Fourier space. Each factor is unitary, producing excellent norm conservation.

14.8 Path integrals and stationary action

The propagator can be represented formally as a sum over paths

$$K(x_1, t_1; x_0, t_0) = \int_{x(t_0)=x_0}^{x(t_1)=x_1} \mathcal{D}x \, \exp\left(\frac{i}{\hbar} S[x]\right). \tag{14.18}$$

When $S \gg \hbar$, rapidly oscillating phases cancel except near stationary-action paths. Classical mechanics emerges by stationary phase, not because nonclassical paths are absent [17].

14.9 Imaginary time and statistical mechanics

Set $t = -i\tau$. Then Schrödinger evolution becomes

$$-\hbar \partial_\tau \psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]\psi, \tag{14.19}$$

which resembles diffusion with killing by V. The thermal density operator is

$$\widehat{\rho}_\beta = \frac{e^{-\beta \widehat{H}}}{\operatorname{Tr} e^{-\beta \widehat{H}}}. \tag{14.20}$$

Imaginary-time path integrals map quantum statistical mechanics to classical statistical fields in one additional dimension. This is a deep, precise bridge between amplitudes, diffusion, and Boltzmann weights.

Bridge to the next theory

The first part has moved from deterministic trajectories to probability and amplitude transport. The second begins with the historical observation that forced physicists to combine microscopic mechanics with stochastic motion: Brownian motion.

Part II

From Brownian motion to generative flows

15

Brownian motion: experiment, atoms, and probability

15.1 From an observation to a theory of matter

In 1827 Robert Brown observed irregular motion of microscopic particles suspended in fluid and reported it in 1828 [8]. The motion persisted for inorganic particles, so it could not be attributed to life. The mechanism remained unclear for decades because individual molecular collisions were invisible and atomic theory was still contested.

The conceptual breakthrough was to stop predicting each collision. In 1905 Einstein derived a diffusion law from osmotic equilibrium and molecular kinetics; Smoluchowski obtained a closely related theory in 1906 [16, 64]. Langevin's 1908 treatment wrote an explicit random-force equation, creating the dynamical viewpoint developed in the next chapter [37, 40]. Perrin's measurements of displacement distributions and sedimentation provided influential quantitative evidence for atoms and Avogadro's number [56].

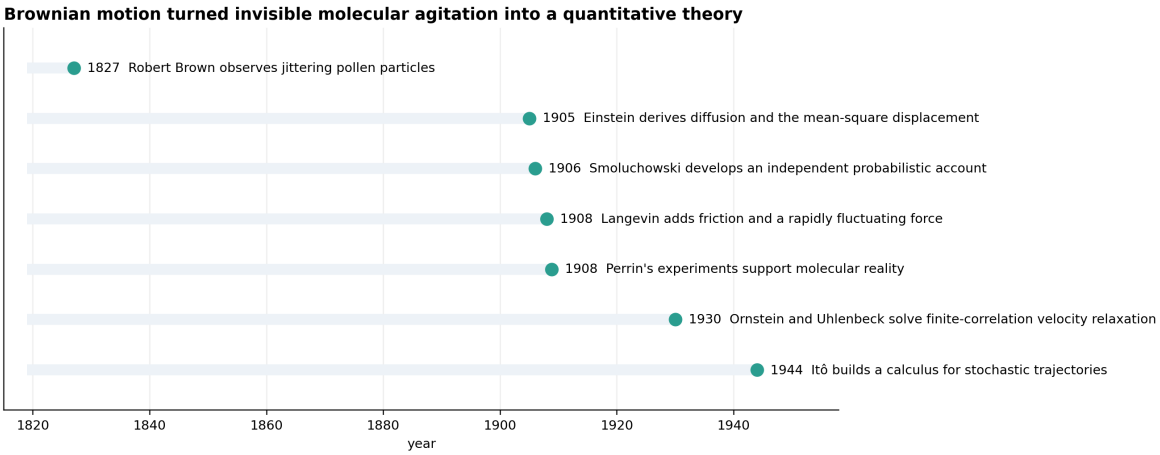


Figure 15.1: Selected milestones connecting Brown's microscopy, Einstein and Smoluchowski's diffusion theories, Langevin's stochastic dynamics, Perrin's experiments, and modern diffusion-based generative modeling.

15.2 Einstein's diffusion argument

Let $p(x,t)$ be the concentration of suspended particles. Fick's law postulates a diffusive flux

$$J_{\text{diff}} = -D\nabla p. \tag{15.1}$$

Conservation gives

$$\partial_t p = D\Delta p. \tag{15.2}$$

For a particle initially at the origin in d dimensions, the fundamental solution is

$$p(x,t) = \frac{1}{(4\pi Dt)^{d/2}} \exp\left(-\frac{\|x\|^2}{4Dt}\right). \tag{15.3}$$

Each coordinate has variance $2Dt$, so

$$\boxed{\mathbb{E}\|X_t - X_0\|^2 = 2dDt.} \tag{15.4}$$

Mean-squared displacement grows linearly, not quadratically as it would for constant ballistic velocity.

Einstein balanced diffusion against drift under an external force F. If mobility is μ, drift velocity is μF and total flux is

$$J = \mu F p - D \nabla p. \tag{15.5}$$

For a conservative force $F = -\nabla U$, equilibrium requires $J = 0$ and $p \propto \mathrm{e}^{-U/(k_B T)}$. Therefore

$$D = \mu k_B T. \tag{15.6}$$

For a sphere of radius a in a fluid with viscosity η, Stokes mobility is $\mu = (6\pi\eta a)^{-1}$, yielding the Stokes–Einstein relation

$$D = \frac{k_B T}{6\pi\eta a}. \tag{15.7}$$

This turns visible random motion into a measurement of molecular-scale constants.

15.3 Random walks and the central limit mechanism

Consider independent steps ξ_k with zero mean and variance σ^2. After n steps,

$$S_n = \sum_{k=1}^{n} \xi_k, \qquad \mathbb{E}[S_n^2] = n\sigma^2. \tag{15.8}$$

Take a time step Δt and spatial step with variance $2D\Delta t$. At fixed $t = n\Delta t$, the central limit theorem gives a Gaussian displacement with variance $2Dt$. Letting $\Delta t \to 0$ produces Brownian motion and the heat equation.

This limit explains universality: microscopic collision details disappear if increments are sufficiently weak, numerous, and short-correlated. When step variances diverge or correlations persist, other limits arise, such as Lévy flights or fractional Brownian motion.

15.4 Diffusive versus ballistic regimes

A real particle has inertia. At very short times its velocity has not decorrelated, so

$$\mathbb{E}[(X_t - X_0)^2] \approx \mathbb{E}[V_0^2] t^2. \tag{15.9}$$

At times long compared with the momentum relaxation time $\tau_m = m/\gamma$, repeated collisions randomize velocity and the mean-squared displacement becomes $2Dt$. The underdamped Langevin equation interpolates between these regimes:

$$\mathbb{E}[(X_t - X_0)^2] = 2D\left[t - \tau_m(1 - \mathrm{e}^{-t/\tau_m})\right] \tag{15.10}$$

for stationary initial velocity in one dimension. Expanding at small t gives $(k_B T/m)t^2$; at large t it gives $2Dt$.

15.5 Sample paths and ensemble laws

A Brownian trajectory is continuous, nowhere differentiable, and has unbounded total variation on every interval. Yet its probability density is smooth for every $t > 0$. This contrast is fundamental: stochastic processes can have extremely rough paths while their ensemble evolution obeys a regular partial differential equation.

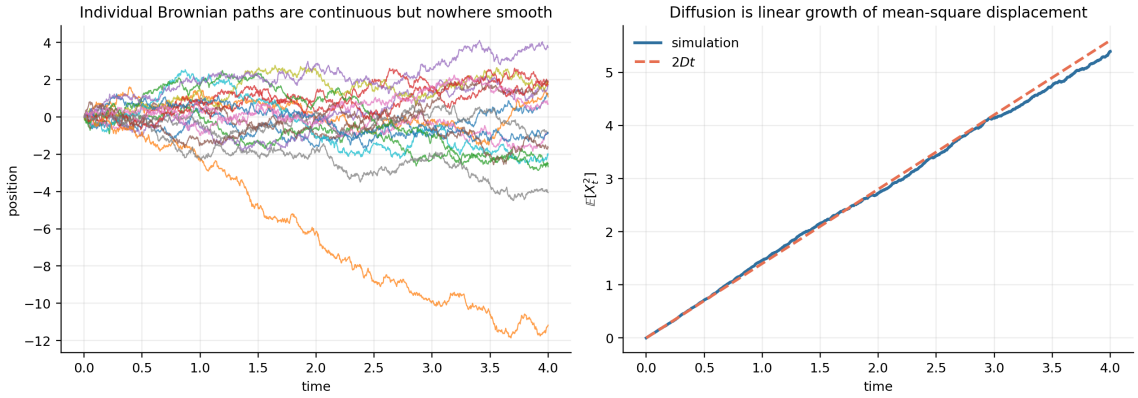


Figure 15.2: Brownian sample paths are irregular, while averaging squared displacement over many paths produces the smooth linear law $\mathbb{E}[X_t^2] = 2Dt$.

Computational experiment

The simulation generates independent Gaussian increments with variance $2D\Delta t$. The final relative error of the empirical mean-squared displacement against $2Dt$ is about 3.68% for the chosen finite ensemble. The interactive plot allows individual trajectories to be hidden or inspected without changing the ensemble statistic.

15.6 Brownian motion as a Markov semigroup

The heat kernel defines transition densities

$$p_t(y \mid x) = \frac{1}{(4\pi Dt)^{d/2}} \exp\left(-\frac{\|y-x\|^2}{4Dt}\right). \tag{15.11}$$

They satisfy the Chapman–Kolmogorov semigroup property

$$p_{t+s}(z \mid x) = \int p_t(z \mid y) p_s(y \mid x)\, \mathrm{d}y. \tag{15.12}$$

The associated operator $P_t f(x) = \mathbb{E}[f(x + \sqrt{2D}W_t)]$ obeys $P_{t+s} = P_t P_s$ and has generator $D\Delta$. Diffusion models deliberately construct a time-inhomogeneous Gaussian Markov process that gradually erases data structure, then learn how to reverse it.

Bridge to the next theory

Einstein derived the law of the density. Langevin supplied a law for each noisy trajectory. Combining inertia, friction, and a random force makes the Boltzmann distribution emerge as the stationary state and exposes the fluctuation–dissipation relation dynamically.

16

Langevin dynamics: inertia, friction, and thermal noise

16.1 The underdamped equation

For a particle of mass m in potential $U(q)$, Langevin dynamics is

$$\mathrm{d}Q_t = \frac{P_t}{m}\mathrm{d}t, \tag{16.1}$$

$$\mathrm{d}P_t = -\nabla U(Q_t)\mathrm{d}t - \gamma P_t \mathrm{d}t + \sqrt{2\gamma m k_B T}\mathrm{d}W_t. \tag{16.2}$$

The three momentum terms have distinct roles:

1. $-\nabla U$ is deterministic force;
2. $-\gamma P$ dissipates kinetic energy and removes memory;
3. the Brownian force injects fluctuations from the bath.

The amplitude $\sqrt{2\gamma m k_B T}$ is fixed so that the stationary momentum variance is $m k_B T$. Friction and noise are not independent modeling choices at equilibrium.

16.2 A bath-of-oscillators derivation

A microscopic route couples a coordinate Q to many harmonic modes (x_j, p_j):

$$H = \frac{P^2}{2m} + U(Q) + \sum_j \left[\frac{p_j^2}{2m_j} + \frac{1}{2}m_j\omega_j^2\left(x_j - \frac{c_j Q}{m_j\omega_j^2}\right)^2\right]. \tag{16.3}$$

The bath equations can be solved exactly and substituted into the system equation, giving a generalized Langevin equation

$$m\ddot{Q}(t) = -U'(Q(t)) - \int_0^t K(t-s)\dot{Q}(s)\,\mathrm{d}s + R(t). \tag{16.4}$$

The memory kernel K depends on the bath spectrum, and the random force R depends on bath initial conditions. If those initial conditions are thermal,

$$\langle R(t)R(s)\rangle = k_B T K(|t-s|). \tag{16.5}$$

A broad, rapidly decorrelating bath makes $K(t) \approx 2m\gamma\delta(t)$, producing Markovian friction and white noise. This derivation shows what the effective stochastic terms summarize: discarded Hamiltonian degrees of freedom.

16.3 Stationary Boltzmann distribution

The Kramers equation is Equation (10.21). Substitute

$$f_*(q,p) = Z^{-1}\exp[-\beta H(q,p)], \qquad H = \frac{p^2}{2m} + U(q). \tag{16.6}$$

The Hamiltonian transport terms cancel because f_* is a function of H. For the momentum terms,

$$\nabla_p \cdot (\gamma p f_*) + \gamma m k_B T \Delta_p f_* = 0, \tag{16.7}$$

using $\nabla_p f_* = -\beta(p/m) f_*$. Hence f_* is stationary.

The stationary probability current is subtler than in the overdamped case because Hamiltonian flow circulates in phase space. Detailed balance includes momentum reversal under time reversal, not simply zero phase-space current everywhere.

16.4 Energy balance

Apply Itô's lemma to $H = P^2/(2m) + U(Q)$ in one dimension:

$$\mathrm{d}H = \frac{P}{m}\mathrm{d}P + U'(Q)\mathrm{d}Q + \frac{1}{2m}(\mathrm{d}P)^2 \tag{16.8}$$

$$= \left[-\frac{\gamma P^2}{m} + \gamma k_B T\right]\mathrm{d}t + \sqrt{\frac{2\gamma k_B T}{m}} P \mathrm{d}W_t. \tag{16.9}$$

Taking expectations,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[H] = -\gamma\left(\mathbb{E}\left[\frac{P^2}{m}\right] - k_B T\right). \tag{16.10}$$

Friction removes kinetic energy when it exceeds equipartition; noise restores it when it is below. At equilibrium $\mathbb{E}[P^2/m] = k_B T$.

16.5 Free-particle velocity correlation

For $U = 0$, momentum is an OU process:

$$\mathrm{d}P_t = -\gamma P_t \mathrm{d}t + \sqrt{2\gamma m k_B T}\mathrm{d}W_t. \tag{16.11}$$

At stationarity,

$$\langle V_t V_0\rangle = \frac{k_B T}{m}\mathrm{e}^{-\gamma t}. \tag{16.12}$$

Integrating velocity gives long-time diffusion coefficient

$$D = \int_0^\infty \langle V_t V_0\rangle \mathrm{d}t = \frac{k_B T}{m\gamma}. \tag{16.13}$$

If friction is instead written as $-\zeta V$, then $\zeta = m\gamma$ and $D = k_B T/\zeta$.

16.6 Numerical splitting

Naive Euler integration can distort the invariant measure. Splitting methods alternate exactly or accurately solvable pieces:

1. kick momentum by the force;
2. drift position using momentum;
3. solve the OU thermostat exactly;
4. reverse the deterministic half-steps.

The BAOAB family is widely used because it has favorable configurational sampling accuracy for many systems. The supplied implementation uses a symmetric kick–drift–OU–drift–kick structure.

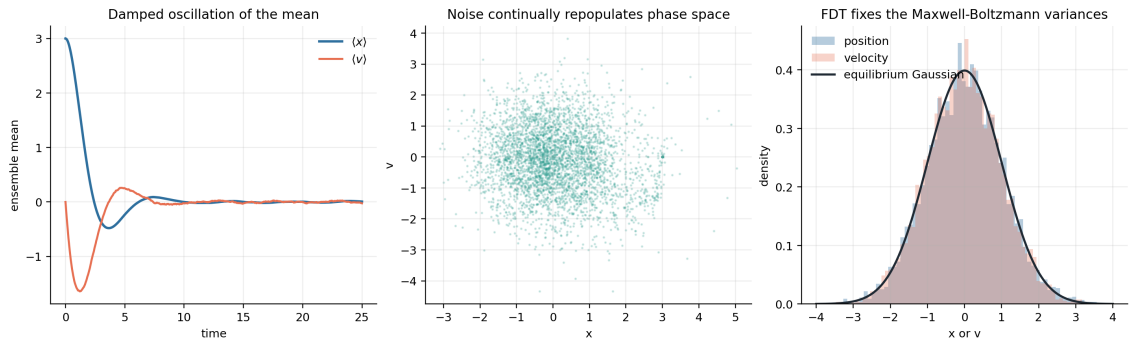


Figure 16.1: Underdamped Langevin trajectories in a potential. Momentum creates inertial loops, friction dissipates them, and noise continually re-excites motion so that the ensemble approaches thermal equilibrium.

16.7 Langevin dynamics as a sampler

Given any energy $U_\theta(x)$, Langevin dynamics can sample approximately from $p_\theta(x) \propto \mathrm{e}^{-U_\theta(x)/T}$. This is the physical origin of Langevin Monte Carlo. Momentum can accelerate exploration across flat directions; friction tunes the transition between nearly Hamiltonian motion and overdamped diffusion. Hamiltonian Monte Carlo removes stochastic forcing during trajectories and restores exactness with a Metropolis correction.

Common pitfall

A discretized Langevin chain does not in general have the exact Boltzmann law as its invariant distribution. Step-size bias disappears as $h \to 0$ or can be corrected by a Metropolis accept/reject step. In machine learning, unadjusted Langevin dynamics is often used despite this bias for computational reasons.

Bridge to the next theory

When momentum relaxes much faster than position, it can be eliminated. The resulting overdamped process moves directly in configuration space and is the cleanest bridge between Boltzmann energies, score fields, and diffusion-model sampling.

17

Overdamped Langevin dynamics and Boltzmann stationarity

17.1 Eliminating fast momentum

Write physical underdamped dynamics as

$$m\ddot{X}_t = -\zeta \dot{X}_t - \nabla U(X_t) + \sqrt{2\zeta k_B T}\,\eta(t). \tag{17.1}$$

When the momentum relaxation time m/ζ is much shorter than the positional timescale, set the inertial term to leading order to obtain

$$\boxed{\mathrm{d}X_t = -\mu \nabla U(X_t)\mathrm{d}t + \sqrt{2\mu k_B T}\mathrm{d}W_t,} \qquad \mu = \zeta^{-1}. \tag{17.2}$$

Rescaling time can set $\mu = 1$. The corresponding Smoluchowski equation is

$$\partial_t p = \mu \nabla \cdot (p \nabla U + k_B T \nabla p). \tag{17.3}$$

17.2 Direct proof of the stationary law

Let

$$\pi(x) = Z^{-1} \mathrm{e}^{-\beta U(x)}, \qquad \beta = (k_B T)^{-1}. \tag{17.4}$$

Then $\nabla \pi = -\beta \pi \nabla U$, so the stationary current is

$$J_\pi = -\mu \pi \nabla U - \mu k_B T \nabla \pi = 0. \tag{17.5}$$

Therefore $\partial_t \pi = -\nabla \cdot J_\pi = 0$. Under confining and irreducibility conditions, this stationary law is unique and the process converges to it.

A second proof writes

$$\partial_t p = \mu k_B T \nabla \cdot \left(p \nabla \log \frac{p}{\pi} \right). \tag{17.6}$$

Then Equation (12.19) shows that $D_{\mathrm{KL}}(p_t \| \pi)$ decreases monotonically. The equilibrium law is the minimizer of free energy and the fixed point of the dynamics.

Intuition

The deterministic force sends particles downhill in energy; thermal diffusion spreads them uphill in probability. At equilibrium these currents cancel at every point. Low temperature emphasizes minima of U; high temperature flattens the distribution.

17.3 Double-well example

Take

$$U(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2. \tag{17.7}$$

There are minima at $x = \pm 1$ and a barrier at $x = 0$. The equilibrium density is bimodal:

$$\pi(x) \propto \exp\left[-\frac{1}{T}\left(\frac{1}{4}x^4 - \frac{1}{2}x^2 \right) \right]. \tag{17.8}$$

At low T, local relaxation inside a well is fast but barrier crossing is rare. Kramers' escape time scales exponentially as

$$\tau_{\text{escape}} \asymp C e^{\Delta U/T} \tag{17.9}$$

with a prefactor determined by curvatures. Thus a process can possess the correct invariant law and still be unusably slow as a sampler.

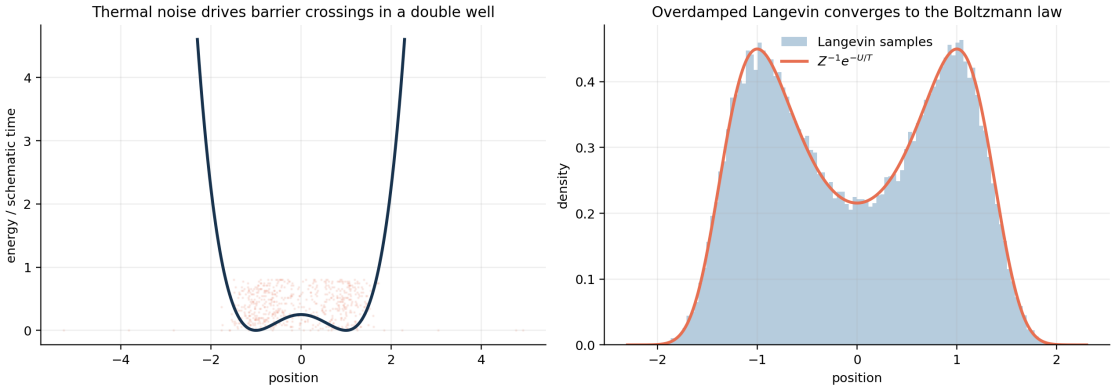


Figure 17.1: Overdamped Langevin relaxation in a double-well energy. Histograms approach the analytic Boltzmann density, while individual paths display metastable residence and occasional barrier crossings.

17.4 Unadjusted Langevin algorithm

Discretizing Equation (17.2) with step h gives

$$X_{k+1} = X_k - h\mu\nabla U(X_k) + \sqrt{2h\mu k_B T}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I). \tag{17.10}$$

This is the unadjusted Langevin algorithm (ULA). If $U = -\log p$ and $T = 1$, the drift is the score $\nabla \log p$. The Metropolis-adjusted Langevin algorithm uses Equation (17.10) as a proposal and corrects its finite-step bias with an acceptance ratio.

In high dimensions, choosing h involves stiffness: narrow directions require small steps, while broad directions mix slowly. Preconditioning replaces scalar mobility by a positive matrix M:

$$\mathrm{d}X_t = -M\nabla U(X_t)\mathrm{d}t + \sqrt{2k_B T M}\mathrm{d}W_t. \tag{17.11}$$

If M depends on x, additional Itô drift terms are generally needed to preserve the intended density.

17.5 Gradient flow in Wasserstein space

Equation (17.3) can be written formally as

$$\partial_t p = \nabla \cdot \left(p \nabla \frac{\delta \mathcal{F}}{\delta p} \right), \tag{17.12}$$

where

$$\mathcal{F}[p] = \int U(x)p(x)\mathrm{d}x + k_B T \int p(x)\log p(x)\mathrm{d}x. \tag{17.13}$$

This is steepest descent of free energy under the geometry induced by optimal transport. The deterministic gradient flow of particles minimizes U and collapses to modes; diffusion adds entropy, making the density rather than a point the optimization object.

17.6 Annealing and simulated annealing

Let temperature decrease in time:

$$\mathrm{d}X_t = -\nabla U(X_t)\mathrm{d}t + \sqrt{2T(t)}\mathrm{d}W_t. \tag{17.14}$$

High temperature crosses barriers; low temperature concentrates near global minima. Convergence guarantees require extremely slow logarithmic cooling in general. Diffusion generative models use a superficially opposite schedule: they first add enough noise to make the distribution simple, then reverse a learned, time-dependent stochastic process to recover all data modes.

Bridge to the next theory

The Ornstein–Uhlenbeck process is the exactly solvable linear Langevin model. Its Gaussian transition kernel, mean reversion, and stationary variance provide both a laboratory for stochastic calculus and the forward noising mechanism behind variance-preserving diffusion models.

18

The Ornstein–Uhlenbeck process

18.1 Exact solution

The scalar OU process is

$$\mathrm{d}X_t = -\theta(X_t - \mu)\mathrm{d}t + \sigma \mathrm{d}W_t, \qquad \theta > 0. \tag{18.1}$$

Multiply by the integrating factor $\mathrm{e}^{\theta t}$:

$$\mathrm{d}(\mathrm{e}^{\theta t} X_t) = \theta\mu \mathrm{e}^{\theta t}\mathrm{d}t + \sigma \mathrm{e}^{\theta t}\mathrm{d}W_t. \tag{18.2}$$

Integrating gives

$$X_t = \mu + (X_0 - \mu)\mathrm{e}^{-\theta t} + \sigma \int_0^t \mathrm{e}^{-\theta(t-s)}\mathrm{d}W_s. \tag{18.3}$$

The stochastic integral is Gaussian. Conditional on $X_0 = x_0$,

$$\mathbb{E}[X_t \mid x_0] = \mu + (x_0 - \mu)\mathrm{e}^{-\theta t}, \tag{18.4}$$

$$\mathrm{Var}(X_t \mid x_0) = \frac{\sigma^2}{2\theta}(1 - \mathrm{e}^{-2\theta t}). \tag{18.5}$$

As $t \to \infty$, the stationary law is

$$X_\infty \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{2\theta}\right). \tag{18.6}$$

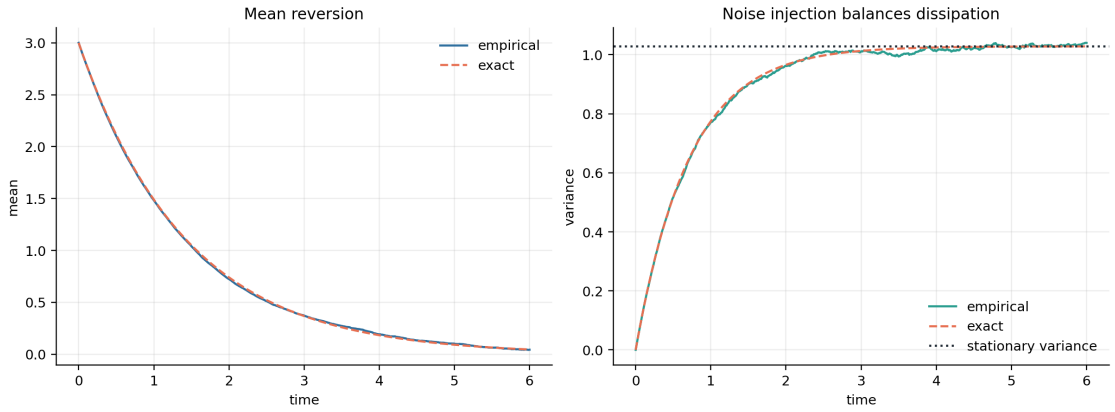


Figure 18.1: Monte Carlo mean and variance of an OU process compared with exact formulas. The process forgets its initial condition exponentially and approaches a stationary Gaussian.

Computational experiment

For the supplied ensemble, the final absolute errors are approximately 1.26×10^{-3} in the mean and 4.26×10^{-3} in the variance. Because the transition law is exact, the experiment separately exposes Monte Carlo error and time-discretization error.

18.2 Stationary covariance and spectrum

If initialized at stationarity, the autocovariance is

$$C(\tau) = \frac{\sigma^2}{2\theta} \mathrm{e}^{-\theta|\tau|}. \tag{18.7}$$

Its power spectral density is the Fourier transform

$$S(\omega) = \frac{\sigma^2}{\theta^2 + \omega^2}. \tag{18.8}$$

Thus the OU process is a low-pass filtered white noise with correlation time θ^{-1}. It is the unique stationary Gaussian Markov process with continuous paths, up to affine transformations [72].

18.3 Multivariate OU dynamics

For $X_t \in \mathbb{R}^d$,

$$\mathrm{d}X_t = AX_t \mathrm{d}t + B\mathrm{d}W_t, \tag{18.9}$$

with eigenvalues of A having negative real parts, the stationary law is Gaussian with zero mean and covariance Σ solving the continuous Lyapunov equation

$$A\Sigma + \Sigma A^\top + BB^\top = 0. \tag{18.10}$$

If A has an antisymmetric component, the process can sustain rotational probability currents even with a stationary Gaussian density. The covariance alone does not diagnose detailed balance.

18.4 Time-inhomogeneous OU and variance-preserving noise

A variance-preserving (VP) diffusion uses

$$\mathrm{d}X_t = -\frac{1}{2}\beta(t)X_t \mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}W_t. \tag{18.11}$$

Define

$$\alpha(t) = \exp\left[-\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s\right], \qquad \sigma_t^2 = 1 - \alpha(t)^2. \tag{18.12}$$

The exact conditional law is

$$X_t \mid X_0 = x_0 \sim \mathcal{N}(\alpha(t)x_0, \sigma_t^2 I). \tag{18.13}$$

If data have unit coordinate variance, the process approximately preserves it while destroying non-Gaussian structure. As $\alpha(t) \to 0$, X_t approaches a standard normal independently of X_0.

18.5 Score of a Gaussian-corrupted distribution

The marginal density is a scaled Gaussian convolution

$$p_t(x) = \int p_0(x_0)\mathcal{N}(x; \alpha_t x_0, \sigma_t^2 I)\,\mathrm{d}x_0. \tag{18.14}$$

Its score can be related to the conditional posterior mean:

$$\nabla_x \log p_t(x) = -\frac{x - \alpha_t \mathbb{E}[X_0 \mid X_t = x]}{\sigma_t^2}. \tag{18.15}$$

Equivalently,

$$\mathbb{E}[X_0 \mid X_t = x] = \frac{x + \sigma_t^2 \nabla \log p_t(x)}{\alpha_t}. \tag{18.16}$$

The score is therefore a denoising vector field: it points toward the posterior mean clean signal after accounting for the Gaussian corruption.

Bridge to the next theory

Every noised marginal p_t defines an energy $E_t(x) = -\log p_t(x)$ and a score $-\nabla E_t(x)$. This turns a diffusion model into a time-indexed family of energy landscapes, with a learned field that avoids computing any partition function.

19

Energy-based models, scores, and denoising

19.1 From Boltzmann physics to a statistical model

An energy-based model (EBM) assigns a scalar energy $E_\theta(x)$ and defines

$$p_\theta(x) = \frac{\mathrm{e}^{-E_\theta(x)}}{Z_\theta}, \qquad Z_\theta = \int \mathrm{e}^{-E_\theta(x)}\mathrm{d}x. \tag{19.1}$$

Temperature can be absorbed into E_θ. Low-energy configurations are likely; high-energy configurations are unlikely. Adding a constant to the energy changes Z_θ but not the normalized density.

The partition function couples every point in state space. For flexible high-dimensional energies, evaluating Z_θ and its gradient is generally intractable. This is the central computational difficulty and also the reason score methods are powerful: spatial derivatives remove the unknown constant.

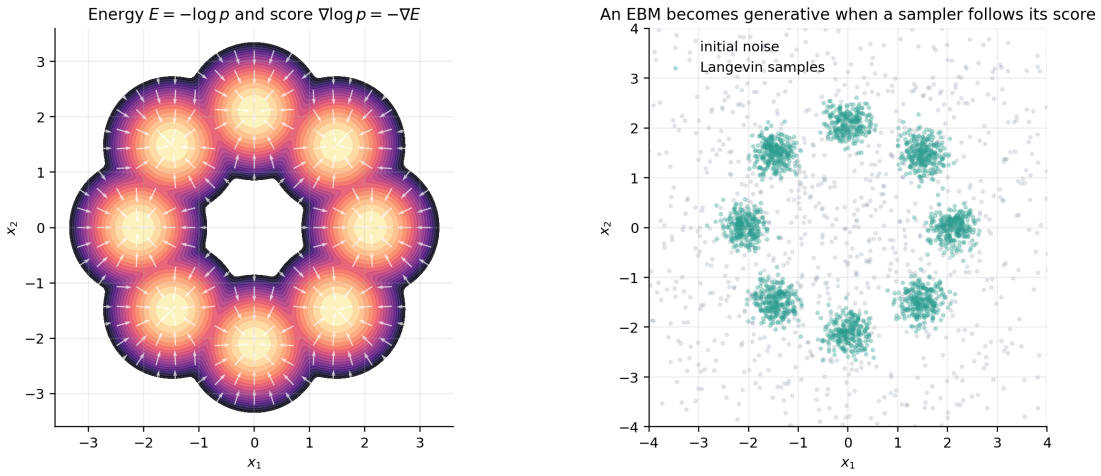


Figure 19.1: An energy function and its Boltzmann density. Probability accumulates in low-energy basins; the score is the negative energy gradient and points uphill in log density.

19.2 Maximum likelihood and the positive–negative phases

For one observation x, the log likelihood is

$$\log p_\theta(x) = -E_\theta(x) - \log Z_\theta. \tag{19.2}$$

Differentiate:

$$\nabla_\theta \log p_\theta(x) = -\nabla_\theta E_\theta(x) - \frac{1}{Z_\theta}\nabla_\theta Z_\theta \tag{19.3}$$

$$= -\nabla_\theta E_\theta(x) + \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta(X)]. \tag{19.4}$$

The data term lowers energy on observations (positive phase); the model expectation raises energy on samples the model currently favors (negative phase). Estimating the negative phase requires sampling from the evolving model, historically using Gibbs or Langevin dynamics [1, 24, 39].

Contrastive divergence starts chains near data and runs only a few steps, trading consistency for speed. Persistent

contrastive divergence reuses model particles. Modern alternatives include noise-contrastive objectives, variational bounds, and score matching.

19.3 The score eliminates the partition function

The score of a density is

$$s_\theta(x) = \nabla_x \log p_\theta(x). \tag{19.5}$$

For Equation (19.1),

$$s_\theta(x) = -\nabla_x E_\theta(x), \tag{19.6}$$

because Z_θ does not depend on x. If we can learn the score directly, we know the local geometry of log density without knowing its global normalization.

A conservative score field is the gradient of a scalar potential. A generic neural vector field need not be exactly conservative, especially when trained approximately. Diffusion samplers only require a useful time-dependent vector field, but integrability matters when interpreting it as one global EBM or estimating likelihoods.

19.4 Explicit score matching

Suppose data have unknown density $p_*(x)$ and model score $s_\theta(x)$. Fisher divergence is

$$\mathcal{J}(\theta) = \frac{1}{2}\mathbb{E}_{p_*}\left[\|s_\theta(X) - \nabla \log p_*(X)\|^2\right]. \tag{19.7}$$

Expanding gives a model term, an unknown data-score norm independent of θ, and a cross term

$$-\mathbb{E}_{p_*}[s_\theta \cdot \nabla \log p_*] = -\int s_\theta \cdot \nabla p_* \, \mathrm{d}x. \tag{19.8}$$

Assuming boundary terms vanish, integration by parts yields

$$-\int s_\theta \cdot \nabla p_* \mathrm{d}x = \int p_* \nabla \cdot s_\theta \mathrm{d}x. \tag{19.9}$$

Therefore, up to a constant independent of θ,

$$\boxed{\mathcal{J}_{\mathrm{SM}}(\theta) = \mathbb{E}_{p_*}\left[\frac{1}{2}\|s_\theta(X)\|^2 + \nabla \cdot s_\theta(X)\right].} \tag{19.10}$$

This objective uses only data samples and model derivatives [25]. Computing divergence exactly can be expensive in high dimension; sliced or Hutchinson trace estimators reduce cost.

19.5 Denoising score matching

Direct data distributions may concentrate near a low-dimensional manifold, making their score singular or poorly behaved. Add Gaussian noise:

$$\widetilde{X} = X + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{19.11}$$

The conditional corruption score is known:

$$\nabla_{\widetilde{x}} \log q_\sigma(\widetilde{x} \mid x) = -\frac{\widetilde{x} - x}{\sigma^2} = -\frac{\varepsilon}{\sigma}. \tag{19.12}$$

Denoising score matching minimizes

$$\mathcal{L}_\sigma(\theta) = \mathbb{E}_{x\sim p_*, \varepsilon}\left[\left\|s_\theta(x + \sigma\varepsilon, \sigma) + \frac{\varepsilon}{\sigma}\right\|^2\right]. \tag{19.13}$$

The regression minimizer is the marginal score

$$s^*(\widetilde{x}, \sigma) = \nabla_{\widetilde{x}} \log p_\sigma(\widetilde{x}), \tag{19.14}$$

not merely the conditional score. This follows from conditional expectation:

$$\mathbb{E}\left[-\frac{\varepsilon}{\sigma} \mid \widetilde{X} = \widetilde{x}\right] = \nabla \log p_\sigma(\widetilde{x}). \tag{19.15}$$

Vincent established the connection between denoising autoencoders and score matching [73].

19.6 Tweedie's formula

For additive Gaussian noise $Y = X + \sigma\varepsilon$,

$$\mathbb{E}[X \mid Y = y] = y + \sigma^2 \nabla_y \log p_Y(y). \tag{19.16}$$

Thus the score is a scaled optimal denoising residual. Predicting noise, clean data, a velocity, or the score are related parameterizations of the same conditional information, with different numerical conditioning across noise levels.

19.7 Noise-conditional score fields

A single low-noise energy landscape can be highly irregular. Convolution with Gaussian noise smooths it. At large noise, modes merge and the score becomes approximately linear toward the global center; at low noise, the field resolves fine local structure.

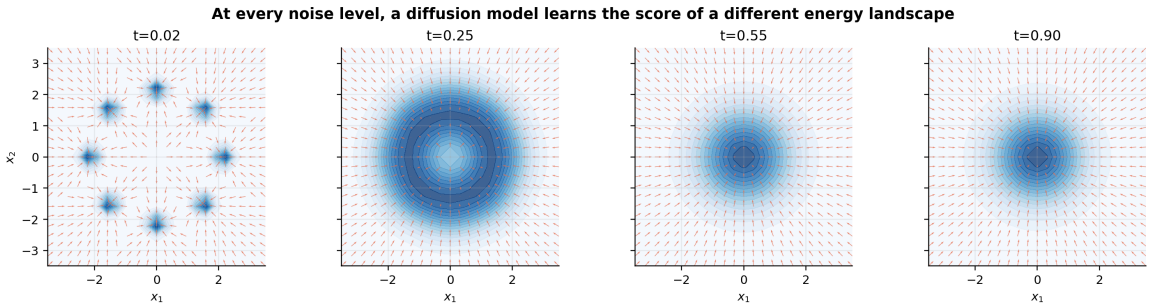


Figure 19.2: Exact scores of the same multimodal target at several noise levels. Gaussian smoothing simplifies the energy landscape and creates a continuation path from a simple prior to a structured data distribution.

Annealed Langevin dynamics starts at high noise and gradually lowers σ, using the score at each level:

$$X_{k+1} = X_k + \epsilon_k s_\theta(X_k, \sigma_k) + \sqrt{2\epsilon_k}\xi_k. \tag{19.17}$$

The schedule helps particles move between modes when the landscape is smooth before refining details [66, 67].

19.8 In what sense is a diffusion model an EBM?

For every time t, the noised marginal p_t defines

$$E_t(x) = -\log p_t(x) + C_t, \qquad s_t(x) = -\nabla E_t(x). \tag{19.18}$$

A score-based diffusion model learns this time-indexed family of energy gradients. It is therefore a continuum of implicit EBMs connected by a known forward diffusion. However, a standard diffusion model is not usually represented by one tractable normalized terminal energy, and a learned vector field may not integrate exactly to a global scalar at every time. Its generative definition is the reverse-time dynamics, not merely an equilibrium density.

Common pitfall

"The score is the gradient of probability" is wrong. It is the gradient of *log* density. This makes it invariant to normalization and changes its units: $s(x)$ has units inverse to x.

Bridge to the next theory

Denoising score matching learns a score at one noise scale. A diffusion model chooses a whole Markov chain or SDE of noise scales, trains all scores jointly, and reverses the process to generate data.

20

Discrete-time diffusion probabilistic models

20.1 A tractable forward Markov chain

A denoising diffusion probabilistic model (DDPM) defines a fixed forward chain

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}), \tag{20.1}$$

with Gaussian transitions

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right). \tag{20.2}$$

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Repeated Gaussian composition gives the closed form

$$\boxed{q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I\right).} \tag{20.3}$$

Hence a random time step can be sampled directly:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{20.4}$$

With a schedule that makes $\bar{\alpha}_T \approx 0$, the terminal distribution is close to standard Gaussian.

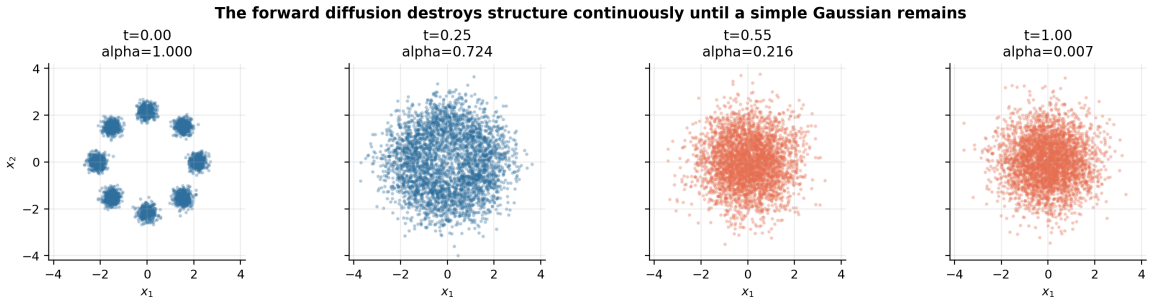


Figure 20.1: Forward noising of an eight-mode target under a variance-preserving process. Modes broaden, merge, and approach a simple Gaussian prior.

20.2 Why the reverse conditional is learnable

The exact reverse marginal $q(x_{t-1} \mid x_t)$ is generally intractable because it depends on the unknown data distribution. But conditioned on x_0, Gaussian algebra gives

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1}; \widetilde{\mu}_t(x_t, x_0), \widetilde{\beta}_t I), \tag{20.5}$$

where

$$\widetilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t, \tag{20.6}$$

$$\widetilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{20.7}$$

A neural reverse model is chosen as

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \tag{20.8}$$

with Gaussian mean and variance predicted or prescribed.

20.3 Variational training objective

The negative log likelihood is bounded using the evidence lower bound:

$$-\log p_\theta(x_0) \le \mathbb{E}_q \left[-\log p(x_T) - \sum_{t=1}^{T} \log p_\theta(x_{t-1} \mid x_t) + \sum_{t=1}^{T} \log q(x_t \mid x_{t-1}) \right]. \tag{20.9}$$

It decomposes into a terminal-prior KL, intermediate denoising KL terms, and a reconstruction term. With common variance choices, each intermediate KL reduces to a weighted squared error between the true posterior mean and the network mean.

Parameterize the mean through predicted noise $\varepsilon_\theta(x_t, t)$:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right). \tag{20.10}$$

The widely used simplified objective is

$$\boxed{\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon} \left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2 \right].} \tag{20.11}$$

Ho, Jain, and Abbeel showed that this reweighted variational objective produces high-quality samples and connects directly to denoising score matching [23].

20.4 Noise prediction, score prediction, and data prediction

For Equation (20.4), the conditional score is

$$\nabla_{x_t} \log q(x_t \mid x_0) = -\frac{\varepsilon}{\sqrt{1-\bar{\alpha}_t}}. \tag{20.12}$$

The optimal noise predictor satisfies

$$s_\theta(x_t, t) = -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1-\bar{\alpha}_t}} \approx \nabla_{x_t} \log q_t(x_t). \tag{20.13}$$

One can instead predict

$$\widehat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\varepsilon_\theta}{\sqrt{\bar{\alpha}_t}}, \tag{20.14}$$

$$v = \sqrt{\bar{\alpha}_t}\varepsilon - \sqrt{1-\bar{\alpha}_t} x_0. \tag{20.15}$$

These parameterizations contain equivalent ideal information but have different signal-to-noise scaling and training behavior.

20.5 Sampling

Start with $x_T \sim \mathcal{N}(0, I)$ and iterate

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z, \qquad z \sim \mathcal{N}(0, I), \tag{20.16}$$

with $z = 0$ at the final step. Hundreds or thousands of neural evaluations were used in early models. Improved schedules, learned variances, higher-order solvers, distillation, and alternative parameterizations reduce this cost [31, 45, 50, 59].

20.6 What the model learns at each time

At large noise, the optimal predictor captures global structure because local detail is erased. At low noise, it estimates fine residuals near the data manifold. A shared time-conditioned network must solve all these regression problems. Time or noise embeddings tell the network which energy landscape it is differentiating.

Training weights decide which signal-to-noise regimes dominate. Uniform sampling in integer t is not uniform in log signal-to-noise ratio. Modern designs often parameterize noise continuously and choose weights to improve conditioning, likelihood, perceptual quality, or distillation stability.

20.7 Latent diffusion and conditioning

For high-dimensional images, diffusion can be applied in a learned autoencoder latent space, reducing spatial cost while retaining semantic information [58]. Conditioning variables c enter the score or noise network. Classifier guidance uses

$$\nabla_x \log p_t(x \mid c) = \nabla_x \log p_t(x) + \nabla_x \log p_t(c \mid x), \tag{20.17}$$

while classifier-free guidance combines conditional and unconditional network outputs. Guidance changes the effective score and typically trades diversity for condition alignment.

Common pitfall

The forward process is not trained and does not gradually "learn to destroy" an image. It is a fixed corruption process. The neural network learns conditional denoising information used to approximate the reverse transitions.

Bridge to the next theory

As step size tends to zero, the forward Gaussian chain becomes an SDE. Time reversal then has an exact formula involving the marginal score, and the same marginals can be generated by a deterministic probability-flow ODE.

21

Score SDEs, reverse time, and the probability-flow ODE

21.1 A continuous forward diffusion

Consider

$$\mathrm{d}X_t = f(X_t, t)\mathrm{d}t + g(t)\mathrm{d}W_t, \qquad t \in [0, T], \tag{21.1}$$

where scalar $g(t) > 0$ is written for clarity. Its density satisfies

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g(t)^2 \Delta p_t. \tag{21.2}$$

The forward process is designed so that p_0 is data and p_T is a tractable prior. Common choices include:

Table 21.1: Canonical score-SDE families.

Family	SDE	Qualitative behavior
Variance preserving	$\mathrm{d}X = -\frac{1}{2}\beta(t)X\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}W$	signal shrinks, variance tends to one
Variance exploding	$\mathrm{d}X = \sqrt{\mathrm{d}[\sigma^2(t)]/\mathrm{d}t}\mathrm{d}W$	signal mean retained, variance grows
Sub-VP	modified VP diffusion coefficient	likelihood-oriented intermediate variance

21.2 Reverse-time SDE from probability current

The forward current is

$$J_t = f p_t - \frac{1}{2} g^2 \nabla p_t = p_t \left(f - \frac{1}{2} g^2 \nabla \log p_t \right). \tag{21.3}$$

To describe the same paths backward, define reverse clock $\tau = T - t$ and $Y_\tau = X_{T-\tau}$. The reverse process must undo the net probability current while retaining diffusion. The reverse-time SDE, expressed while integrating t from T down to 0, is

$$\boxed{\mathrm{d}X_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t) \right] \mathrm{d}t + g(t)\mathrm{d}\overline{W}_t,} \tag{21.4}$$

where $\mathrm{d}t < 0$ in the numerical traversal and $\overline{W}$ is Brownian motion in reverse time. In an increasing reverse clock τ, signs transform accordingly. The score correction is what makes random diffusion run toward, rather than away from, structured high-density regions [4, 22, 68].

For state-dependent diffusion matrix $D(x, t) = GG^\top$, the reverse drift includes both $-D\nabla \log p_t$ and a divergence-of-D term. The scalar, state-independent case avoids this complication.

Why the coefficient is g^2, not $g^2/2$

The forward Fokker–Planck diffusion current is $-\frac{1}{2}g^2\nabla p$. A reverse SDE still has its own diffusion current in reverse time. Reversing the total current requires changing the drift by twice the forward diffusive velocity, hence $-g^2\nabla \log p$, not $-\frac{1}{2}g^2\nabla \log p$.

21.3 Training the time-dependent score

If the perturbation kernel $p_{0t}(x_t \mid x_0)$ is known, train

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,x_t}\left[\lambda(t)\left\|s_\theta(x_t,t) - \nabla_{x_t}\log p_{0t}(x_t \mid x_0)\right\|^2\right]. \tag{21.5}$$

Conditional-expectation projection makes the optimizer the marginal score $\nabla \log p_t$. The weight $\lambda(t)$ controls variance and which noise levels matter most.

21.4 The probability-flow ODE

Rewrite Equation (21.2) as a continuity equation:

$$\partial_t p_t = -\nabla\cdot(f p_t) + \frac{1}{2}g^2\nabla\cdot(p_t\nabla\log p_t) \tag{21.6}$$

$$= -\nabla\cdot\left[p_t\left(f - \frac{1}{2}g^2\nabla\log p_t\right)\right]. \tag{21.7}$$

Therefore the deterministic ODE

$$\boxed{\frac{\mathrm{d}X_t}{\mathrm{d}t} = f(X_t,t) - \frac{1}{2}g(t)^2\nabla\log p_t(X_t)} \tag{21.8}$$

has exactly the same one-time marginal densities p_t as the forward SDE. Integrating it backward with the learned score generates samples deterministically from a random initial prior draw.

The factor $1/2$ now appears because the ODE has no additional diffusion current. Its entire velocity must equal the Fokker–Planck current divided by density.

21.5 Likelihood through divergence

For an ODE $\dot{X}_t = v_\theta(X_t,t)$, the instantaneous change of variables is

$$\frac{\mathrm{d}}{\mathrm{d}t}\log p_t(X_t) = -\nabla\cdot v_\theta(X_t,t). \tag{21.9}$$

Thus

$$\log p_0(x_0) = \log p_T(x_T) + \int_0^T \nabla\cdot v_\theta(X_t,t)\,\mathrm{d}t, \tag{21.10}$$

with signs adjusted to integration direction. Hutchinson estimators approximate the divergence using vector–Jacobian products. This provides likelihood evaluation for score models, at the cost of an ODE solve and trace estimates.

21.6 Exact mixture experiment

The package uses an eight-component Gaussian mixture for which p_t and $\nabla\log p_t$ are analytic. This isolates numerical integration from score approximation.

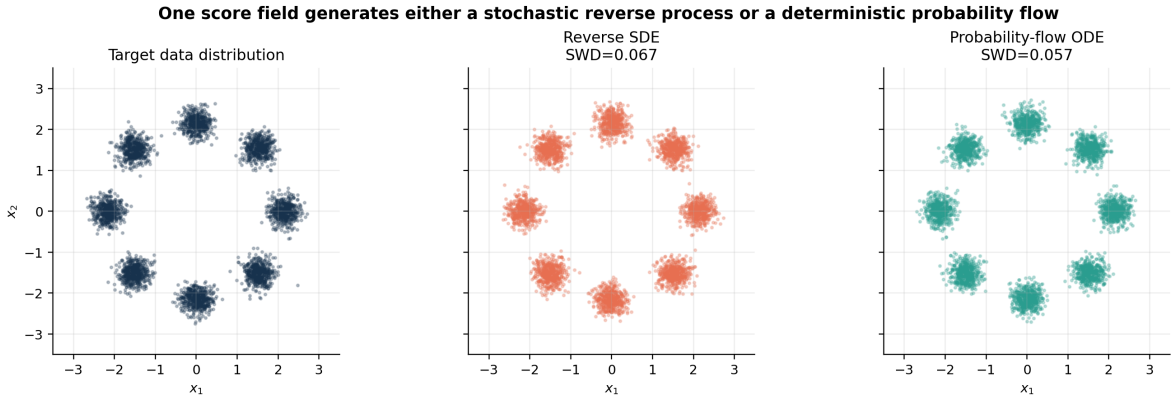


Figure 21.1: Samples generated from the same Gaussian prior by the exact-score reverse SDE and probability-flow ODE. Both recover the multimodal target through different path laws.

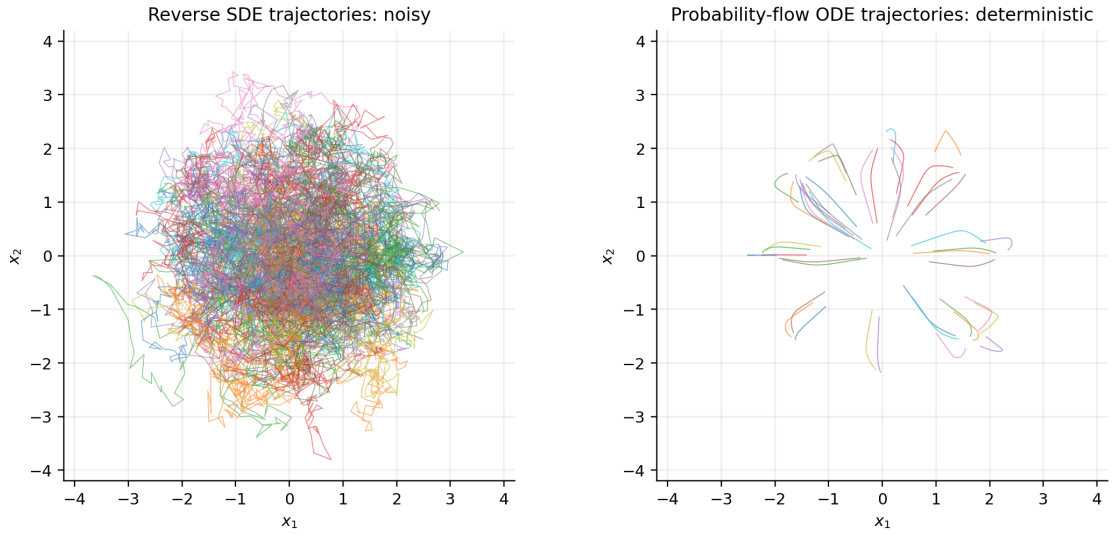


Figure 21.2: Individual reverse trajectories. The stochastic sampler jitters while modes form; the probability-flow ODE transports each initial point deterministically.

Computational experiment

Against independent target samples, sliced Wasserstein distances are approximately 0.0670 for the exact-score reverse SDE and 0.0568 for the probability-flow ODE in the provided run. These finite-sample values should not be read as a universal ranking: they depend on discretization, sample count, and metric projections. With exact continuous integration, both methods have the same target marginal.

21.7 A learned score experiment

A small multilayer perceptron is trained by continuous denoising score matching on the same mixture. Reverse-SDE samples using this approximate score achieve sliced Wasserstein distance about 0.1672, worse than the exact-score baselines but clearly recovering the ring structure.

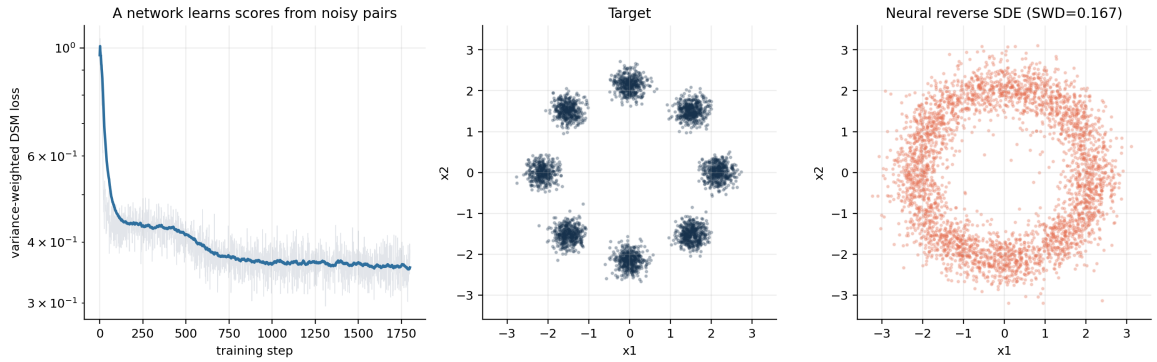


Figure 21.3: Training loss and generated samples for a deliberately small neural score model. The gap to exact-score sampling makes statistical approximation error visible.

21.8 Numerical solvers and predictor–corrector sampling

Euler–Maruyama is the simplest reverse-SDE integrator. Predictor–corrector samplers alternate a reverse-SDE predictor with Langevin corrector steps targeting the current p_t. Probability-flow ODEs permit adaptive Runge–Kutta and specialized diffusion solvers. High-order methods exploit semilinear structure and score parameterizations to reduce neural evaluations [31, 45].

Discretization error is not ordinary local error alone: the score can grow rapidly near low noise, and a small vector-field bias may be amplified when integrating backward. Noise schedules and preconditioning are therefore numerical-analysis choices as much as modeling choices.

Common pitfall

The probability-flow ODE shares the SDE's marginal density at each time, not its path distribution. Multi-time correlations, stochasticity, and trajectory geometry differ. "Equivalent" must always specify which object is equivalent.

Bridge to the next theory

The probability-flow ODE starts from an SDE and extracts its current velocity. Flow matching reverses the logic: choose a probability path or stochastic interpolant, derive a compatible conditional velocity, and learn that velocity directly by regression.

22

Flow matching, continuous normalizing flows, and stochastic interpolants

22.1 Deterministic transport

A time-dependent vector field $v_t(x)$ defines

$$\dot{X}_t = v_t(X_t), \tag{22.1}$$

and transports density by

$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0. \tag{22.2}$$

If the ODE map is invertible, it is a continuous normalizing flow (CNF). Likelihoods follow Equation (21.9), and samples follow the forward ODE from a simple base distribution [11, 53].

Traditional CNFs train by likelihood and repeatedly integrate the ODE during optimization. Flow matching instead constructs target velocities at random space–time points and trains by supervised regression, avoiding simulation in the training loop.

22.2 Conditional probability paths

Let Z denote conditioning information, often a pair of endpoints. Suppose we can sample a conditional path density $p_t(x \mid Z)$ and know a conditional vector field $u_t(x \mid Z)$ satisfying

$$\partial_t p_t(x \mid Z) + \nabla \cdot [p_t(x \mid Z) u_t(x \mid Z)] = 0. \tag{22.3}$$

The marginal path is

$$p_t(x) = \mathbb{E}_Z[p_t(x \mid Z)]. \tag{22.4}$$

A compatible marginal velocity is

$$v_t(x) = \mathbb{E}[u_t(X_t \mid Z) \mid X_t = x]. \tag{22.5}$$

Indeed,

$$\partial_t p_t = -\nabla \cdot \mathbb{E}_Z[p_t(x \mid Z) u_t(x \mid Z)] \tag{22.6}$$

$$= -\nabla \cdot [p_t(x) v_t(x)]. \tag{22.7}$$

Thus conditional velocities average to the correct marginal transport.

22.3 The flow-matching regression theorem

Train v_θ with

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t, Z, X_t \sim p_t(\cdot \mid Z)} \left[\| v_\theta(X_t, t) - u_t(X_t \mid Z) \|^2 \right]. \tag{22.8}$$

For each (x, t), squared-error regression is minimized by the conditional expectation in Equation (22.5). Therefore the learned field transports the marginal path even though the target used during training depends on latent conditioning variables [43].

22.4 Linear interpolation and rectified flow

Sample $X_0 \sim p_0$ from a base distribution and $X_1 \sim p_1$ from data. For the straight interpolant

$$X_t = (1-t)X_0 + tX_1, \tag{22.9}$$

the conditional velocity is simply

$$u_t(X_t \mid X_0, X_1) = X_1 - X_0. \tag{22.10}$$

The marginal velocity is

$$v_t(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]. \tag{22.11}$$

Even though individual conditional paths are straight, the marginal field can curve because many endpoint pairs pass through the same region. Reflow or rectification procedures update the coupling to make trajectories straighter and easier to integrate [44].

Independent endpoint coupling is easy but may create crossing paths and high conditional variance. Better couplings, including optimal-transport-inspired ones, can shorten trajectories and simplify regression.

22.5 Gaussian conditional paths

A broad family is

$$X_t = \alpha_t X_1 + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{22.12}$$

with conditional velocity

$$\dot{X}_t = \dot{\alpha}_t X_1 + \dot{\sigma}_t \varepsilon. \tag{22.13}$$

Using $\varepsilon = (X_t - \alpha_t X_1)/\sigma_t$,

$$u_t(x \mid X_1) = \frac{\dot{\sigma}_t}{\sigma_t} x + \left(\dot{\alpha}_t - \frac{\dot{\sigma}_t}{\sigma_t}\alpha_t\right) X_1. \tag{22.14}$$

Regressing this target is closely related to predicting noise, data, score, or the v-parameterization in diffusion models.

For a Gaussian perturbation path, the marginal score and posterior data mean are linked by Equation (18.15); consequently the flow velocity can be written using the score. The probability-flow ODE is one particular flow-matching field associated with a diffusion path.

22.6 Stochastic interpolants

A stochastic interpolant generalizes Equation (22.9) by combining endpoints and optional noise:

$$X_t = I(t, X_0, X_1) + \gamma(t) Z. \tag{22.15}$$

Conditional expectations of its time derivative and noise define velocity and score fields. The framework can generate deterministic ODEs and families of forward/backward SDEs sharing chosen marginals. It unifies flow matching, score models, Schrödinger bridges, and transport constructions under one interpolation calculus [2, 3].

22.7 Connection to non-equilibrium statistical mechanics

For an SDE current,

$$J = p\left(f - \frac{1}{2}g^2 \nabla \log p\right), \tag{22.16}$$

the current velocity is $v = J/p$. This separates the motion of probability mass into a deterministic drift and an osmotic term generated by density gradients. Flow models learn a current velocity directly. Score models learn the thermodynamic force $\nabla \log p$ and combine it with a prescribed drift and diffusion.

Both are non-equilibrium protocols: a time-dependent field transports one distribution into another while entropy, energy, and probability currents change. The distinction is that generative training engineers the path and learns its

driving field rather than observing a naturally relaxing material system. Language such as "thermodynamic" should identify the actual mathematical correspondence, not imply that a neural sampler literally dissipates physical heat.

22.8 Exact flow-matching experiment

For independent Gaussian base X_0 and Gaussian-mixture data endpoint X_1, the conditional expectation in Equation (22.5) can be evaluated analytically by mixture conditioning. The resulting ODE transports the base into the target.

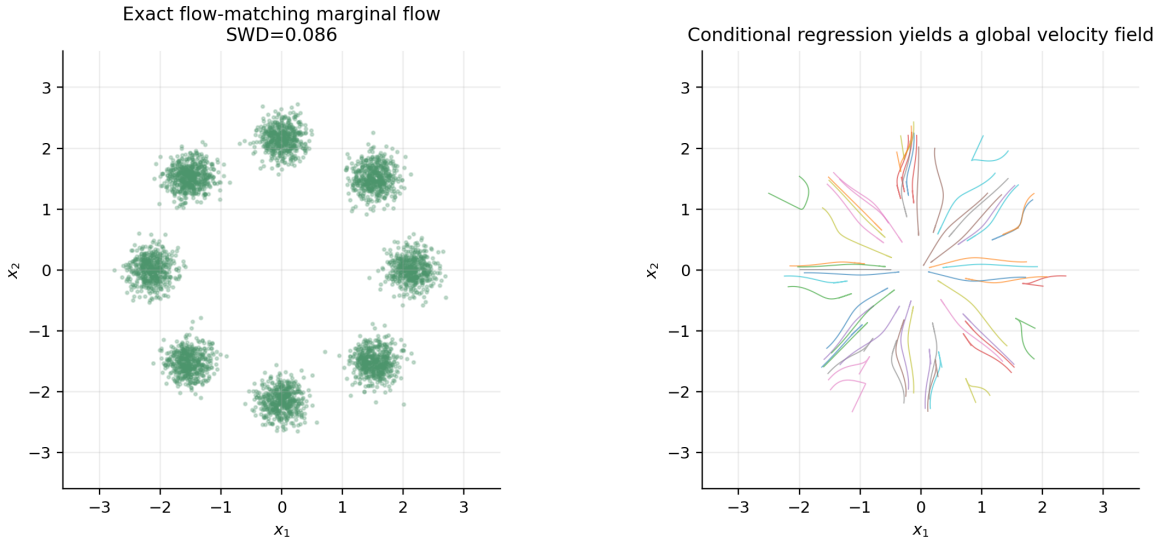


Figure 22.1: Exact conditional flow matching on the toy mixture. The deterministic path differs from the probability-flow ODE but reaches the same target distribution.

Computational experiment

The exact flow-matching ODE reaches sliced Wasserstein distance about 0.0864 from the target in the supplied finite-step run. Open `interactive/deterministic_transport_trajectories.html` to compare its particle trajectories with the probability-flow ODE. Endpoint agreement does not imply identical intermediate paths.

22.9 Schrödinger bridges and controlled diffusion

Given a reference diffusion and endpoint distributions, the Schrödinger bridge finds the path measure closest in relative entropy to the reference while matching both endpoints. It can be viewed as a stochastic optimal transport problem and solved using forward/backward potentials, iterative proportional fitting, or learned controls [13, 41, 61]. In a small-noise limit it connects to optimal transport; at nonzero noise it preserves stochasticity and entropy.

Common pitfall

Flow matching does not require straight trajectories. It requires a probability path and a compatible conditional velocity target. "Rectified flow," Gaussian flow matching, optimal-transport paths, and diffusion paths are different constructions within the broader idea.

Bridge to the next theory

We now consolidate the models, experiments, and literature into a practical map: what each objective learns, which differential equation it solves, what is exact, and where contemporary research remains active.

23

A practical and literature map of diffusion and flow models

23.1 One table, four generative viewpoints

Table 23.1: Closely related generative formulations.

Method	Learned object	Evolution at sampling	Core identity
Energy-based model	$E_\theta(x)$ or $-\nabla E_\theta$	Langevin or MCMC at one target density	$p \propto \mathrm{e}^{-E}$
DDPM	noise, data, or velocity at discrete t	learned Gaussian reverse chain	Gaussian posterior and ELBO
Score SDE	$\nabla \log p_t$	reverse SDE or probability-flow ODE	reverse drift $f - g^2 \nabla \log p_t$
Flow matching	current velocity v_t	deterministic ODE	$\partial_t p + \nabla \cdot (pv) = 0$

The objects are transformable under suitable paths. On a Gaussian corruption path, an optimal noise predictor determines the score; the score determines the probability-flow velocity; posterior endpoint predictions determine conditional flow velocities. Their practical differences arise from parameterization, weighting, path choice, approximation error, and numerical solver.

23.2 Historical development

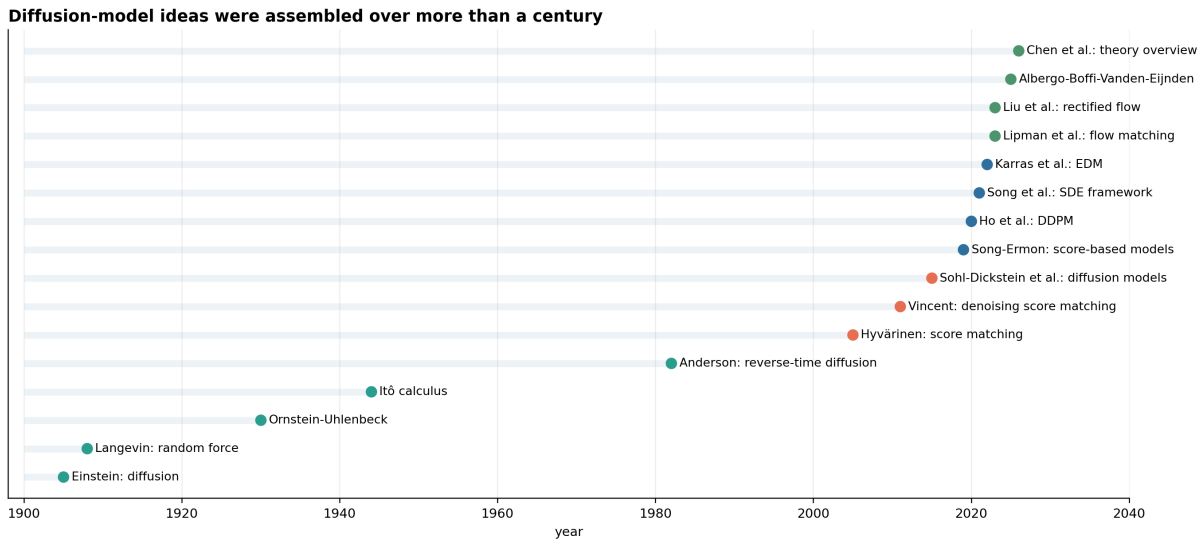


Figure 23.1: A selective literature timeline from stochastic mechanics and score matching to diffusion probabilistic models, score SDEs, improved samplers, flow matching, and stochastic interpolants.

A compact reading sequence is:

1. Anderson for reverse-time diffusion and Haussmann–Pardoux for mathematical time reversal [4, 22];
2. Hyvärinen and Vincent for score matching and denoising score matching [25, 73];
3. Sohl-Dickstein et al. for diffusion probabilistic modeling and Ho et al. for the modern DDPM objective [23, 65];
4. Song et al. for the SDE unification, reverse SDE, and probability-flow ODE [68];
5. Karras et al. and Lu et al. for design-space analysis and fast solvers [31, 45];
6. Lipman et al. for flow matching, Liu et al. for rectified flow, and Albergo et al. for stochastic interpolants [2, 43, 44];
7. Mézard and Montanari for the broader statistical-physics language of graphical models, cavity fields, phases, and message passing [47].

A recent mathematical overview can help compare rapidly evolving formulations, but primary papers should be consulted for exact assumptions and conventions [10].

23.3 Measured experiment summary

Table 23.2: Selected measured outputs from the bundled experiments.

Quantity	Value	Interpretation
Explicit Euler maximum energy error	3.795e+10	unstable baseline
Velocity-Verlet maximum energy error	0.0008	structure-preserving
Optimized/free classical action ratio	1	should approach 1
Finite-bath KL to canonical law	0.002607	reservoir approximation
Hamiltonian phase-area ratio	1	Liouville: should be 1
Damped phase-area ratio	0.1511	dissipative contraction
OU final variance error	0.004257	SDE vs Fokker-Planck
FDT normalized-curve RMSE	0.0218	correlation vs response
Brownian MSD relative error	0.03682	simulation vs $2Dt$
Quantum norm drift	3.89e-13	unitarity check
Reverse-SDE sliced Wasserstein	0.067	exact score
Probability-flow sliced Wasserstein	0.05676	exact score
Flow-matching sliced Wasserstein	0.08644	exact velocity
Neural-score sliced Wasserstein	0.1672	learned score

These numbers are reproducibility checks, not benchmark claims. They use low-dimensional analytic targets, modest sample sizes, and fixed random seeds so that every concept can be inspected. A result is valuable here when it falsifies an implementation bug or reveals a structural distinction.

23.4 How to read sampling error

For a generated empirical measure $\widehat{p}$ and target samples from p, no single scalar metric is complete. The package uses sliced Wasserstein distance: project both samples onto many random one-dimensional directions, compute one-dimensional Wasserstein distances, then average. It is interpretable and cheap, but it can miss rare high-dimensional discrepancies.

Visual plots can reveal mode collapse, but they can also conceal density errors. Log likelihood requires normalized models or probability-flow divergence. Kernel and classifier-based distances depend on representation and bandwidth. A rigorous evaluation should combine several diagnostics and report Monte Carlo uncertainty.

23.5 Three distinct sources of generative error

For diffusion and flow models, separate:

1. **path mismatch**: the terminal forward law is not exactly the chosen prior, or the designed path is poorly conditioned;
2. **field approximation**: the neural score or velocity differs from the population-optimal field;
3. **solver error**: finite steps approximate the learned differential equation.

The analytic mixture experiments remove field approximation. The neural-score experiment restores it. Changing integration steps while holding the exact score fixed isolates solver error. This decomposition is more informative than attributing all failure to "the model."

23.6 Low-noise irregularity and spectral bias

At low noise, an empirical data distribution convolved with a narrow Gaussian develops one sharp basin around each sample. Its score can vary on scale σ and have magnitude of order $1/\sigma$. A finite neural network trained by gradient methods often learns smoother, lower-frequency structure first. This spectral bias can make the learned score resemble the population-smoothed geometry before it resolves individual training examples.

This offers one mechanism for a generalization–memorization time gap: at intermediate training, the score is a smooth interpolation that supports realistic generation; at prolonged training and very low noise, the network may fit empirical irregularities more closely. The statement is architecture-, optimization-, and data-dependent, so it should be tested by measuring score spectra, nearest-neighbor behavior, and held-out likelihood rather than treated as a universal theorem.

23.7 Thermodynamic analogies: exact and limited

The following correspondences are mathematically exact under their assumptions:

- $E = -\log p + C$ converts normalized densities to energies;
- Langevin drift uses $-\nabla E = \nabla \log p$;
- Fokker–Planck is a continuity equation with diffusion current;
- equilibrium overdamped Langevin decreases relative free energy;
- probability-flow velocity is current divided by density.

Other claims are analogies. A learned reverse diffusion is a driven, time-inhomogeneous generative protocol, not necessarily a physical thermal relaxation. Its "temperature" is a chosen noise schedule, and its neural score can violate conservative integrability. Precision about these limits makes the statistical-mechanics viewpoint stronger, not weaker.

23.8 Exercises

Mechanics and action

1. Derive the brachistochrone Euler–Lagrange equation using travel time as the action. Identify the cyclic coordinate and its first integral.
2. For a pendulum, numerically compare explicit RK4 and a symplectic method over 10^5 periods. Plot energy error and phase area.

Equilibrium and stochastic mechanics

3. Starting from two Einstein solids, derive the most probable energy split and show that equality of entropy slopes is equality of temperature.
4. Use Itô's lemma to derive the SDE for $Y_t = X_t^2$ when X_t is OU. Obtain closed ODEs for the first two moments.
5. Prove that the overdamped Langevin KL derivative is minus relative Fisher information.

Spin glasses and inference

6. Enumerate several $N = 16$ SK disorder samples. Plot the distribution of ground-state energy and overlap histograms. Which observations are finite-size dependent?
7. Derive the naive mean-field equation for SK and identify the missing self-reaction that motivates the TAP correction.

Diffusion and flow models

8. Starting from the VP Fokker–Planck equation, derive the probability-flow ODE and verify every factor of $1/2$.

9. Train noise-, data-, score-, and velocity-parameterized networks on the toy mixture under matched weighting. Compare gradient norms across log-SNR.
10. Replace independent endpoint coupling in flow matching by an approximate optimal assignment. Compare path length, velocity variance, and solver steps.
11. Estimate probability-flow likelihood with a Hutchinson divergence estimator and compare it with the exact Gaussian-mixture likelihood.

23.9 Open research directions

Important active directions include faster distillation, adaptive stochastic solvers, path and coupling design, likelihood–perception tradeoffs, theoretical generalization at low noise, diffusion on manifolds and discrete structures, symmetry-equivariant scores, posterior sampling for inverse problems, and principled combinations of diffusion with energy or reward guidance. The stochastic-interpolant viewpoint suggests treating drift, score, velocity, and endpoint coupling as a joint design space rather than isolated model families.

23.10 The complete conceptual chain

The route can now be compressed into ten equations:

$$m\ddot{q} = F, \tag{23.1}$$

$$\delta \int L\mathrm{d}t = 0, \tag{23.2}$$

$$\dot{q} = \nabla_p H, \quad \dot{p} = -\nabla_q H, \tag{23.3}$$

$$p_\beta(z) \propto \mathrm{e}^{-\beta H(z)}, \tag{23.4}$$

$$\partial_t \rho + \nabla \cdot (\rho v) = 0, \tag{23.5}$$

$$\partial_t p = -\nabla \cdot (ap) + \tfrac{1}{2}\nabla\nabla : (Dp), \tag{23.6}$$

$$\mathrm{d}X = -\nabla U \mathrm{d}t + \sqrt{2T}\mathrm{d}W, \tag{23.7}$$

$$s_t(x) = \nabla \log p_t(x), \tag{23.8}$$

$$\mathrm{d}X = [f - g^2 s_t]\mathrm{d}t + g\mathrm{d}\overline{W}, \tag{23.9}$$

$$\dot{X} = v_t(X), \qquad \partial_t p_t + \nabla \cdot (p_t v_t) = 0. \tag{23.10}$$

Each line changes what is represented: acceleration, whole paths, phase flow, uncertainty, density transport, stochastic broadening, thermal sampling, local log-density geometry, reverse diffusion, and deterministic generative transport. The theories are not interchangeable, but they are connected by explicit limiting operations and adjoint equations.

Intuition

The final unifying picture is a moving probability density. Mechanics supplies vector fields; statistical mechanics supplies distributions; stochastic mechanics supplies diffusion; machine learning estimates the score or current required to steer one distribution into another.

Part III

Appendices

A

Mathematical toolkit

This appendix fixes notation and collects identities used throughout the book. It is intended as a working reference rather than a substitute for analysis, probability, or differential geometry.

A.1 Vectors, gradients, and Jacobians

For $f : \mathbb{R}^d \to \mathbb{R}$, the gradient is the column vector

$$\nabla f = (\partial_1 f, \dots, \partial_d f)^\top. \tag{A.1}$$

For $F : \mathbb{R}^d \to \mathbb{R}^m$, the Jacobian has entries

$$(DF)_{ij} = \frac{\partial F_i}{\partial x_j}. \tag{A.2}$$

The divergence of a vector field $v : \mathbb{R}^d \to \mathbb{R}^d$ is

$$\nabla \cdot v = \operatorname{tr}(Dv) = \sum_i \partial_i v_i. \tag{A.3}$$

The Hessian is $(\nabla^2 f)_{ij} = \partial_i \partial_j f$, and the Laplacian is $\Delta f = \operatorname{tr}(\nabla^2 f)$.

For a differentiable curve $x(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t), t) = \partial_t f + \nabla f \cdot \dot{x}. \tag{A.4}$$

For a matrix $A(t)$ with nonzero determinant, Jacobi's formula is

$$\frac{\mathrm{d}}{\mathrm{d}t} \log |\det A(t)| = \operatorname{tr}(A^{-1} \dot{A}). \tag{A.5}$$

Applied to a flow Jacobian, this produces the divergence identity used in continuous normalizing flows.

A.2 Taylor expansion and orders

For $f : \mathbb{R}^d \to \mathbb{R}$,

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + O(\|h\|^3). \tag{A.6}$$

In ordinary calculus, $h = O(\Delta t)$ makes the quadratic term $O(\Delta t^2)$. In Itô calculus, a stochastic increment is $h = O(\sqrt{\Delta t})$, so $h^2 = O(\Delta t)$ and the Hessian term remains.

A.3 Integration by parts

In one dimension,

$$\int_a^b u(x) v'(x) \mathrm{d}x = [uv]_a^b - \int_a^b u'(x) v(x) \mathrm{d}x. \tag{A.7}$$

In d dimensions,

$$\int_\Omega f \nabla \cdot v \, \mathrm{d}x = \int_{\partial\Omega} f v \cdot n \, \mathrm{d}S - \int_\Omega \nabla f \cdot v \, \mathrm{d}x. \tag{A.8}$$

Most score-matching and Fokker–Planck derivations assume that the boundary term vanishes through decay, periodicity, or reflecting conditions. This assumption should always be checked.

A.4 Probability, conditional expectation, and projection

For random variables X, Y,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]. \tag{A.9}$$

Conditional expectation is an L^2 projection. Among measurable functions $g(Y)$,

$$g^*(Y) = \mathbb{E}[X \mid Y] \tag{A.10}$$

minimizes $\mathbb{E}\|X - g(Y)\|^2$. This single theorem explains why denoising regression recovers a marginal score and why conditional flow matching recovers a marginal velocity.

Variance decomposes as

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]). \tag{A.11}$$

The first term is irreducible conditional uncertainty; the second is variability that a predictor can explain.

A.5 Gaussian identities

If $X \sim \mathcal{N}(m, \Sigma)$, then

$$\log p(x) = C - \frac{1}{2}(x-m)^\top \Sigma^{-1}(x-m), \qquad \nabla_x \log p(x) = -\Sigma^{-1}(x-m). \tag{A.12}$$

For jointly Gaussian (X, Y),

$$\mathbb{E}[X \mid Y = y] = m_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - m_Y), \tag{A.13}$$

with conditional covariance

$$\Sigma_{X|Y} = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}. \tag{A.14}$$

For $Z \sim \mathcal{N}(0, 1)$,

$$\mathbb{E}[\mathrm{e}^{tZ}] = \mathrm{e}^{t^2/2}. \tag{A.15}$$

This moment-generating identity produces replica interactions after Gaussian disorder averaging.

A.6 Entropy, KL divergence, and Fisher divergence

For densities p, q,

$$D_{\mathrm{KL}}(p\|q) = \int p(x) \log \frac{p(x)}{q(x)} \mathrm{d}x \geq 0. \tag{A.16}$$

It is asymmetric and may be infinite. Differential entropy is

$$h(p) = -\int p \log p, \tag{A.17}$$

which depends on units; KL divergence is invariant under smooth invertible coordinate changes.

The relative Fisher information is

$$I(p\|q) = \int p(x) \left\| \nabla \log \frac{p(x)}{q(x)} \right\|^2 \mathrm{d}x. \tag{A.18}$$

For Langevin relaxation, $\mathrm{d}D_{\mathrm{KL}}(p_t\|q)/\mathrm{d}t$ is minus a constant times this quantity.

A.7 Legendre transforms

For a convex function $f(x)$,

$$f^*(p) = \sup_x \{p \cdot x - f(x)\}. \tag{A.19}$$

At a smooth optimum, $p = \nabla f(x)$ and $x = \nabla f^*(p)$. If f is strictly convex, the gradients are inverse maps. Thermodynamic potentials and the Hamiltonian are Legendre transforms adapted to changing controlled variables.

A.8 Functional derivatives

For a functional $F[\rho]$, define its first variation by

$$F[\rho + \varepsilon\eta] - F[\rho] = \varepsilon \int \frac{\delta F}{\delta \rho}(x)\eta(x)\mathrm{d}x + o(\varepsilon). \tag{A.20}$$

Examples are

$$\frac{\delta}{\delta\rho(x)} \int U\rho \mathrm{d}x = U(x), \tag{A.21}$$

$$\frac{\delta}{\delta\rho(x)} \int \rho \log \rho \mathrm{d}x = 1 + \log \rho(x). \tag{A.22}$$

The additive constant in a functional derivative is irrelevant under the normalization constraint $\int \rho = 1$.

A.9 Fourier transforms

Use

$$\widehat{f}(k) = \int_{\mathbb{R}^d} f(x)\mathrm{e}^{-\mathrm{i}k\cdot x}\mathrm{d}x, \qquad f(x) = \frac{1}{(2\pi)^d} \int \widehat{f}(k)\mathrm{e}^{\mathrm{i}k\cdot x}\mathrm{d}k. \tag{A.23}$$

Then $\widehat{\nabla f} = \mathrm{i}k\widehat{f}$ and $\widehat{\Delta f} = -\|k\|^2\widehat{f}$. The heat equation solution is

$$\widehat{p}(k,t) = \mathrm{e}^{-D\|k\|^2 t}\widehat{p}(k,0), \tag{A.24}$$

showing explicitly that diffusion damps high spatial frequencies first. This smoothing underlies both low-noise difficulty and high-noise simplicity in score models.

A.10 Ordinary and stochastic differential equations

An ODE $\dot{x} = f(x,t)$ defines one path per initial point under regularity. An Itô SDE

$$\mathrm{d}X = a\mathrm{d}t + B\mathrm{d}W \tag{A.25}$$

defines a distribution over paths. Three notions of numerical convergence are common:

- pathwise or strong: $\mathbb{E}\|X_T - X_T^h\|$;
- weak: $|\mathbb{E}\varphi(X_T) - \mathbb{E}\varphi(X_T^h)|$;
- distributional metrics such as Wasserstein distance.

Generative sampling is usually evaluated distributionally, but pathwise properties matter for coupling and reproducibility.

A.11 Notation dictionary

Symbol	Meaning
Q	configuration manifold
$q, \dot{q}$	generalized coordinate and velocity
p	canonical momentum, or a probability density when context is clear
H, L, S	Hamiltonian, Lagrangian, and action
$z = (q,p)$	phase-space point
ρ, p_t	density or time-indexed density

J	probability current; also symplectic matrix when displayed as a matrix
W_t	standard Brownian motion
a, B, D	SDE drift, diffusion factor, and $D = BB^\top$
s_t	score $\nabla \log p_t$
E_θ	model energy
Z	partition function
β	inverse temperature, or diffusion schedule when written $\beta(t)$
α_t, σ_t	signal and noise coefficients
v_t	deterministic probability-flow or flow-matching velocity

B

Reproducibility and experiment catalog

B.1 Package layout

Path	Purpose
`src/physicsdiff/`	reusable mechanics, statistical-mechanics, stochastic, quantum, and diffusion functions
`experiments/generate_all.py`	primary figure and metric generator
`experiments/generate_diffusion_only.py`	exact-score and exact-velocity diffusion experiments
`experiments/train_toy_score_model.py`	PyTorch denoising-score training experiment
`experiments/assemble_metrics.py`	consolidates metrics into JSON, CSV, and LaTeX
`notebooks/`	source notebook, executed notebook, and HTML rendering
`interactive/`	offline Plotly gallery and local JavaScript bundle
`figures/`	static publication-quality figures used by this PDF
`results/`	machine-readable outputs and trained toy checkpoint
`book/`	LaTeX source and bibliography
`tests/`	numerical and invariant checks

B.2 Recreate the environment

From the package root:

```
python -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
python -m pip install --upgrade pip
python -m pip install -e .
```

Run all lightweight tests with

```
pytest -q
```

Generate the figures and deterministic experiments with

```
python experiments/generate_all.py
python experiments/generate_diffusion_only.py
python experiments/train_toy_score_model.py
python experiments/assemble_metrics.py
```

The neural training script uses a fixed seed but small floating-point differences across hardware are expected. The exact analytic experiments should agree within solver and Monte Carlo tolerances.

B.3 Experiment index

ID	Figure	Scientific check
00	roadmap	conceptual dependencies across the book
01	projectile	Newtonian integration and energy
02	integrator energy	instability versus symplectic bounded error
03	action minimization	discrete stationary action
04	Legendre transform	convex dual geometry
05	pendulum portrait	energy levels and separatrix
06	thermal bath	microcanonical reservoir to canonical law
07	standard map	regular islands and chaotic regions
08	continuity	flux conservation
09	Liouville volume	incompressible versus dissipative flow
10	OU Fokker–Planck	analytic density evolution
11	quadratic variation	Itô correction mechanism
12	FDT	equilibrium correlation versus response
13	spin glass	exact finite SK landscape and overlaps
14	quantum packet	unitary split-step evolution
15	history	Brownian and generative-model timeline
16	Brownian MSD	$2Dt$ law
17	underdamped Langevin	inertial thermalization
18	overdamped Boltzmann	stationary energy law and metastability
19	OU moments	exact mean and variance
20	EBM	energy-density-score relation
21	forward diffusion	Gaussian corruption path
22	reverse comparison	reverse SDE versus probability-flow ODE
23	trajectories	stochastic versus deterministic reverse paths
24	flow matching	learned current velocity viewpoint
25	score fields	smoothing across noise scales
26	literature	selective conceptual chronology
27	neural score	approximation error and training loss

B.4 Core numerical invariants

The tests emphasize identities that should not depend on visual judgement:

- velocity Verlet keeps harmonic energy error bounded;
- the optimized free-particle action matches the straight-path value;
- Hamiltonian linear flow has unit determinant;
- OU empirical moments match their analytic solution;
- overdamped Langevin histograms approach the Boltzmann reference;
- split-step Schrödinger evolution preserves norm;
- analytic mixture scores agree with finite-difference log-density gradients;
- reverse SDE, probability-flow ODE, and flow-matching samples recover all mixture modes.

B.5 Interactive figures

Open `interactive/index.html` in a modern browser. The gallery works offline because `plotly.min.js` is included locally. It contains:

- selectable Brownian paths;
- a zoomable pendulum phase portrait;
- Hamiltonian and damped phase-volume transport;
- time-controlled overdamped relaxation;
- forward diffusion at selectable noise times;
- generated sample comparisons;
- deterministic probability-flow and flow-matching trajectories.

The PDF contains static snapshots for portability and archival stability.

B.6 Randomness and reproducibility

Every script constructs a local pseudo-random generator from an explicit seed. Re-running on the same software stack should reproduce arrays closely, but exact bitwise identity is not guaranteed across BLAS libraries, CPUs, GPUs, or PyTorch versions. Scientific reproducibility is assessed through invariant quantities and statistical tolerances rather than screenshots alone.

B.7 Extending the package

Natural extensions include:

1. replace the toy mixture by image latent vectors or a manifold-valued dataset;
2. add MALA and Hamiltonian Monte Carlo comparisons for the EBM;
3. implement adaptive reverse-SDE and DPM solvers;
4. compare score, noise, data, and velocity parameterizations under equal compute;
5. add belief propagation and approximate message passing experiments connected to the spin-glass chapter;
6. estimate free energies using Jarzynski and bridge sampling.

C

Glossary of concepts

Term	Meaning
Action	Functional $S[q] = \int L\mathrm{d}t$ whose stationary paths satisfy the equations of motion.
Annealed disorder	Random variables equilibrated or averaged inside the partition function; contrast with quenched disorder.
Boltzmann distribution	Equilibrium density proportional to $\mathrm{e}^{-\beta E}$.
Brownian motion	Continuous Gaussian process with independent increments and variance proportional to elapsed time.
Canonical ensemble	Equilibrium ensemble at fixed temperature, volume, and particle number.
Canonical momentum	$p_i = \partial L/\partial \dot{q}^i$; it may differ from kinetic momentum.
Chaos	Sensitive dependence on initial conditions, usually associated with positive Lyapunov exponents; not synonymous with ergodicity.
Conditional flow matching	Regression on a velocity conditioned on sampled latent endpoints or paths, whose conditional expectation is a marginal transport field.
Continuity equation	$\partial_t \rho + \nabla \cdot (\rho v) = 0$, expressing local conservation under deterministic transport.
Detailed balance	Pairwise reversal balance implying equilibrium time-reversal symmetry and zero irreversible current.
Diffusion model	Generative model built by reversing a progressive stochastic corruption process or an associated probability path.
Dissipation	Conversion of resolved mechanical energy into unresolved degrees of freedom; macroscopically associated with entropy production.
Energy-based model	Density represented as $p_\theta \propto \mathrm{e}^{-E_\theta}$.
Ensemble	Probability distribution over microscopic states consistent with macroscopic controls.
Ergodicity	Property that invariant sets are trivial, enabling long-time averages to equal ensemble averages almost surely.
Euler–Lagrange equation	$\mathrm{d}(\partial L/\partial \dot{q})/\mathrm{d}t - \partial L/\partial q = 0$.
Fluctuation–dissipation theorem	Relation connecting equilibrium correlation fluctuations to linear response or dissipative coefficients.
Fokker–Planck equation	Forward PDE for the density of an Itô diffusion.
Flow matching	Training method that regresses a time-dependent velocity field for a prescribed probability path.
Free energy	Thermodynamic potential balancing internal energy and entropy; also a variational functional minimized by equilibrium.
Frustration	Incompatibility preventing all local interactions from being simultaneously satisfied.
Hamiltonian	Legendre transform $H = p \cdot \dot{q} - L$, generating phase-space dynamics.
Itô lemma	Stochastic chain rule including a Hessian correction from quadratic variation.
Langevin dynamics	Stochastic dynamics combining deterministic force, dissipation, and noise.
Legendre transform	Duality operation replacing a variable by its conjugate slope.
Liouville theorem	Hamiltonian phase-space volume is preserved; the density is constant along Hamiltonian trajectories.
Macrostate	Coarse description specified by a small number of observables.
Microstate	Complete microscopic state at the model's resolution.
Mixing	Decay of correlations or asymptotic independence of transported sets; stronger than ergodicity.
Ornstein–Uhlenbeck process	Linear mean-reverting Gaussian Markov diffusion.

Partition function	Normalizer $Z = \sum \mathrm{e}^{-\beta E}$ or $\int \mathrm{e}^{-\beta H}$ generating equilibrium thermodynamics.
Probability current	Flux J satisfying $\partial_t p = -\nabla \cdot J$.
Probability-flow ODE	Deterministic ODE with the same one-time marginals as a specified diffusion.
Quenched disorder	Random environment held fixed while the dynamical variables equilibrate.
Replica	Independent copy of dynamical variables sharing the same quenched disorder.
Replica symmetry breaking	Hierarchical non-equivalence of overlaps among pure states in a glassy phase.
Reverse-time SDE	Diffusion whose drift contains the marginal score and reproduces a forward diffusion in reverse.
Score	Spatial log-density gradient $\nabla_x \log p(x)$.
Score matching	Objective for learning a density's score without its normalization constant.
Stationary action	Vanishing first variation of the action under admissible endpoint-fixed perturbations.
Stationary distribution	Distribution invariant under the dynamics; it need not satisfy detailed balance.
Stochastic interpolant	Random path construction connecting endpoint distributions and inducing velocity and score fields.
Symplectic	Preserving the canonical two-form; the defining geometry of Hamiltonian flow.
TAP equations	Mean-field self-consistency equations with an Onsager reaction correction.
Thermal bath	Large environment exchanging energy with a smaller system and fixing its temperature.
Typicality	Concentration of macroscopic observables for most microscopic states under an ensemble.
Velocity field	Deterministic state-space field whose flow transports particles and probability density.

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